MATHEMATICS

SEQUENCE & PROGRESSION

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SEQUENCE & PROGRESSION

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AP & AM BASED

1. There are n A.M's between 3 and 54, such that the 8th mean: $(n-2)^{th}$ mean:: 3: 5. The value of n is :

   यदि 3 और 54 के मध्य n समान्तर माध्य इस प्रकार हैं कि 8 वाँ माध्यः $(n-2)^{th}$ वाँ माध्य :: 3: 5 हो, तो n का मान है :

   (A) 12 (B) 16 (C) 18 (D) 20

Ans. B

Sol. $\frac{(54-3)}{n+1} = d \quad ; \quad d = \frac{51}{n+1}$

$$\frac{A_8}{A_{n-2}} = \frac{3}{5} \quad \Rightarrow \quad \frac{3+8\frac{51}{n+1}}{3+(n-2)\frac{51}{n+1}} = \frac{3}{5} \quad \Rightarrow \quad \frac{3n+3+408}{3n+3+51n-102} = \frac{3}{5}$$

$$\Rightarrow \quad 15n + 2055 = 162n - 297 \quad \Rightarrow \quad 147\,n = 2352 \quad ; \quad n = 16$$

2. Sum of the series $S = 1^2 - 2^2 + 3^2 - 4^2 + ... - 2002^2 + 2003^2$ is :

   (A) 2007006 (B) 1005004 (C) 2000506 (D) None of these

   श्रेणी $S = 1^2 - 2^2 + 3^2 - 4^2 + ... - 2002^2 + 2003^2$ का योगफल है :

   (A) 2007006 (B) 1005004 (C) 2000506 (D) इनमें से कोई नहीं

Ans. A

Sol. $S = 1^2 - 2^2 + 3^2 - 4^2 + ...- 2002^2 + 2003^2 = \sum_{n=1}^{2003} n^2 - 2\ [2^2 + 4^2 + ...+ 2002^2]$

$$= \left[\sum_{n=1}^{1001} n^2\right] = \frac{2003 \times 2004 \times 4007}{6} - \frac{8(1001)(1002)(2003)}{6}$$

$$= \frac{2003 \times 2004}{6}\ [4007 - 4004] = 2007006 \text{ Ans}$$

3. If $1, \log_9(3^{1-x}+2), \log_3(4.3^x - 1)$ are in A.P. then x equals :

   यदि $1, \log_9(3^{1-x}+2), \log_3(4.3^x - 1)$ स.श्रे. में है तो x होगा :

   (A) $\log_3 4$ (B) $1 - \log_3 4$ (C) $1 - \log_4 3$ (D) $\log_4 3$

Ans. B

Sol. $1, \log_9(3^{1-x}+2), \log 3(4\cdot 3^x-1)$ are in A.P.

$2\log_9(3^{1-x}+2) = 1+\log_3(4\cdot 3^x-1)$

$\log_3(3^{1-x}+2) = \log_3 3(4\cdot 3^x-1)$

$3^{1-x}+2 = 12\cdot 3^x - 3$

$$\frac{3}{3^x} = 12\cdot 3^x - 5$$

Let $3^x = t$

$12t^2 - 5t - 3 = 0$

$(3t+1)(4t-3) = 0$

$$t = -\frac{1}{3}, t = \frac{5}{4}$$

$$3^x = -\frac{1}{3}$$ Not possible

$$3^x = \frac{3}{4}$$

$$\log_3 3^x = \log\left(\frac{3}{4}\right)$$

$x = 1 - \log_3 4$

4. If the third term of an A.P. is 7 and its 7th term is 2 more than three times of its 3rd term, then sum of its first 20 terms is :

यदि एक समान्तर श्रेणी का तीसरा पद 7 तथा इसका 7वाँ पद इसके तीसरे पद के तीन गुने से 2 अधिक है, तो इसके प्रथम 20 पदों का योग होगा :

(A) 228 (B) 74 (C) 740 (D) 1090

Ans. C

Sol. $T_3 = 7$,

$a + 2d = 7$ ...(1)

$T_7 = 2 + 3 \times T_3 = 2 + 3 \times T$

$T_7 = 2 + 21 = 23$

$a + 6d = 23$ ...(2)

(1) – (2) : $(2d - 6d) = (7 - 23)$

$-4d = -16 \Rightarrow d = 4 \Rightarrow a = -1$

$$S_{20} = \frac{20}{2}\left[2\times(-1)+(20-1)4\right]$$

$= 10[-2 + 76]$

$= 740$

5. Let T_r be the rth term of an A.P. whose first term is a and common difference is d. If for some positive integers m, n, $m \neq n$, $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then $a - d$ equals :

माना किसी समान्तर श्रेणी का r वाँ पद T_r है। पहला पद a तथा सार्व अन्तर d है। यदि किन्हीं दो धन पूर्णांकों m तथा n,के लिए ($m \neq n$) यदि $T_m = \frac{1}{n}$ एवं $T_n = \frac{1}{m}$ हो, तो a – d बराबर है :

(A) 0 (B) 1 (C) 1/mn (D) $\frac{1}{m}+\frac{1}{n}$

Ans. A

Sol. $T_r = a + (r-1)d$

$T_m = \frac{1}{n} = a + (m-1)d$ ...(1)

$T_n = \frac{1}{m} = a + (n-1)d$ ...(2)

Solving (1), (2), we get

$a = \frac{1}{mn} = d$

$\Rightarrow a - d = 0$

6. Let $a_1, a_2, a_3, ...$ be terms of an A.P. If $\frac{a_1 + a_2 + ... + a_p}{a_1 + a_2 + ... + a_q} = \frac{p^2}{q^2}$, $p \neq q$ then $\frac{a_6}{a_{21}}$ equals :

माना $a_1, a_2, a_3, ...$ एक समान्तर श्रेणी के पद है। यदि $\frac{a_1 + a_2 + ... + a_p}{a_1 + a_2 + ... + a_q} = \frac{p^2}{q^2}$, $p \neq q$ तो $\frac{a_6}{a_{21}}$ का मान है :

(A) $\frac{7}{2}$ (B) $\frac{2}{7}$ (C) $\frac{11}{41}$ (D) $\frac{41}{11}$

Ans. C

Sol. $\because \frac{(S_m)_{IAP}}{(S_n)_{IIAP}} = \frac{m^2}{n^2}$ then $\frac{T_m}{T_n} = \frac{2m-1}{2n-1}$

so $\frac{a_6}{a_{21}} = \frac{2\times 6 - 1}{2\times 21 - 1} = \frac{11}{41}$

7. The number of natural numbers < 300 that are divisible by 6 but not by 9 is :

300 से कम प्राकृत संख्याओं की संख्या होगी, जो कि 6 से विभाजित है लेकिन 9 से नहीं :

(A) 49 (B) 37 (C) 33 (D) 16

Ans. C

Sol. Divisible by 6 $\Rightarrow$ Common difference = 6
First tem = a = 6
$T_n = a + (n-1)d = 6 + (n-1)\,6$
$6 + (n-1)\,l < 300$
$1 + n - 1 < 50$
$n < 50 \Rightarrow n = 49$
Divisible by 6, 9 $\Rightarrow$ Common difference = LCM (6, 9) = 18
First term = 18
$T_m = 18 + (m-1)18 < 300$
$18m < 300$
$m < 16{\cdot}66 \Rightarrow m = 16$
Total terms divisible by 6 but not 9 = 49 – 16 = 33

8. A person is to count 4500 currency notes. Let a_n denote the number of notes he counts in the nth minute. If $a_1 = a_2 = ... a_{10} = 150$ and $a_{10}, a_{11}, ...$ are in AP with common difference of -2, then the time taken by him to count all notes is : **[AIEEE-2010]**

(A) 24 minutes (B) 34 minutes (C) 125 minutes (D) 135 minutes

वह व्यक्ति को 4500 मुद्रा नोट गिनने है। माना nवें मिनट में वह a_n नोट गिनता है। यदि $a_1 = a_2 = ... a_{10} = 150$ तथा $a_{10}, a_{11}, ...$ एक समांतर श्रेणी में है जिसका सार्वअन्तर –2 है, तो उसे सभी नोटों को गिनने में समय लगेगा : **[AIEEE-2010]**

(A) 24 मिनट (B) 34 मिनट (C) 125 मिनट (D) 135 मिनट

Ans. B

Sol. Total currency = 4500
$a_1 = a_2 = ... = a_{10} = 150$
We count $10 \times 150 = 1500$ currency in 10 minutes
Now, total currency left 4500 – 1500 = 3000
$a_{11} = 150 - 2 = 148$
$a_{11}, a_{12}, a_{13}, ..., a_n ... = 148, 146, 144, ...$

$$S_n = \frac{n}{2}\left[2\times 148 + (n-1)(-2)\right]$$

$3000 = n\,[148 - n + 1]$
$3000 = 149n - n^2$
$n^2 - 149n + 3000 = 0$

$$n = \frac{149 \pm \sqrt{101}}{2}$$

$n_1 = 125$ not possible $n_2 = 24$
So, total minutes = 24 + 10 = 34

9. Let S_1 be the sum of first 2n terms of an arithmetic progression. Let S_2 be the sum of first 4n terms of the same aritmetic progression. If $(S_2 - S_1)$ is 1000, then the sum of the first 6n terms of the arithmetic progression is equal to:

[JEE(Main)Mar.2021]

माना एक समांतर श्रेढ़ी के प्रथम 2n पदों का योगफल S_1 है। माना उसी समांतर श्रेढ़ी के प्रथम 4n पदों का योगफल S_2 है। यदि $(S_2 - S_1) = 1000$ है, तो इस समांतर श्रेढ़ी के प्रथम 6n पदों का योग बराबर है : **[JEE(Main)Mar.2021]**

(A) 7000 (B) 5000 (C) 3000 (D) 1000

Ans. C

Sol. $S_1 = \frac{2n}{2}\left(2a + (2n-1)d\right) = n\left(2a + (2n-1)d\right)$

$S_2 = \frac{4n}{2}\left(2a + (4n-1)d\right) = 2n\left(2a + (4n-1)d\right)$

$S_2 - S_1 = 1000$

$2n\,(2a + (4n-1)d) - n\,(2a + (2n-1)d) = 1000$

$n\,[4a + (4n-1)2d - 2a - (2n-1)d] = 1000$

$n\,[2a + d\,(8n - 2 - 2n + 1)] = 1000$

$n\,[2a + d\,(6n-1)] = 1000$

$\frac{6n}{2}\left(2a + (6n-1)d\right) = 3000$

$S = 3000$

10. Let a_n be the nth term of an A.P. If $\sum_{r=1}^{100} a_{2r} = \alpha$ and $\sum_{r=1}^{100} a_{2r-1} = \beta$, then the common difference of the A.P. is :

[AIEEE-20011]

माना एक समांतर श्रेणी का n वां पद a_n है। यदि $\sum_{r=1}^{100} a_{2r} = \alpha$ तथा $\sum_{r=1}^{100} a_{2r-1} = \beta$ है, तो समांतर श्रेणी का सार्व अन्तर है :

[AIEEE-20011]

(A) $\frac{\alpha - \beta}{200}$ (B) $\alpha - \beta$ (C) $\frac{\alpha - \beta}{100}$ (D) $\beta - \alpha$

Ans. C

Sol. $\sum_{r=1}^{100} a_{2r} = \alpha$, $\sum_{r=1}^{100} a_{2r-1} = \beta$

$a_2 + a_4 + a_6 + \ldots\ldots + a_{200} = \alpha$ ...(1)

$a_1 + a_3 + a_5 + ... + a_{199} = \beta$ ...(2)
$(a_2 - a_1) + (a_4 - a_3) + ... + (a_{200} - a_{199}) = \alpha - \beta$
$d + d + ... + d = \alpha - \beta$
(100 times) $= \alpha - \beta$

$$d = \frac{\alpha - \beta}{100}$$

11. For any three positive real numbers a, b and c, $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then : **[JEE(Main)-2017]**

(A) a, b and c are in A.P. (B) a, b and c are in G.P.

(C) b, c and a are in G.P. (D) b, c and a are in A.P.

किन्हीं तीन धनात्मक वास्तविक संख्याओं a, b तथा c के लिए $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$ है, तो : **[JEE(Main)-2017]**

(A) a, b तथा c समान्तर श्रेढ़ी है (B) a, b तथा c गुणोतर श्रेढ़ी है

(C) b, c तथा a गुणोतर श्रेढ़ी है (D) b, c तथा a समान्तर श्रेढ़ी है

Ans. D

Sol. $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$
$9((5a^2) + b^2) + (5c)^2 - 3(5a)(5c) = 3(5a)(3b) + (3b)(5c)$
$(15a)^2 + (3b)^2 + (5c)^2 - (15a)(5c) - (15a)(3b) - (3b)(5a) = 0$
$(15a - 3b)^2 + (3b - 5c)^2 + (5c - 15a)^2 = 0$
$15a = 3b = 5c = k$

$$a = \frac{k}{15}, b = \frac{k}{3}, c = \frac{k}{5}$$

$$a = \left(\frac{k}{15}\right)45, b = \left(\frac{k}{3}\right)45, c = \left(\frac{k}{5}\right)45$$

$a = 3k$, $b = 15k$, $c = 9k$
$2c = a + b$
$\Rightarrow$ b,c, a are in A.P.

12. Let $a_1, a_2, a_3, ... a_{49}$ be in A.P. such that $\sum_{k=0}^{12} a_{4k+1} = 416$ and $a_9 + a_{43} = 66$. If $a_1^2 + a_2^2 + ... + a_{17}^2 = 140m$, then m is equal to : **[JEE(Main)-2018]**

माना $a_1, a_2, a_3, ... a_{49}$ एक समांतर श्रेढ़ी में ऐसे है कि $\sum_{k=0}^{12} a_{4k+1} = 416$ तथा $a_9 + a_{43} = 66$ है। यदि $a_1^2 + a_2^2 + ... + a_{17}^2 = 140m$ है, तो m बराबर है : **[JEE(Main)-2018]**

(A) 34 (B) 33 (C) 66 (D) 68

Ans. A

Sol. a_1, a_2, a_{49} ..... $\rightarrow$ A.P.

$$\sum_{k=0}^{12} a_{4k+1} = a_1 + a_5 + a_9 + ... + a_{49} = 416$$

$a_9 + a_{43} = 66$

$(a_1 + 8d) + (a_1 + 42d) = 66$

$2a_1 + 50d = 66$

$a_1 + 25d = 33$ ...(1)

$a_1 + (a_1 + 4d) + (a_1 + 8d) + (a_1 + 12d) + ... + (a_1 + 48d) = 416$

$13a_1 + 4d\,[1 + 2 + 3 + ... + 12) = 416$

$$13a_1 + 4d \times \left(\frac{12 \times 13}{2}\right) = 416$$

$13a_1 + 312d = 416$

$a_1 + 24d = 32$ ...(2)

(1) – (2) : $d = 1 \Rightarrow a_1 = 32 - 24 = 8 \Rightarrow a_1 = 8$

Now,

$$a_1^2 + a_2^2 + ... + a_{17}^2 = 140m$$

$$a_1^2 + \left(a_1^2 + d^2 + 2a_1d\right) + \left(a_1^2 + 4d^2 + 4a_1d\right) + ... + \left(a_1^2 + 256d^2 + 32a_1d\right) = 140m$$

$$17a_1^2 + \left(2a_1d\right)\left[1 + 2 + 3 + ... + 16\right] + d^2\left[1^2 + 2^2 + 3^2 + ... + 16^2\right] = 140m$$

$$17a_1^2 + \left(2a_1d\right)\left[\frac{16 \times 17}{2}\right] + d^2\left[\frac{16 \times 17 \times 33}{6}\right] = 140m$$

$$17 \times 64 + 16\left[\frac{16 \times 17}{2}\right] + \left[\frac{16 \times 17 \times 33}{6}\right] = 140m$$

$4760 = 140m \Rightarrow m = 34$

GP & GM BASED

13. If $x > 1, y > 1, z > 1$ are in GP, then $\frac{1}{1 + \ell n\, x}, \frac{1}{1 + \ell n\, y}, \frac{1}{1 + \ell n\, z}$ are in :

(A) AP (B) HP (C) GP (D) None of these

यदि $x > 1, y > 1, z > 1$ गु.श्रे. में हो, तो $\frac{1}{1 + \ell n\, x}, \frac{1}{1 + \ell n\, y}, \frac{1}{1 + \ell n\, z}$ किस श्रेढ़ी में हैं :

(A) स.श्रे. (B) ह.श्रे. (C) गु.श्रे. (D) इनमें से कोई नहीं

Ans. B

Sol. $x > 1, y > 1, z > 1$
x, y, z are in G.P.
$\Rightarrow \ln x, \ln y, \ln z$ are in A.P.
$\Rightarrow 1+\ln x, 1 + \ln y, 1 + \ln z$ are in A.P.

$\Rightarrow \frac{1}{1+\ln x}, \frac{1}{1+\ln y}, \frac{1}{1+\ln z}$ are in H.P.

14. For a sequence $\{a_n\}$, $a_1 = 2$ and $\frac{a_{n+1}}{a_n} = \frac{1}{3}$. Then $\sum_{r=1}^{20} a_r$ is :

(A) $\frac{20}{2}[4 + 19 \times 3]$ (B) $3\left(1-\frac{1}{3^{20}}\right)$ (C) $2(1 - 3^{20})$ (D) None of these

किसी अनुक्रम $\{a_n\}$ के लिए $a_1 = 2$ और $\frac{a_{n+1}}{a_n} = \frac{1}{3}$ हो, तो $\sum_{r=1}^{20} a_r$ का मान है :

(A) $\frac{20}{2}[4 + 19 \times 3]$ (B) $3\left(1-\frac{1}{3^{20}}\right)$ (C) $2(1 - 3^{20})$ (D) इनमें से कोई नहीं

Ans. B

Sol. $a_1 = 2$; $a_{n+1} = \frac{a_n}{3}$; $a_2 = \frac{a_1}{3} = \frac{2}{3}$; $a_3 = \frac{a_2}{3} = \frac{2}{3^2}$

$$a_1 + a_2 + \ldots\ldots\ldots\ldots\ldots + a_{20} = 2 + \frac{2}{3} + \frac{2}{3^2} + \ldots\ldots = \frac{2.\left[1-\left(\frac{1}{3}\right)^{20}\right]}{1-\frac{1}{3}} = 3\left(1-\frac{1}{3^{20}}\right)$$

15. α, β be the roots of the equation $x^2 - 3x + a = 0$ and γ, δ the roots of $x^2 - 12x + b = 0$ and numbers $\alpha, \beta, \gamma, \delta$ (in this order) form an increasing G.P., then :

α, β समीकरण $x^2 - 3x + a = 0$ के मूल एवं γ, δ समीकरण $x^2 - 12x + b = 0$ के मूल हैं और संख्याएँ $\alpha,\beta,\gamma,\delta$ (इसी क्रम में) वृद्धमान गु.श्रे. बनाती हैं, तो :

(A) $a = 3, b = 12$ (B) $a = 12, b = 3$ (C) $a = 2, b = 32$ (D) $a = 4, b = 16$

Ans. C

Sol. $\alpha + \beta = 3$, $\alpha\beta = a$; $\gamma + \delta = 12$, $\gamma\delta = b$
$\alpha , \beta , \gamma , \delta$ are in G.P. Let r be the common ratio so $\alpha(1+ r) = 3$
$\alpha r^2(1+ r) = 12 \quad \Rightarrow \quad r^2 = 4 \quad \Rightarrow \quad r = 2$

so $\alpha = 1$ $\Rightarrow$ so $a = 2$, $b = 32$ Ans

Hindi. $\alpha + \beta = 3$, $\alpha\beta = a$; $\gamma + \delta = 12$, $\gamma\delta = b$

α , β , γ , δ गुणोत्तर श्रेढ़ी में है। माना कि r सार्वअनुपात है।

अतः $\alpha(1+r) = 3$ $\Rightarrow$ $\alpha r^2(1+r) = 12 \Rightarrow$ $r^2 = 4$ $\Rightarrow$ $r = 2$

अतः $\alpha = 1$. अतः $a = 2$, $b = 32$ Ans

16. If the sum of n terms of a G.P. (with common ratio r) beginning with the pth term is k times the sum of an equal number of terms of the same series beginning with the qth term, then the value of k is :

यदि एक गु.श्रे. (सार्वअन्तर r) के p वें पद से प्रारम्भ होने वाले n पदों का योगफल, उसी श्रेढ़ी के q वें पद से प्रारम्भ होने वाले समान पदों के योग का k गुना हो, तो k का मान है :

(A) $r^{p/q}$ (B) $r^{q/p}$ (C) r^{p-q} (D) r^{p+q}

Ans. C

Sol. $\frac{ar^{p-1}(r^n-1)}{r-1} = k \cdot \frac{ar^{q-1}(r^n-1)}{r-1}$; $r^{p-1} = k \cdot r^{q-1}$; $k = r^{p-q}$

17. In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4th, 6th and 8th terms is equal to : **[JEE(Main)Feb.2021]**

एक वर्धमान गुणोत्तर श्रेणी में, दूसरे तथा छठे पदों का योगफल $\frac{25}{2}$ है तीसरे तथा पाँचवें पदों का गुणनफल 25 है, तो चौथे, छठे तथा आठवें पदों का योगफल है : **[JEE(Main)Feb.2021]**

(A) 35 (B) 30 (C) 32 (D) 26

Ans. A

Sol. Let first term = a and common ratio = r.

Given $ar + ar^5 = \frac{25}{2}$ .....(1)

and $ar^2 \cdot ar^4 = 25 \Rightarrow ar^3 = 5$ .....(2)

From (1) & (2) ; $r = \sqrt{2}$ and $a = \frac{5}{2\sqrt{2}}$.

$t_4 + t_6 + t_8 = ar^3 + ar^5 + ar^7$

$= ar^3(1 + r^2 + r^4)$

$= 5(1 + 2 + 4) = 35$

18. In a geometric progression consisting of positive terms, each term equals the sum of the next two terms. Then the common ratio of this progression equals :

यदि किसी धनात्मक गुणोत्तर श्रेणी का प्रत्येक पद आने वाले पदों का योग हो, तो सार्वअन्तर होगा :

(A) $\frac{1}{2}\left(1-\sqrt{5}\right)$ (B) $\frac{1}{2}\sqrt{5}$ (C) $\sqrt{5}$ (D) $\frac{1}{2}\left(\sqrt{5}-1\right)$

Ans. D

Sol. $a = ar + ar^2 \quad \Rightarrow \quad r^2 + r - 1 = 0 \quad \Rightarrow \quad r = \frac{-1 \pm \sqrt{5}}{2}$

$\Rightarrow \quad r = \frac{\sqrt{5}-1}{2}$ (–ve not permissible)

(ऋणात्मक संभव नहीं है)

19. Three positive numbers form an increasing G.P. If the middle term in this G.P. is doubled, the new numbers are in A.P. Then the common ratio of the G.P. is : **[JEE(Main)-2014]**

तीन धनात्मक संख्याएँ एक वर्द्धमान गुणोत्तर श्रेणी बनाती है, यदि इसके मध्य को दुगना कर दिया जाये तो वे एक समान्तर श्रेणी में होंगे, तो गुणोत्तर श्रेणी का सार्वअन्तर होगा : **[JEE(Main)-2014]**

(A) $2-\sqrt{3}$ (B) $2+\sqrt{3}$ (C) $\sqrt{2}+\sqrt{3}$ (D) $3+\sqrt{2}$

Ans. B

Sol. $\frac{a}{r}$, a, ar are in G.P.

$\frac{a}{r}$, 2a, ar are in A.P.

$2(2a) = \frac{a}{r} + ar$

$4 = \frac{1}{r} + r$

$r^2 + 1 = 4r$

$r^2 - 4r + 1 = 0$

$r = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}$

$\because$ Our sequence in increasing G.P., so $r > 1$

$r = 2 + \sqrt{3}$

20. If m is the A.M. of two distinct real numbers l and n (l, n > 1) and G_1, G_2 and G_3 are three geometric means between l and n, then $G_1^4 + 2G_2^4 + G_3^4$ equals : **[JEE(Main)-2015]**

यदि दो विभिन्न वास्तविक संख्याओं l तथा n (l, n > 1) का समांतर माध्य (A.M.) m है और l तथा n के बीच तीन गुणोतर माध्य (G.M.) G_1, G_2 तथा G_3 हैं, तो $G_1^4 + 2G_2^4 + G_3^4$ बराबर है : **[JEE(Main)-2015]**

(A) $4l^2mn$ (B) $4lm^2n$ (C) $4lmn^2$ (D) $4l^2m^2n^2$

Ans. B

Sol. $m = \frac{n+\ell}{2}$ ...(1)

ℓ, G, G_2, G_3, n are in G.P.

$\Rightarrow G_1 = \ell r, G_2 = \ell r^2, G_3 = \ell r^3, n = \ell r^4$

$n = \ell r^4 \Rightarrow r = \left(\frac{n}{\ell}\right)^{\frac{1}{4}}$

Now $G_1^4 + 2G_2^4 + G_3^4 = \left[\ell\left(\frac{n}{\ell}\right)^{\frac{1}{4}}\right]^4 + 2\left[\ell^4\left(\frac{n}{\ell}\right)^{\frac{2}{4}}\right]^4 + \left[\ell\left(\frac{n}{\ell}\right)^{\frac{3}{4}}\right]^4$

$= \ell^4\left(\frac{n}{\ell}\right) + 2\left[\ell^4\left(\frac{n}{\ell}\right)^2\right] + \left[\ell^4\left(\frac{n}{\ell}\right)^3\right]$

$= n\ell^3 + 2\ell^2n^2 + \ell n^3$

$= n\ell\left(\ell^2 + 2\ell n + n^2\right)$

$= n\ell\left(\ell + n\right)^2$

$= n\ell\left(2m\right)^2$ (By equation (1))

$= 4m^2n\ell$

$= 4\ell m^2n$

21. If the 2nd, 5th and 9th terms of a non-constant A.P. are in G.P., then the common ratio of this G.P. is : **[JEE(Main)-2016]**

यदि एक अचरेतर समान्तर श्रेढ़ी का 2nd, 5th तथा 9th पद एक गुणोत्तर श्रेढ़ी में है, तो उस गुणोत्तर श्रेढ़ी का सार्व अनुपात है : **[JEE(Main)-2016]**

(A) 1 (B) 7/4 (C) 8/5 (D) 4/3

Ans. D

Sol. T_2, T_5, T_9 are in G.P.
$(A + d), (a + 4d), (a + 8d)$ are in G.P.

$$r = \frac{a+4d}{a+d} = \frac{a+8d}{a+4d}$$

$\Rightarrow (a + 4d)^2 = (a + d)(a + 8d)$
$a^2 + 16d^2 + 8ad = a^2 + 8d^2 + 9ad$
$8d^2 - ad = 0$
$d(8d - a) = 0$
$d \neq 0 \Rightarrow a = 8d$

$$\text{Now} \Rightarrow r = \frac{a+4d}{a+d} = \frac{12d}{9d} = \frac{4}{3}$$

AM GM HM

22. If A, G and H are respectively the A.M., G.M. & H.M. of three positive numbers a, b, and c, then the equation whose roots are a, b, and c is given by :

यदि तीन धनात्मक संख्याओं a,b,c का समान्तर माध्य, गुणोत्तर माध्य एवं हरात्मक माध्य क्रमशः A, G एवं H हो, तो वह समीकरण जिसके मूल a,b,c हैं, होगी :

(A) $x^3 - 3Ax^2 + 3G^3x - G^3 = 0$ (B) $x^3 - 3Ax^2 + 3(G^3/H)x - G^3 = 0$
(C) $x^3 + 3Ax^2 + 3(G^3/H)x - G^3 = 0$ (D) $x^3 - 3Ax^2 - 3(G^3/H)x + G^3 = 0$

Ans. B

Sol. $AM = A = \frac{a+b+c}{3}$; $GM = G = (abc)^{1/3}$

$$HM = H = \frac{3abc}{ab+bc+ca} = \frac{3G^3}{ab+bc+ca}$$

Equation whose roots are a,b,c $\Rightarrow x^3 - (a + b + c)x^2 + (\Sigma ab)x - abc = 0$

$$\Rightarrow x^3 - 3Ax^2 + \frac{3G^3}{H}.x - G^3 = 0 \text{ Ans}$$

Hindi. समान्तर माध्य $= A = \frac{a+b+c}{3}$; गुणोत्तर माध्य $= G = (abc)^{1/3}$

$$\text{हरात्मक माध्य} = H = \frac{3abc}{ab+bc+ca} = \frac{3G^3}{ab+bc+ca}$$

समीकरण जिसके मूल a,b,c हो, निम्न है –

$\Rightarrow x^3 - (a + b + c)x^2 + (\Sigma ab)x - abc = 0$ $\Rightarrow x^3 - 3Ax^2 + \frac{3G^3}{H}.x - G^3 = 0$ Ans

23. If $x_i > 0$, i = 1, 2, ..., 50 and $x_1 + x_2 + .. + x_{50} = 50$, then the minimum value of $\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_{50}}$ equal to :

यदि $x_i > 0$, i = 1, 2, ..., 50 और $x_1 + x_2 + .. + x_{50} = 50$ हो, तो $\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_{50}}$ का न्यूनतम मान है :

(A) 50 (B) $(50)^2$ (C) $(50)^3$ (D) $(50)^4$

Ans. A

Sol. $x_1 + x_2 + x_3 + ... + x_{50} = 50$

AM ≥ HM

$$\frac{x_1 + x_2 + ... + x_{50}}{50} \geq \frac{1}{\frac{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + ... + \frac{1}{x_{50}}}{50}}$$

$$\Rightarrow \frac{x_1 + x_2 + ... + x_{50}}{50} \geq \frac{50}{\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_{50}}} \Rightarrow \frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_{50}} \geq 50$$

so min value of $\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_{50}} = 50$ अतः $\frac{1}{x_1} + \frac{1}{x_2} + ... + \frac{1}{x_{50}}$ का न्यूनतम मान = 50

24. If A_1, A_2 be two A.M.s and G_1, G_2 be two G.M.s betwee n a and b, then $\frac{A_1 + A_2}{G_1 G_2}$ is equal to :

यदि a और b के मध्य दो समान्तर माध्य A_1, A_2 और दो गुणोत्तर माध्य G_1, G_2 हो, तो $\frac{A_1 + A_2}{G_1 G_2}$ का मान है :

(A) $\frac{a+b}{2ab}$ (B) $\frac{2ab}{a+b}$ (C) $\frac{a+b}{ab}$ (D) $\frac{a+b}{\sqrt{ab}}$

Ans. C

Sol. a, A_1, A_2, b are in A.P.

$A_1 = a + d$, $A_2 = a + 2d$, $b = a + 3d \Rightarrow \frac{b-a}{3} = d$

$$A_1 + A_2 = 2a + 3d = 2a + 3\left(\frac{b-a}{3}\right)$$

$$= 2a + b - a = a + b$$

a, G_1, G_2, b are in G.P.
$G_1 = ar$, $G_2 = ar^2$, $b = ar^3$
$G_1G_2 = a^2r^3 = ab$

Now $\frac{A_1 + A_2}{G_1G_2} = \frac{a+b}{ab}$

25. If P, Q be the A.M., G.M. respectively between any two rational numbers a and b, then P – Q is equal to :

किन्हीं दो परिमेय संख्याओं a एवं b के मध्य गुणोत्तर माध्य एवं समान्तर माध्य क्रमशः P एवं Q हो, तो P – Q का मान है :

(A) $\frac{a-b}{a}$ (B) $\frac{a+b}{2}$ (C) $\frac{2ab}{a+b}$ (D) $\left(\frac{\sqrt{a}-\sqrt{b}}{\sqrt{2}}\right)^2$

Ans. D

Sol. $P = \frac{a+b}{2}$, $Q = \sqrt{ab}$; $P - Q = \frac{a+b-2\sqrt{ab}}{2} = \frac{\left(\sqrt{a}-\sqrt{b}\right)^2}{2}$

26. If $\alpha \in \left(0, \frac{\pi}{2}\right)$ then $\sqrt{x^2+x} + \frac{\tan^2\alpha}{\sqrt{x^2+x}}$ is always greater than or equal to :

यदि $\alpha \in \left(0, \frac{\pi}{2}\right)$ हो, तो $\sqrt{x^2+x} + \frac{\tan^2\alpha}{\sqrt{x^2+x}}$ का न्यूनतम मान है :

(A) $2\tan\alpha$ (B) 1 (C) 2 (D) $\sec^2\alpha$

Ans. A

Sol. $\sqrt{x^2+x} + \frac{\tan^2\alpha}{\sqrt{x^2+x}} \geq 2\sqrt{\left(\sqrt{x^2+x}\right)\times\left(\frac{\tan^2\alpha}{\sqrt{x^2+x}}\right)} = 2\tan\alpha$

HP & HM BASED

27. The harmonic mean of the roots of the equation $\left(5+\sqrt{2}\right)x^2 - \left(4+\sqrt{5}\right)x + 8 + 2\sqrt{5} = 0$ is :

समीकरण $\left(5+\sqrt{2}\right)x^2 - \left(4+\sqrt{5}\right)x + 8 + 2\sqrt{5} = 0$ के मूलों का हरात्मक माध्य है :

(A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. $\left(5+\sqrt{2}\right)x^2-\left(4+\sqrt{5}\right)x+\left(8+2\sqrt{5}\right)=0$

$$\alpha+\beta=\frac{+\left(4+\sqrt{5}\right)}{\left(5+\sqrt{2}\right)} \qquad \alpha\beta=\frac{8+2\sqrt{5}}{5+\sqrt{2}}$$

$$H=\frac{2\alpha\beta}{\alpha+\beta}=\frac{2\left(8+2\sqrt{5}\right)}{4+\sqrt{5}}=4$$

28. If $a_1, a_2, ... a_n$ are in H.P., then the expression $a_1a_2 + a_2a_3 + ... + a_{n-1}a_n$ is equal to :

यदि $a_1, a_2, ... a_n$ हरात्मक श्रेणी में है, तो व्यंजक $a_1a_2 + a_2a_3 + ... + a_{n-1}a_n$ का मान है :

(A) $(n-1)(a_1-a_n)$ (B) na_1a_n (C) $(n-1)a_1a_n$ (D) $n(a_1-a_n)$

Ans. C

Sol. $\frac{1}{a_1}, \frac{1}{a_2}, ..., \frac{1}{a_n}$ are in A.P. समान्तर श्रेढ़ी में है।

$$\Rightarrow \quad \frac{1}{a_2}-\frac{1}{a_1}=.....=\frac{1}{a_n}-\frac{1}{a_{n-1}}=d$$

$$\Rightarrow \quad a_1a_2=\frac{a_1-a_2}{d},\ a_2a_3=\frac{a_2-a_3}{d}.....,\ \frac{a_{n-1}-a_n}{d}=a_{n-1}-a_n$$

$$\Rightarrow \quad a_1a_2+a_2a_3+.....+a_{n-1}a_n=\frac{a_1-a_2}{d}+.....+\frac{a_{n-1}-a_n}{d}=\frac{a_1-a_n}{d}=(n-1)a_1a_n$$

29. If $x=\sum_{n=0}^{\infty}a^n, y=\sum_{n=0}^{\infty}b^n, z=\sum_{n=0}^{\infty}c^n$ where a, b, c are in A.P. and $|a|<1, |b|<1, |c|<1$ then x, y, z are in :

(A) GP (B) AP

(C) Arithmetic - Geometric Progression (D) HP

यदि $x=\sum_{n=0}^{\infty}a^n, y=\sum_{n=0}^{\infty}b^n, z=\sum_{n=0}^{\infty}c^n$ जबकि a, b, c समान्तर श्रेणी में है तथा $|a|<1, |b|<1$ तथा $|c|<1$ तो x, y तथा z:

(A) गुणोत्तर श्रेणी में है (B) समान्तर श्रेणी में है

(C) समान्तर गुणोत्तर श्रेणी में है (D) हरात्मक श्रेणी में है

Ans. D

Sol. $x=\frac{1}{1-a}, y=\frac{1}{1-b}, z=\frac{1}{1-c}$ $\Rightarrow$ a, b, c are in A.P. समान्तर श्रेढ़ी में है।

$\Rightarrow$ 1 – a, 1 – b, 1 – c are also in A.P. भी समान्तर श्रेढ़ी में है।

$\Rightarrow$ $\frac{1}{1-a}, \frac{1}{1-b}, \frac{1}{1-c}$ are in H.P. हरात्मक श्रेढ़ी में है।

AGP

30. The sum of infinite series $1-\frac{3}{2}+\frac{5}{4}-\frac{7}{8}+.....$ is :

श्रेढ़ी $1-\frac{3}{2}+\frac{5}{4}-\frac{7}{8}+.....$ के अनन्त पदों का योगफल है :

(A) 2/9 (B) 2/3 (C) – 2/9 (D) 9/2

Ans. A

Sol. $S=1-\frac{3}{2}+\frac{5}{4}-\frac{7}{8}+.....$

$\frac{S}{2}=\frac{1}{2}-\frac{3}{4}+\frac{5}{8}+.....$

$\frac{3S}{2}=1-\frac{2}{2}+\frac{2}{4}-\frac{2}{8}+....$

$=1-2\left(\frac{\frac{1}{2}}{1+\frac{1}{2}}\right)$

$\frac{3S}{2}=1-2\left(\frac{1}{3}\right)=\frac{+1}{3}$

$S=\frac{2}{9}$

31. If $(10)^9+2(11)^1(10)^8+3(11)^2(10)^7+....+10(11)^9=k(10)^9$, then k is equal to : **[JEE(Main)-2014]**

यदि $(10)^9+2(11)^1(10)^8+3(11)^2(10)^7+....+10(11)^9=k(10)^9$ है, तो k बराबर है : **[JEE(Main)-2014]**

(A) 100 (B) 110 (C) 121/10 (D) 441/100

Ans. A

Sol. $(10)^9+2(11)^1(10)^8+3(11)^2(10)^7+.....+10(11)^9=k(10)^9$

$$1+2\left(\frac{11}{10}\right)+3\left(\frac{11}{10}\right)^2+.....+10\left(\frac{11}{10}\right)^9=k \quad ...(1)$$

$$0+\left(\frac{11}{10}\right)+2\left(\frac{11}{10}\right)^2+.....+9\left(\frac{11}{10}\right)^9+10\left(\frac{11}{10}\right)^{10}=\frac{11k}{10} \quad ...(2)$$

(2) – (1)

$$\frac{k}{10}=10\left(\frac{11}{10}\right)^{10}-\left[1+\left(\frac{11}{10}\right)+\left(\frac{11}{10}\right)^2+....+\left(\frac{11}{10}\right)^9\right]$$

$$\frac{k}{10}=10\left(\frac{11}{10}\right)^{10}-\left[1\cdot\left(\frac{\left(\frac{11}{10}\right)^{10}-1}{\frac{11}{10}-1}\right)\right]$$

$$\frac{k}{10}=10\left(\frac{11}{10}\right)^{10}-\left[10\left(\frac{11}{10}\right)^{10}-10\right]$$

$$\frac{k}{10}=10$$

k = 100

MISCELLANEOUS SEQUENCES

32. The sum of the series $1^3-2^3+3^3-.....+9^3=$

श्रेणी $1^3-2^3+3^3-.....+9^3$ का योग है :

(A) 300 (B) 125 (C) 425 (D) 0

Ans. C

Sol. $S=1^3-2^3+3^3-4^3+.....+9^3$

$r=1$

$S=(1^3+3^3+5^3+7^3+9^3)-(2^3+4^3+6^3+8^3)$

$=(1^3+3^3+5^3+7^3+9^3)-8(1^3+2^3+3^3+4^3)$

$=(1^3+2^3+3^3+4^3+5^3+6^3+7^3+8^3+9^3)-(2^3+4^3+6^3+8^3)-8\,(1^3+2^3+3^3+4^3)$

$$=\left[\frac{9(10)}{2}\right]^2-16\left[\frac{4\times5}{2}\right]^2=425$$

33. If $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....$ upto $\infty=\frac{\pi^2}{6}$, then $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+.....=$

(A) $\pi^2/12$ (B) $\pi^2/24$ (C) $\pi^2/8$ (D) None of these

यदि $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....=\frac{\pi^2}{6}$ हो, तो $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+..... =$

(A) $\pi^2/12$ (B) $\pi^2/24$ (C) $\pi^2/8$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\because \quad I=\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....\infty=\frac{\pi^2}{6}$

Let $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+.....=A \quad \because \quad I=\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+.....\infty\right)+\frac{1}{2^2}\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....\infty\right)$

$\Rightarrow \quad I=A+\frac{I}{4}\Rightarrow A=\frac{3I}{4}=\frac{3}{4}\times\frac{\pi^2}{6} \quad \Rightarrow A=\frac{\pi^2}{8}$

HIndi $\because \quad I=\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....\infty=\frac{\pi^2}{6}$

माना $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+.....=A \quad \because \quad I=\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+.....\infty\right)+\frac{1}{2^2}\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....\infty\right)$

$\Rightarrow \quad I=A+\frac{I}{4}\Rightarrow A=\frac{3I}{4}=\frac{3}{4}\times\frac{\pi^2}{6} \quad \Rightarrow A=\frac{\pi^2}{8}$

34. The sum of first 9 terms of the series $\frac{1^3}{1}+\frac{1^3+2^3}{1+3}+\frac{1^3+2^3+3^3}{1+3+5}+.....$ is : **[JEE(Main)2015]**

श्रेणी $\frac{1^3}{1}+\frac{1^3+2^3}{1+3}+\frac{1^3+2^3+3^3}{1+3+5}+.....$ के प्रथम 9 पदों का योग है : **[JEE(Main)-2015]**

(A) 71 (B) 96 (C) 142 (D) 192

Ans. B

Sol. $T_n=\frac{1^3+2^3+3^3+....+n^3}{1+3+5+.....+(2n-1)}=\frac{\left[\frac{n(n+1)}{2}\right]}{n^2}$

$T_n=\left(\frac{n+1}{2}\right)^2$

$S_9=T_1+T_2+T_3+....+T_9$

$$= \frac{1}{(2)^2}\left[(2)^2+(3)^2+(4)^2+.....+(10)^2\right]$$

$$= \frac{1}{4}\left[\left[(1)^2+(2)^2+(3)^2+.....+(10)^2\right]-1\right]$$

$$= \frac{1}{4}\left[\left\{\frac{10\times11\times21}{6}\right\}-1\right] = 96$$

35. If the sum of the first ten terms of the series $\left(1\frac{3}{5}\right)^2+\left(2\frac{2}{5}\right)^2+\left(3\frac{1}{5}\right)^2+4^2+\left(4\frac{4}{5}\right)^2+.....$, is $\frac{16}{5}$m, then m is :

[JEE(Main)-2016]

यदि श्रेणी $\left(1\frac{3}{5}\right)^2+\left(2\frac{2}{5}\right)^2+\left(3\frac{1}{5}\right)^2+4^2+\left(4\frac{4}{5}\right)^2+.....$, के प्रथम दस पदों का योग $\frac{16}{5}$m है, तो m बराबर है :

[JEE(Main)-2016]

(A) 100 (B) 99 (C) 102 (D) 101

Ans. D

Sol. $\left(1\frac{3}{5}\right)^2+\left(2\frac{2}{5}\right)^2+\left(3\frac{1}{5}\right)^2+4^2+\left(4\frac{4}{5}\right)^2+.....$

$$\left(\frac{8}{5}\right)^2+\left(\frac{12}{5}\right)^2+\left(\frac{16}{5}\right)^2+(4)^2+\left(\frac{24}{5}\right)^2+.....$$

$$\left(\frac{8}{5}\right)^2+\left(\frac{12}{5}\right)^2+\left(\frac{16}{5}\right)^2+\left(\frac{20}{5}\right)^2+\left(\frac{24}{5}\right)^2+.....$$

$$\left(\frac{8}{5}\right)^2+\left(\frac{12}{5}\right)^2+.....+\left(\frac{44}{5}\right)^2=\frac{16}{5}m$$

$$\left(\frac{4}{5}\right)^2\left[(2)^2+(3)^2+.....+(11)^2\right]=\frac{16}{5}m$$

$$\frac{1}{5}\left[\frac{11\times12\times23}{6}-1\right]=m \Rightarrow m=101$$

36. If, for a positive integer n, the quadratic equation, $x(x+1)+(x+1)(x+2)+.....+(x+\overline{n-1})(x+n)=10n$ has two consecutive integral solutions, then n is equal to : **[JEE(Main)2017]**

यदि किसी घनपूर्णांक n के लिए, द्विघाती समीकरण $x(x+1)+(x+1)(x+2)+.....+(x+\overline{n-1})(x+n)=10n$ के दो क्रमित पूर्णांकीय हल है, तो n बराबर है : **[JEE(Main)-2017]**

(A) 10 (B) 11 (C) 12 (D) 9

Ans. B

Sol. $\sum_{r=1}^{n}\left(x+(r-1)\right)(x+r)=10n$

$$\sum_{r=1}^{n}\left[x^2+x(2r-1)+\left(r^2-r\right)\right]=10n$$

$$\left(x^2-x\right)n+n(n+1)x+\frac{n(n+1)(2n+1)}{6}-\frac{n(n+1)}{2}=10n$$

$$10=x^2-x+1(n+1)x+\left(\frac{n^2-1}{3}\right)$$

$x^2+nx+\left(\frac{n^2-31}{3}\right)=0$ α , $\alpha+1$ are roots of this quadratic equation.

$2\alpha+1=-n \quad \Rightarrow \alpha=-\left(\frac{n+1}{2}\right)$

$$\alpha(\alpha+1)=\frac{n^2-31}{3}$$

$$-\left(\frac{n+1}{2}\right)\left(\frac{1-n}{2}\right)=\frac{n^2-31}{3}$$

$3(n^2-1)=4n^2-124$
$n^2-121=0$
$n=\pm 11$
$\Rightarrow n=11$

37. Let A be the sum of the first 20 terms and B be sum of the first 40 terms of the series $1^2+2.2^2+3^2+2.4^2+5^2+2.6^2+.....$ If $B-2A=100\lambda$, then λ is equal to : **[JEE(Main)-2018]**

माना श्रेणी $1^2+2.2^2+3^2+2.4^2+5^2+2.6^2+.....$ के प्रथम 20 पदों का योग A है तथा प्रथम 40 पदों का योग B है। यदि $B-2A=100\lambda$, तो λ बराबर है : **[JEE(Main)-2018]**

(A) 464 (B) 496 (C) 232 (D) 248

Ans. D

Sol. $1^2+2\cdot2^2+3^2+2\cdot4^2+5^2+2\cdot6^2+.....$

$S_{20} = A$, $S_{40} = B$

$A = S_{20} = 1^2 + 2\cdot2^2 + 3^2 + ... + 19^2 + 2\cdot(20)^2$

$= [1^2 + 3^2 + 5^2 + ... + 19^2] + [2\cdot2^2 + 2\cdot4^2 + ... + 2\cdot(20)^2]$

$= [1^2 + 3^2 + 5^2 + ... + 19^2] + 2\cdot2^2\,[1 + 2^2 + 3^2 + 10^2]$

$= [1^2 + 2^2 + 3^2 + ... + 19^2 + 20^2] + 8[1^2 + 2^2 + ... + 10^2] - [2^2 + 4^2 + 6^2 + ... + 20^2]$

$$A = \left(\frac{20\times21\times41}{6}\right) + 4\left(\frac{10\times11\times21}{6}\right)$$

$B = 1^2 + 2\cdot2^2 + ... + (39)^2 + (40)^2$

$= [1^2 + 2^2 + 3^2 + ... + (40)^2] + 2\cdot2^2\,[1 + 2^2 + 3^2 + ... + (20)^2] - 4[1 + 2^2 + ... + (20)^2]$

$$B = \left(\frac{40\times41\times81}{6}\right) + 4\left(\frac{20\times21\times41}{6}\right)$$

$$B - 2A = \left(\frac{40\times41\times81}{6}\right) + 4\left(\frac{20\times21\times41}{6}\right) - 2\left(\frac{20\times21\times41}{6}\right) - 8\left(\frac{10\times11\times21}{6}\right)$$

$= 24800 = 100\lambda \Rightarrow \lambda = 248$

38. The coefficient of x^8 in the polynomial $(x-1)(x-2)(x-3) ..... (x-10)$ is :

बहुपद $(x-1)(x-2)(x-3) ..... (x-10)$ में x^8 का गुणांक होगा :

(A) 2640 (B) 1320 (C) 1370 (D) 2740

Ans. B

Sol. $f(x) = (x-1)(x-2)(x-3) ..... (x-9)(x-10)$

$x^{10} - [1 + 2 + ... + 10)x^9 + (1\cdot2 + 1\cdot3 + 1\cdot4 + ...)\,x^8 + ...$

Coefficient of $x^8 = (1\cdot2 + 1\cdot3 + ... + 1\cdot10) + (2\cdot3 + ... + 2\cdot10) + (3\cdot4 + ... + 3\cdot10) + ... + (9\cdot10)$

$(1 + 2 + 3 + ... + 10)^2 = (1^2 + 2^2 + ... + 10^2) + 2\,[(1\cdot2 + ... + 1\cdot10) + ... + (9\cdot10)]$

$$\left(\frac{10\times11}{2}\right)^2 = \frac{10\times11\times21}{6} + 2\ [\text{Coefficient of } x^8]$$

$$\frac{2640}{2} = \text{coefficient of } x^8$$

Coeff. of $x^8 = 1320$

39. Find the value of $\frac{2}{1^3} + \frac{6}{1^3+2^3} + \frac{12}{1^3+2^3+3^3} + \frac{20}{1^3+2^3+3^3+4^3} + ...$ upto ∞ terms :

$\frac{2}{1^3} + \frac{6}{1^3+2^3} + \frac{12}{1^3+2^3+3^3} + \frac{20}{1^3+2^3+3^3+4^3} + ...$ से ∞ पदों तक का मान होगा :

(A) 2 (B) $\frac{1}{2}$ (C) 4 (D) $\frac{1}{4}$

Ans. C

Sol. $S=\frac{2}{1^3}+\frac{6}{1^3+2^3}+\frac{12}{1^3+2^3+3^3}+\frac{20}{1^3+2^3+3^3+4^3}+.....$

$$T_n=\frac{n(n+1)}{\left(\frac{n(n+1)}{2}\right)^2}=\frac{4}{n(n+1)}$$

$$S=\sum_{n=1}^{\infty}\frac{4}{n(n+1)}=4\sum_{n=1}^{\infty}\frac{(n+1)-n}{n(n+1)}$$

$$S=4\sum_{n=1}^{\infty}\left[\frac{1}{n}-\frac{1}{n+1}\right]=4$$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

AP & AM BASED

1. If $A_1, A_2, A_3, \ldots\ldots A_{51}$ are arithmetic means inserted between the numbers a and b, then find the value of $\left(\frac{b+A_{51}}{b-A_{51}}\right)-\left(\frac{A_1+a}{A_1-a}\right)$.

यदि a व b के मध्य $A_1, A_2, A_3, \ldots\ldots A_{51}$ है तो $\left(\frac{b+A_{51}}{b-A_{51}}\right)-\left(\frac{A_1+a}{A_1-a}\right)$ का मान बताइये।

Ans. 102

Sol. $a, A_1, A_2, \ldots\ldots\ldots, A_{51}, b$ are in A.P.

$b = a + 52d$

$$\left(\frac{b+A_{51}}{b-A_{51}}\right)-\left(\frac{A_1+a}{A_1-a}\right)=\left(\frac{b+a+51d}{b-a-51d}\right)-\left(\frac{2a+d}{d}\right)$$

$$=\left(\frac{a+52d+a+51d}{d}\right)-\left(\frac{2a+d}{d}\right)$$

$$=\frac{2a+103d-2a-d}{d}$$

$= 102$

2. Let $a_1, a_2, a_3.....a_{11}$ be real numbers satisfying $a_1 = 15$, $27 - 2a_2 > 0$ and $a_k = 2a_{k-1} - a_{k-2}$ for $k = 3, 4..... 11$. If $\frac{a_1^2+a_2^2+.....+a_{11}^2}{11}=90$, then the value of $\frac{a_1+a_2+.....+a_{11}}{11}$ is equal to. **[JEE-2010]**

यदि $a_1, a_2, a_3.....a_{11}$ वास्तविक संख्याऐं है तथा $a_1 = 15$, $27 - 2a_2 > 0$ और $a_k = 2a_{k-1} - a_{k-2}$ जहाँ $k = 3, 4..... 11$. यदि $\frac{a_1^2+a_2^2+.....+a_{11}^2}{11}=90$ तो $\frac{a_1+a_2+.....+a_{11}}{11}$ का मान बताइये। **[JEE-2010]**

Ans. 0

Sol. $a_1 = 15$

$\frac{a_k+a_{k-2}}{2} = a_{k-1}$ for $k = 3, 4, ..., 11$ के लिए $\Rightarrow$ $a_1, a_2, ..., a_{11}$ are in AP समान्तर श्रेढी में है।

$a_1 = a = 15$

$$\frac{a_1^2 + a_2^2 + ... + a_n^2}{11} = 90 \quad \Rightarrow \quad \frac{(15)^2 + (15+d)^2 + ... + (15+10d)^2}{11} = 90$$

$\Rightarrow \quad 7d^2 + 30d + 27 = 0 \quad \Rightarrow \quad d = -3 \text{ or } -\frac{9}{7}$

Since चूंकि $27 - 2a_2 > 0 \quad \Rightarrow \quad a_2 < \frac{27}{2} \quad \Rightarrow \quad d = -3$

$$\frac{a_1 + a_2 + ... + a_{11}}{11} = \frac{11}{2}\ \frac{[30 + 10(-3)]}{11} = 0$$

3. Let $a_1, a_2, a_3, ..., a_{100}$ be an arithmetic progression with $a_1 = 3$ and $S_p = \sum_{i=1}^{p} a_i$, $1 \le p \le 100$. For any integer n with $1 \le n \le 20$, let $m = 5n$. If $\frac{S_m}{S_n}$ does not depend on n, then a_2 is. **[JEE-2011]**

माना $a_1, a_2, a_3, ..., a_{100}$ एक समान्तर श्रेणी में हो जिसके लिये $a_1 = 3$ तथा $S_p = \sum_{i=1}^{p} a_i$, $1 \le p \le 100$, किसी पूर्णांक n के लिये $1 \le n \le 20$. यदि $m = 5n$ तथा $\frac{S_m}{S_n}$, n के मान पर निर्भर नहीं करता है तो a_2 होगा। **[JEE-2011]**

Ans. 3 or 9

Sol. $\frac{S_m}{S_n} = \frac{S_{5n}}{S_n} = \frac{\frac{5n}{2}[6 + (5n-1)d]}{\frac{n}{2}[6 + (n-1)d]} = \frac{5[(6-d) + 5nd]}{[(6-d) + nd]}$; $d = 6$ or $d = 0$

Now if $d = 0$ then $a_2 = 3$ else $a_2 = 9$ for single choice more appropriate choice is 9, but in principal, question seems to have an error.

$\therefore \quad a_2 = 3 + 6 = 9$

GP & GM BASED

4. The 1st, 2nd and 3rd terms of an arithmetic series are a, b and a^2 where 'a' is negative. The 1st, 2nd and 3rd terms of a geometric series are a, a^2 and b find the

(a) value of a and b

(b) sum of infinite geometric series if it exists. If no then find the sum to n terms of the G.P.

(c) sum of the 40 term of the arithmetic series.

एक समानान्तर श्रेणी के प्रथम, द्वितीय तथा तृतीय पद क्रमशः a, b तथा a^2 है जहाँ 'a' एक ऋणात्मक संख्या है। एक गुणोत्तर श्रेणी के प्रथम तीन पद a, a^2 तथा b है तो

(a) a एवं b का मान निकालिये।

(b) अनन्त गुणोतर श्रेणी का योग निकालिये (यदि अस्तित्व में हो) यदि नही तो G.P के n पदों का योग निकालिये।

(c) समान्तर श्रेणी के 40 पदों का योग निकालिये।

Ans. (a) $a = -\frac{1}{2},\ b = -\frac{1}{8}$; (b) $-\frac{1}{3}$; (c) $\frac{545}{2}$

Sol. (a) a, b, a^2 are in A.P. $(a < 0)$

a, a^2, b are in G.P.

$\Rightarrow 2b = a + a^2$

$(a^2)^2 = ab$

$$a^4 = a\left(\frac{a+a^2}{2}\right)$$

$$a^2\left[a^2 - \left(\frac{a+1}{2}\right)\right] = 0$$

$a^2 = 0 \Rightarrow$ Not possible

$$a^2 - \left(\frac{a+1}{2}\right) = 0$$

$2a^2 - a - 1 = 0$

$2a^2 - 2a + a - 1 = 0$

$2a(a-1) + (a-1) = 0$

$$a = -\frac{1}{2},\ a = 1 \text{ (reject)}$$

$$a = -\frac{1}{2}$$

$$2b = a + a^2 = \frac{-1}{2} + \frac{1}{4} = \frac{-1}{4}$$

$$b = \frac{-1}{8}$$

(b) Sum of infinite G.P. $= \dfrac{a}{1-\left(\dfrac{a^2}{a}\right)} = \dfrac{a}{1-a}$

$$= \frac{\left(-\frac{1}{2}\right)}{1+\frac{1}{2}} = -\frac{1}{3}$$

(c) Sum of 40 terms of A.P.

$$S = \frac{40}{2}\left[2\left(-\frac{1}{2}\right)+39\left(-\frac{1}{8}+\frac{1}{2}\right)\right]$$

$$= 20\left[-1+39\left(\frac{-1+4}{8}\right)\right]$$

$$= 20\left[\frac{-8+39\times 3}{8}\right]$$

$$= \frac{545}{2}$$

5. For n = 1, 2, 3, ..., let $A_n = \left(\frac{3}{4}\right)-\left(\frac{3}{4}\right)^2+\left(\frac{3}{4}\right)^3+.....+(-1)^{n-1}\left(\frac{3}{4}\right)^n$, and $B_n = 1 - A_n$,

Find the smallest natural number n_0 such that $B_n > A_n$ for all $n \geq n_0$. **[JEE-2006]**

यदि $A_n = \left(\frac{3}{4}\right)-\left(\frac{3}{4}\right)^2+\left(\frac{3}{4}\right)^3+.....+(-1)^{n-1}\left(\frac{3}{4}\right)^n$ तथा $B_n = 1 - A_n$ है तो वो प्राकृतिक संख्या n_0 बताइए जिसके लिये $B_n > A_n$. $\forall$ $n > n_0$. **[JEE-2006]**

Ans. $n_0 = 6$

Sol. $a_n = \frac{3}{4}-\left(\frac{3}{4}\right)^2+\left(\frac{3}{4}\right)^3+.....(-1)^{n-1}\left(\frac{3}{4}\right)^n$, $b_n = 1 - a_n = \frac{3}{4}\left[\frac{1-\left(-\frac{3}{4}\right)^n}{\left(1+\frac{3}{4}\right)}\right] = \frac{3}{7}\left[1-\left(-\frac{3}{4}\right)^n\right]$

$b_n > a_n \quad \Rightarrow \quad 1 - a_n > a_n$

$2a_n < 1 \quad \Rightarrow \quad \frac{6}{7}-\frac{6}{7}\left(-\frac{3}{4}\right)^n < 1$

$-\frac{6}{7}\left(-\frac{3}{4}\right)^n < \frac{1}{7} \quad \Rightarrow \quad -3(-3)^n < 2^{2n-1} \quad \Rightarrow \quad (-1)^{n+1}\,3^{n+1} < 2^{2n-1}$

For n even inequality is always valid

For n, odd ; $n + 1 \rightarrow$ even

$3^{n+1} < 2^{2n-1}$; $n = 1,\ 3^2 < 2^1$, (in valid)

$n = 3,\ 3^4 < 2^5,\ 81 < 32$ (in valid) ; $n = 5,\ 3^6 < 2^9$ (in valid)

$n = 7,\ 3^8 < 2^{13},\ 6561 < 8192$ (valid) ; min. natural number $n_0 = 6$

6. If $x = 1 + a + a^2 + a^3 + \ldots$ to ∞ ($|a| < 1$) and $y = 1 + b + b^2 + b^3 + \ldots$ to ∞ ($|b| < 1$), then prove that $1 + ab + a^2b^2 + a^3b^3 + \ldots \infty = \dfrac{xy}{x + y - 1}$.

यदि $x = 1 + a + a^2 + a^3 + \ldots$ से ∞ ($|a| < 1$) तक तथा $y = 1 + b + b^2 + b^3 + \ldots$ से ∞ ($|b| < 1$) तक है, तो सिद्ध कीजिए कि $1 + ab + a^2b^2 + a^3b^3 + \ldots$ से ∞ तक $= \dfrac{xy}{x + y - 1}$ है।

Ans.

Sol. $x = 1 + a + a^2 + \ldots\ldots = \dfrac{1}{1-a}$

$y = 1 + b + b^2 + \ldots\ldots = \dfrac{1}{1-b}$

$1 + ab + a^2b^2 + \ldots. = \dfrac{1}{1-ab}$

$$\frac{xy}{x+y-1} = \frac{\left(\frac{1}{1-a}\right)\left(\frac{1}{1-b}\right)}{\left(\frac{1}{1-a}\right)+\left(\frac{1}{1-b}\right)-1} = \frac{1}{(1-b)+(1-a)-(1-a)(1-b)}$$

$$= \frac{1}{2-a-b-(1-a-b+ab)}$$

$$= \frac{1}{1-ab} \qquad \text{H.P.}$$

7. There are 25 trees at equal distances of 5 m in a line with a well , the distance of the well from the nearest tree being 10 m . A gardner waters all the trees separately starting from the well and he returns to the well after watering each tree to get water for the next . Find the total distance , the gardner will cover in order to water all the trees.

यदि किसी रेखा में 25 वृक्ष एक दूसरे से 5 m की दूरी पर है तथा इसी रेखा में एक कुआँ है जो सबसे नजदीकी पेड़ से 10 m दूर है। एक बागवान कुऐं से एक वृक्ष को पानी देकर, दूसरे पेड़ के लिये पानी लाने के लिए कुऐं पर जाता है तथा इसी प्रकार प्रत्येक वृक्ष को पानी देता है। तो वह कुल दूरी बताओं जो बागवान सभी पेड़ों को पानी देने में तय करता है।

Ans. 3500

Sol. $T_{25} = 10 + 24 \times 5 = 130$

$$\text{Total distance} = 2 \times \frac{25}{2}[2 \times 10 + (24)5]$$

$= 25\ [20 + 120]$

= 25 [140]
= 3500

8. The sum of infinite number of terms of a G.P. is 4 and the sum of their cubes is 192. Find the series.

एक अन्नत पदो वाली गु. श्रे. के पदो का योग 4 है। और उनके घनो का योग 192 है। श्रेणी ज्ञात कीजिए।

Ans. 6, –3, 3/2, ...

Sol. $\frac{a}{1-r} = 4 \quad \Rightarrow \quad \frac{a^3}{1-r^3} = 192 \quad \Rightarrow \quad \frac{(1-r)^3}{1-r^3} = \frac{192}{(4)^3} \quad \Rightarrow \quad \frac{(1-r)^2}{1+r+r^2} = 3$

$\Rightarrow \quad 1 + r^2 - 2r = 3 + 3r + 3r^2 \quad \Rightarrow \quad 2r^2 + 5r + 2 = 0 \quad \Rightarrow \quad (2r + 1)(r + 2) = 0$

$r = -1/2,\ r = -2$(rejected) When $r = -1/2$, $a = 6$ so series is 6, –3, 3/2 .....

Hindi. $\frac{a}{1-r} = 4 \quad \Rightarrow \quad \frac{a^3}{1-r^3} = 192 \quad \Rightarrow \quad \frac{(1-r)^3}{1-r^3} = \frac{192}{(4)^3} \quad \Rightarrow \quad \frac{(1-r)^2}{1+r+r^2} = 3$

$\Rightarrow \quad 1 + r^2 - 2r = 3 + 3r + 3r^2 \quad \Rightarrow \quad 2r^2 + 5r + 2 = 0 \quad \Rightarrow \quad (2r + 1)(r + 2) = 0$

$r = -1/2,\ r = -2$ (अस्वीकार्य) जब $r = -1/2$, $a = 6$ अतः श्रेणी 6, –3, 3/2 ..... है।

AM GM HM

9. The harmonic mean of two numbers is 4. The airthmetic mean A & the geometric mean G satisfy the relation $2A + G^2 = 27$. Find the two numbers.

दो संख्याओं का हरात्मक माध्य 4 है। उनका समान्तर माध्य 'A' एवं गुणोतर माध्य 'G' है जो सम्बन्ध $2A + G^2 = 27$ को संतुष्ट करते है। दोनों संख्याऐं निकालिए।

Ans. 6, 3

Sol. $H = 4 = \frac{2ab}{a+b} \Rightarrow a + b = \frac{ab}{2}$

$A = \frac{a+b}{2}, \qquad G = \sqrt{ab}$

$2A + G^2 = a + b + ab = 27$

$\frac{ab}{2} + ab = 27$

$\frac{3ab}{2} = 27$

$ab = 18 \Rightarrow a + b = 9$

$x^2 - (a + b)x + ab = 0$

$x^2 - 9x + 18 = 0$

$(x - 6)(x - 3) = 0$

$x = 6, 3$

10. The AM of two numbers exceeds their GM by 15 & HM by 27. Find the numbers.

दो संख्याओं का स.मा. उनके गु. मा. से 15 बढ़ा है एवं ह.मा. से 27 बढ़ा है। दोनों संख्याऐं निकालिए।

Ans. 120, 30

Sol. $\frac{a+b}{2} = \sqrt{ab} + 15$, $\frac{a+b}{2} = \frac{2ab}{a+b} + 27$

$2\sqrt{ab} = (a+b) - 30$, $(a+b)^2 = \left(2\sqrt{ab}\right)^2 + 54(a+b)$

$(a + b)^2 = [(a + b) - 30]^2 + 54 (a + b)$
Let $a + b = x$
$x^2 = x^2 + 900 - 60x + 54x$
$6x = 150$
$x = 150$

$a + b = 150 \quad \Rightarrow \frac{150}{2} 15 = \sqrt{ab}$

$60 = \sqrt{ab}$
$ab = 3600$
$y^2 + \{-(a + b)\} y + ab = 0$
$y^2 - 150y + 3600 = 0$
$(y - 120) (y - 30) = 0$
$y = 30, 120$

11. If a, b, c are positive real numbers, then prove that $[(1 + a) (1 + b) (1 + c)]^7 > 7^7 a^4 b^4 c^4$. **[JEE-2004]**

यदि a, b, c धनात्मक वास्तविक संख्याऐं है तो सिद्ध कीजिये। $[(1 + a) (1 + b) (1 + c)]^7 > 7^7 a^4 b^4 c^4$ **[JEE-2004]**

Ans.

Sol. $(1 + a) (1 + b) (1 + c) = (1 + a + b + ab) (1 + c)$
$= 1 + a + b + ab + c + ac + bc + abc$
$= 1 + a + b + c + ab + bc + ca + abc$
A.M. $\geq$ G.M.

$\frac{a+b+c+ab+bc+ca+abc}{7} \geq \left[a^4b^4c^4\right]^{\frac{1}{7}}$

$(a + b + c + ab + bc + ca + abc) \geq 7\left(a^4b^4c^4\right)^{\frac{1}{7}}$

$1 + a + b + c + ab + bc + ca + abc \geq 7\left(a^4b^4c^4\right)^{\frac{1}{7}}$

$[(1 + a) (1 + b) (1 + c)]^7 \geq 7^7(a^4b^4c)$

12. The minimum value of the sum of real numbers a^{-5}, a^{-4}, $3a^{-3}$, 1, a^8 and a^{10} with $a > 0$ is. **[JEE-2011]**

वास्तविक संख्याऐं a^{-5}, a^{-4} , $3a^{-3}$, 1, a^8 तथा a^{10} जहाँ $a > 0$ का योग का न्यूनतम मान होगा। **[JEE-2011]**

Ans. 8

Sol. A.M. $\geq$ G.M.

$$\frac{\frac{1}{a^5}+\frac{1}{a^4}+\frac{1}{a^3}+\frac{1}{a^3}+\frac{1}{a^3}+1+a^8+a^{10}}{8} \geq \left(\frac{1}{a^5}\cdot\frac{1}{a^4}\cdot\frac{1}{a^3}\cdot\frac{1}{a^3}\cdot\frac{1}{a^3}.1.a^8.a^{10}\right)^{1/8}$$

$$\Rightarrow \quad \frac{1}{a^5}+\frac{1}{a^4}+\frac{3}{a^3}+1+a^8+a^{10} \geq 8(1)^{1/8}$$

$$\Rightarrow \quad \text{minimum value of } \frac{1}{a^5}+\frac{1}{a^4}+\frac{3}{a^3}+1+a^3+a^{10} = 8$$

Hindi A.M. $\geq$ G.M.

$$\frac{\frac{1}{a^5}+\frac{1}{a^4}+\frac{1}{a^3}+\frac{1}{a^3}+\frac{1}{a^3}+1+a^8+a^{10}}{8} \geq \left(\frac{1}{a^5}\cdot\frac{1}{a^4}\cdot\frac{1}{a^3}\cdot\frac{1}{a^3}\cdot\frac{1}{a^3}.1.a^8.a^{10}\right)^{1/8}$$

$$\Rightarrow \quad \frac{1}{a^5}+\frac{1}{a^4}+\frac{3}{a^3}+1+a^8+a^{10} \geq 8(1)^{1/8}$$

$$\Rightarrow \quad \frac{1}{a^5}+\frac{1}{a^4}+\frac{3}{a^3}+1+a^3+a^{10} \text{ का } a = 1 \text{ पर न्यूनतम मान} = 8$$

HP & HM BASED

13. If the 10th term of an HP is 21 and 21st term of the same HP is 10, then find the 210th term.

यदि एक HP का 10वाँ पद 21 है एवं 21वाँ पद 10 है तो 210वाँ पद बताइये।

Ans. 1

Sol. $T_{10} = 21,\ T_{21} = 10$

$$t_{10} = \frac{1}{21}, t_{21} = \frac{1}{10}$$

$$\frac{1}{a}+9d = \frac{1}{21} \qquad ...(1)$$

$$\frac{1}{a}+20d = \frac{1}{10} \qquad ...(2)$$

$(1)-(2)$

$$-11d = \frac{1}{21}-\frac{1}{10}$$

$$-11d = \frac{10-21}{210}$$

$$d = \frac{1}{210} \Rightarrow \frac{1}{a} = \frac{1}{210}$$

$$t_{210} = \frac{1}{210} + (210-1) \times \frac{1}{210} = 1$$

$$T_{210} = 1$$

14. The pth term T_p of H.P. is q(p + q) and q^{th} term T_q is p(p + q) when p > 2, q > 2. Prove that

एक HP का pवाँ पद T_p = q(p + q) है तथा $q^{वाँ}$ पद T_q = p(p + q) है जब p > 2, q > 2 है। सिद्ध कीजिये।

(a) $T_{p+q} = pq$; (b) $T_{pq} = p + q$; (c) $T_{p+q} > T_{pq}$

Ans.

Sol. $T_p = q\,(p+q) \Rightarrow t_p = \frac{1}{q(p+q)} = \frac{1}{a} + (p-1)d$ ...(1)

$T_q = p\,(p+q) \Rightarrow t_q = \frac{1}{p(p+q)} = \frac{1}{a} + (q-1)d$ ...(2)

$$(1)-(2): \left(\frac{1}{p+q}\right)\left(\frac{1}{q}-\frac{1}{p}\right) = (p-q)d$$

$$\frac{1}{(p+q)pq} = d = \frac{1}{a}$$

$$\text{(a) } t_{p+q} = \frac{1}{(p+q)pq} + (p+q-1)\frac{1}{(p+q)(pq)}$$

$$t_{p+q} = \frac{1}{pq}$$

$$\Rightarrow T_{p+q} = pq$$

$$\text{(b) } t_{pq} = \frac{1+(pq-1)}{pq(p+q)} = \frac{1}{p+q}$$

$T_{pq} = p + q$

(c) $T_{p+q} = pq > p + q$ ($\because$ p > 2, q > 2)

AGP

15. Let ' σ ' denotes the sum of the infinite series $\sum_{n=1}^{\infty}\left(\frac{n^2+2n+3}{2^n}\right)$. Compute the value of $(1^3+2^3+3^3+.....+\sigma^3)$.

माना ' σ ' अनन्त श्रेणी $\sum_{n=1}^{\infty}\left(\frac{n^2+2n+3}{2^n}\right)$ का योग है। तो $(1^3 + 2^3 + 3^3 + ... + \sigma^3)$ के मान की गणना कीजिये।

Ans. 8281

Sol. $\sigma=\sum_{n=1}^{\infty}\left(\frac{n^2+2n+3}{2^n}\right)$

$$\sigma=\frac{6}{2}+\frac{11}{2^2}+\frac{18}{2^3}+\frac{27}{2^4}+..... \quad ...(1)$$

$$\frac{\sigma}{2}=0+\frac{6}{2^2}+\frac{11}{2^3}+\frac{18}{2^4}+..... \quad ...(2)$$

$(1)-(2)$

$$\frac{\sigma}{2}=3+\frac{5}{2^2}+\frac{7}{2^3}+\frac{9}{2^4}+.... \quad ...(3)$$

$$\frac{\sigma}{4}=0+\frac{3}{2}+\frac{5}{2^3}+\frac{7}{2^4}+.... \quad ...(4)$$

$(3)-(4)$

$$\frac{\sigma}{4}=\left(3+\frac{5}{4}-\frac{3}{2}\right)+\left(\frac{2}{2^3}+\frac{2}{2^4}+.....\right)$$

$$=\frac{11}{4}+\frac{1}{2^2}\left(1+\frac{1}{2}+\frac{1}{2^2}+.....\right)$$

$$\frac{\sigma}{4}=\frac{11}{4}+\frac{1}{4}\left(\frac{1}{1-\frac{1}{2}}\right)$$

$$\sigma=11+2$$

$$\sigma=13$$

Now $1^3+2^3+3^3+....+13^3=\left(\frac{13\times14}{2}\right)^2$

$= (91)^2$

$= 8281$

16. If total number of runs scored in *n* matches is $\left(\frac{n+1}{4}\right)(2^{n+1}-n-2)$ where $n > 1$, and the runs scored in the kth match are given by $k\cdot 2^{n+1-k}$, where $1 \le k \le n$. Find n. **[JEE-2005]**

यदि n मैंचो में अर्जित कुल रनों की संख्या $\left(\frac{n+1}{4}\right)(2^{n+1}-n-2)$ जहाँ $n > 1$ है एवं kवें मैच में अर्जित रनों को यदि $k\cdot 2^{n+1-k}$ से निरूपित किया जाता है जहाँ $1 \le k \le n$ है। n का मान निकालिए। **[JEE-2005]**

Ans. $n = 7$

Sol. Total run in n matches $= \left(\frac{n+1}{4}\right)\left(2^{n+1}-n-2\right)$

$T_k = k\cdot 2^{n+1-k}$

$$S_n = \sum_{k=1}^{n}\left(2^{n+1}\right)\left(\frac{k}{2^k}\right)$$

$$S_n = 2^{n+1}\left[\frac{1}{2}+\frac{1}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+....+\frac{n}{2^n}\right]$$

$$S = \frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+\frac{4}{2^4}+.....+\frac{n}{2^n} \qquad ...(1)$$

$$\frac{S}{2} = 0+\frac{1}{2^2}+\frac{2}{2^3}+\frac{3}{2^4}+.....+\frac{n-1}{2^n}+\frac{n}{2^{n+1}} \qquad ...(2)$$

$(1)-(2)$

$$\frac{S}{2} = \frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}+.....+\frac{1}{2^n}-\frac{n}{2^{n+1}}$$

$$S = \left(1+\frac{1}{2}+\frac{1}{2^2}+....+\frac{1}{2^{n-1}}\right)\frac{n}{2^n}$$

$$S = \frac{1\left(\frac{1}{2}\right)^n-1}{\frac{1}{2}-1}-\frac{n}{2^n}$$

$$S = \frac{1-2^n}{2^n\left(-\frac{1}{2}\right)}-\frac{n}{2^n}$$

$$S = \frac{2^n-1}{2^{n-1}}-\frac{n}{2^n}$$

$$S = \frac{2^{n+1} - 2 - n}{2^n}$$

$$S_n = 2^{n+1}\left(\frac{2^{n+1} - n - 2}{2^n}\right) = 2\left(2^{n+1} - n - 2\right)$$

$$= \left(\frac{n+1}{4}\right)\left(2^{n+1} - n - 2\right)$$

$$\Rightarrow 2 = \frac{n+1}{4} \Rightarrow n = 7$$

MISCELLANEOUS SEQUENCES

17. If $S_1, S_2, S_3, \ldots S_n, \ldots$ are the sums of infinite geometric series whose first terms are 1, 2, 3, ... n, ... and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\ldots\ldots, \frac{1}{n+1}, \ldots\ldots$ respectively, then find the value of $\sum_{r=1}^{2n-1} S_r^2$.

यदि $S_1, S_2, S_3, \ldots S_n, \ldots$ अनन्त गुणोतर श्रेणी के योग है जिनके प्रथम पद 1, 2, 3, ... n, ... एवं सर्वानुपात $\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\ldots\ldots, \frac{1}{n+1}, \ldots\ldots$ क्रमशः है तो $\sum_{r=1}^{2n-1} S_r^2$ का मान निकालिए।

Ans. $\frac{n\left(2n+1(4n+1)\right)}{3} - 1$

Sol. $S_1 = \frac{1}{1-\frac{1}{2}}, \quad S_2 = \frac{2}{1-\frac{1}{3}}, \quad S_3 = \frac{3}{1-\frac{1}{4}}, \ldots\ldots$

$$S_1 = \frac{2}{1}, S_2 = \frac{2\cdot 3}{2}, S_3 = \frac{3\cdot 4}{3} \ldots\ldots\ldots$$

$$S_1 = 2,\ S_2 = 3,\ S_3 = 4 \ldots\ldots\ldots\ldots$$

$$\sum_{r=1}^{2n-1} (Sr)^2 = S_1^2 + S_2^2 + S_3^2 + \ldots\ldots\ldots + S_{2n-1}^2$$

$$= (2)^2 + (3)^2 + (4)^2 + \ldots\ldots\ldots + (2n)^2$$

$$= \frac{2n\left(2n+1\right)\left(4n+1\right)}{6} - 1$$

18. Find the nth term and the sum to n terms of the sequence :

निम्न श्रेणीयों का nवाँ पद एवं n पदों का योग बताइए :

(i) $1+5+13+29+61+.....$ (ii) $6+13+22+33+.....$

Ans. (i) $2^{n+1}-3;\ 2^{n+2}-4-3n$ (ii) $n^2+4n+1;\ (1/6)\,n\,(n+1)\,(2n+13)+n$

Sol. $S = 1 + 5 + 13 + 29 + 61 + ... T_n$ $S = 1 + 5 + 13 + 29 + ...T_n$

$0 = 1 + 4 + 8 + 16 + 32 + ... T_n$ $T_n = 1+4(2^{n-1}-1) = 2^{n+1}-3$

$S_n = \Sigma(2^{n+1}-3) = (2^2 + 2^3 + ... 2^{n+1}) -3n \quad = 2^2\left(\frac{2^n-1}{2-1}\right)-3n = 2^{n+2}-4-3n$

(b) $S = 6 + 13 + 22 + 33 + ... T_n$ $S = 6 + 13 + 22 + ... T_n$

$T_n = 6 + 7 + 9 + 11 .....$ $= 1 + 5 + 7 + 9 + 11 .....$

$= 1 + \frac{n}{2}(10+(n-1)2) = n^2 + 4n + 1$ $S_n = \Sigma T_n = \frac{1}{6}n(n+1)(2n+13)+n$

19. If the sum $\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+\sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}}+.....+\sqrt{1+\frac{1}{(1999)^2}+\frac{1}{(2000)^2}}$ equal to $n-\frac{1}{n}$ where $n \in N$. Find n.

यदि योग $\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+\sqrt{1+\frac{1}{3^2}+\frac{1}{4^2}}+.....+\sqrt{1+\frac{1}{(1999)^2}+\frac{1}{(2000)^2}}$ का मान $n-\frac{1}{n}$ जहाँ $n \in N$ के बराबर है तो n का मान बताइऐ।

Ans. $n = 2000$

Sol. $T_r = \sqrt{1+\frac{1}{r^2}+\frac{1}{(r+1)^2}}$

$= \sqrt{\frac{\left(r(r+1)\right)^2+(r+1)^2+r^2}{\left(r(r+1)\right)^2}}$

$T_r = \frac{r(r+1)+1}{r(r+1)} = 1+\frac{1}{r(r+1)}$

$S = \sum_{r=1}^{1999} T_r = \sum_{r=1}^{1999}\left(1+\frac{1}{r(r+1)}\right)$

$= 1999 + \sum_{r=1}^{1999}\left(\frac{(r+1)-r}{r(r+1)}\right)$

$$= 1999 + \left(1 - \frac{1}{1999+1}\right)$$

$$n - \frac{1}{n} = 2000 - \frac{1}{2000}$$

$$n = 2000$$

20. The series of natural numbers is divided into groups (1); (2, 3, 4); (5, 6, 7, 8, 9); ... and so on. Show that the sum of the numbers in the nth group is $(n-1)^3 + n^3$.

प्राकृत संख्याओं की श्रेणीयों को निम्नलिखित समूह (1); (2, 3, 4); (5, 6, 7, 8, 9); ... में बांटा जाता है, तो सिद्ध कीजिए कि nवें समूह में संख्याओं का योग $(n-1)^3 + n^3$ है।

Ans.

Sol. (1), (2,3,4), (5,6,7,8,9), ...

Total number of digits in T_n term

1,3,5,7,....., $(1 + (n-1)2) = 2n - 1$

First term of T_n term

$S = 1 + 2 + 5 + 10 + ... + T_n$ ...(1)

$S = 0 + 1 + 2 + 5 + ... + T_{n-1} + T_n$ ...(2)

(1) – (2)

$$0 = 1 + \underbrace{1 + 3 + 5 + ... T_n}_{(n-1)\text{ term}}$$

$$T_n = 1 + \left(\frac{n-1}{2}\right)\left(2 + (n-2)\times 2\right)$$

$n = 1 + (n-1)^2$

n^{th} group

$(1 + (n-1)^2, 2 + (n-1)^2, ... (2n-1)$ terms

$$S = \left(\frac{2n-1}{2}\right)\left[2\left(1 + (n-1)^2\right) + (2n-2)\right]$$

$= (2n-1)\,[1 + (n-1)^2 + n - 1]$

$= (2n-1)\,[(n-1)^2 + n]$

$S = (2n-1)\,[n^2 + 1 - n]$

$= (2n-1)\,[n^2 - n + 1]$

$= 2n^3 - 2n^2 + 2n - n^2 + n - 1$

$= 2n^3 - 3n^2 + 3n - 1$

$S = (n-1)^3 + n^3$

21. Sum the following series to n terms and to infinity :

निम्नलिखित श्रेणीयों के n पदों एवं अनन्त पदों का योग बताइए :

(i) $\frac{1}{1.4.7}+\frac{1}{4.7.10}+\frac{1}{7.10.13}+.....$ (ii) $\sum_{r=1}^{n} r(r+1)(r+2)(r+3)$

(iii) $\sum_{r=1}^{n} \frac{1}{4r^2-1}$ (iv) $\frac{1}{2.4}+\frac{1.3}{2.4.6}+\frac{1.3.5}{2.4.6.8}+.....$

Ans. (i) $s_n = (1/24) - [1/\{6(3n+1)(3n+4)\}]$; $s_\infty = 1/24$ (ii) $(1/5)\, n(n+1)(n+2)(n+3)(n+4)$

(iii) $n/(2n+1)$ (iv) $S_n = \left[\frac{1}{2} - \frac{1.3.5.....(2n-1)(2n+1)}{2.4.6.....(2n)(2n+2)}\right]; S_\infty = \frac{1}{2}$

Sol. (a) $\frac{1}{1.4.7}+\frac{1}{4.7.10}+\frac{1}{7.10.13}+...$

$$T_n = \frac{1}{(3n-2)(3n+1)(3n+4)}$$

$$= \frac{1}{3n+1}\left(\frac{1}{3n-2} - \frac{1}{(3n+4)}\right)\frac{1}{6}$$

$$= \left(\frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)}\right)\frac{1}{6}$$

$$S_n = \frac{1}{6}\left[\frac{1}{1.4} - \frac{1}{4.7} + \frac{1}{4.7} - \frac{1}{7.10} + \frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)}\right]$$

$$= \frac{1}{6}\left[\frac{1}{4} - \frac{1}{(3n+1)(3n+4)}\right]$$

now $S_\infty = \frac{1}{24}$

(b) $T_n = n(n+1)(n+2)(n+3)$

$$= \frac{1}{5}[n(n+1)(n+2)(n+3)((n+4)-(n-1))]$$

$$= \frac{1}{5}[n(n+1)(n+2)(n+3)(n+4)$$

$$-(n-1)(n)(n+2)(n+3)]$$

$$\Sigma T_n = \left(\frac{1}{5}\right) n(n+1)(n+2)(n+3)(n+4)$$

(c) $T_n = \frac{1}{4n^2-1} = \frac{1}{2}\left[\frac{1}{2n-1} - \frac{1}{2n+1}\right]$

$$S_n = \frac{1}{2}\left[1 - \frac{1}{2n+1}\right]$$

$$S_\infty = \frac{1}{2}$$

(d) $T_n = \frac{1\cdot3\cdot5.....(2n-1)[(2n+2)-(2n+1)]}{2\cdot4\cdot6.....(2n+2)}$

$$= \frac{1\cdot3\cdot5.....(2n-1)}{2\cdot4\cdot6.....2n} - \frac{1\cdot3\cdot5.....(2n+1)}{2\cdot4\cdot6.....(2n+2)}$$

$$= \left(\frac{1}{2} - \frac{1.3}{2.4}\right) + \left(\frac{1.3}{2.4} - \frac{1.3.5}{2.4.6}\right) + ... + \left(\frac{1.3.....(2n-1)}{2.4.6.....2n}\right) - \frac{1.3.5.....(2n+1)}{2.4.6.....(2n+2)}$$

$$S_n = \frac{1}{2} - \frac{1\cdot3\cdot5.....(2n+1)}{2\cdot4\cdot6.....(2n+2)},$$

$$S_n = \frac{1}{2} - \frac{1}{2^{n+1}}\left(\frac{1\cdot3\cdot5.....(2n+1)}{1.2.3.....(n+1)}\right)$$

$$S_\infty = \frac{1}{2}$$

22. If $\sum_1^{20} r^3 = a, \sum_1^{20} r^2 = b$ then sum of products of 1, 2, 3, 4 ..... 20 taking two at a time is.

यदि $\sum_1^{20} r^3 = a, \sum_1^{20} r^2 = b$ है, तो 1, 2, 3, 4 ..... 20 को क्रमवार युग्म (दो एक साथ) का गुणनफल होगा।

Ans. $\frac{a-b}{2}$

Sol. $\sum_{r=1}^{20} {}^3 = a$, $\sum_{r=1}^{20} r^2 = b$

$a = 1 + 2^3 + 3^3 + 4^3 + ... + (20)^3 = \left[\frac{20\times 21}{2}\right]^2$

$b = 1 + 2^2 + 3^2 + 4^2 + ... + (20)^2 = \left[\frac{20\times 21\times 41}{b}\right]$

$(1 + 2 + 3 + ... + 20)^2 = (1^2 + 2^2 + 3^2 + ... + 20^2) + 2\,[1\cdot 2 + 1\cdot 3 + ...+ 1\cdot 20 + 2\cdot 3 + ... + 19\cdot 20]$

$\left(\frac{20\times 21}{2}\right)^2 = \left(\frac{20\times 21\times 41}{b}\right) + 2[\alpha]$

$a = b + 2\alpha$

$\alpha = \frac{a-b}{2}$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 23 to 25

Let A_1, G_1, H_1 denote the arithmetic, geometric and harmonic means, respectively, of two distinct positive numbers. For $n \geq 2$, let A_{n-1} and H_{n-1} have arithmetic, geometric and harmonic means as A_n, G_n, H_n respectively.

माना A_1, G_1, H_1 क्रमशः दो भिन्न धनात्मक संख्याओं के समान्तर, गुणोत्तर एवं हरात्मक माध्य है। $n \geq 2$ के लिये माना A_{n-1} एवं H_{n-1} के समान्तर, गुणोत्तर एवं हरात्मक माध्य क्रमशः A_n, G_n, H_n है।

23. Which one of the following statements is correct?

(A) $G_1 > G_2 > G_3 > .....$

(B) $G_1 < G_2 < G_3 < .....$

(C) $G_1 = G_2 = G_3 = ...$

(D) $G_1 < G_3 < G_5 < .....$ and $G_2 > G_4 > G_6 > .....$

निम्नलिखित में से कौनसा कथन सत्य है ?

(A) $G_1 > G_2 > G_3 > .....$

(B) $G_1 < G_2 < G_3 < .....$

(C) $G_1 = G_2 = G_3 = ...$

(D) $G_1 < G_3 < G_5 < .....$ तथा $G_2 > G_4 > G_6 > .....$

Ans. C

Sol. $A_n = \frac{A_{n-1} + H_{n-1}}{2}$ ...(1)

$G_n = \sqrt{(A_{n-1})(H_{n-1})}$ ...(2)

$H_n = \frac{2A_{n-1}\cdot H_{n-1}}{A_{n-1} + H_{n-1}}$ ...(3)

$G_n^2 = (A_n)(H_n)$ ...(4)

By equation (2)

$G_n^2 = A_{n-1}H_{n-1} = A_nH_n$

$\Rightarrow G_n$ is independent from n

$\Rightarrow G_1 = G_2 = G_3 = ...$

24. Which one of the following statements is correct?

(A) $A_1 > A_2 > A_3 > .....$

(B) $A_1 < A_2 < A_3 < .....$

(C) $A_1 > A_3 > A_5 > .....$ and $A_2 < A_4 < A_6 < .....$

(D) $A_1 < A_3 < A_5 < .....$ and $A_2 > A_4 > A_6 > .....$

निम्नलिखित में से कौनसा कथन सत्य है ?

(A) $A_1 > A_2 > A_3 > .....$

(B) $A_1 < A_2 < A_3 < .....$

(C) $A_1 > A_3 > A_5 > .....$ तथा $A_2 < A_4 < A_6 < .....$

(D) $A_1 < A_3 < A_5 < .....$ तथा $A_2 > A_4 > A_6 > .....$

Ans. A

Sol. By (1)

$$A_n = \frac{A_{n-1} + H_{n-1}}{2}$$

$\because A_{n-1} \geq H_{n-1}$

$\because$ numbers are distinct, so $A_{n-1} > H_{n-1}$

$$\Rightarrow A_n < \frac{A_{n-1} + A_{n-1}}{2}$$

$A_n < A_{n-1}$

$A_1 > A_2 > A_3 > .....$

25. Which one of the following statements is correct?

(A) $H_1 > H_2 > H_3 > .....$

(B) $H_1 < H_2 < H_3 < .....$

(C) $H_1 > H_3 > H_5 > .....$ and $H_2 < H_4 < H_6 < .....$

(D) $H_1 < H_3 < H_5 < .....$ and $H_2 > H_4 > H_6 > .....$

निम्नलिखित में से कौनसा कथन सत्य है ?

(A) $H_1 > H_2 > H_3 > .....$

(B) $H_1 < H_2 < H_3 < .....$

(C) $H_1 > H_3 > H_5 > .....$ तथा $H_2 < H_4 < H_6 < .....$

(D) $H_1 < H_3 < H_5 < .....$ तथा $H_2 > H_4 > H_6 > .....$

Ans. B

Sol. $H_n = \dfrac{2}{\dfrac{1}{A_{n-1}} + \dfrac{1}{H_{n-1}}}$ $\Rightarrow A_n < \dfrac{A_{n-1} + A_{n-1}}{2}$

$\dfrac{1}{A_{n-1}} + \dfrac{1}{H_{n-1}} = \dfrac{2}{H_n}$

$\because A_{n-1} > H_{n-1}$

$\dfrac{1}{H_{n-1}} > \dfrac{1}{A_{n-1}}$

$\dfrac{2}{H_{n-1}} > \dfrac{1}{A_{n-1}} + \dfrac{1}{H_{n-1}}$

$\dfrac{2}{H_{n-1}} > \dfrac{2}{H_n}$

$H_n > A_{n-1}$

$H_1 < H_2 < H_3 < .....$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

26. In an A.P., let T_r denote r^{th} term from beginning, $T_p = \dfrac{1}{q(p+q)}$, $T_q = \dfrac{1}{p(p+q)}$, then :

(A) T_1 = common difference (B) $T_{p+q} = \dfrac{1}{pq}$

(C) $T_{pq} = \dfrac{1}{p+q}$ (D) $T_{p+q} = \dfrac{1}{p^2q^2}$

समान्तर श्रेणी में माना T_r, rवें पद को निरूपित करता है (प्रारम्भ से), $T_p = \dfrac{1}{q(p+q)}$, $T_q = \dfrac{1}{p(p+q)}$, तो :

(A) T_1 = सार्वअन्तर (B) $T_{p+q} = \dfrac{1}{pq}$

(C) $T_{pq} = \dfrac{1}{p+q}$ (D) $T_{p+q} = \dfrac{1}{p^2q^2}$

Ans. ABC

Sol. $T_r = a + (r-1)d$

$$T_p = a + (p-1)d = \frac{1}{q(p+q)} \quad ...(1)$$

$$T_q = a + (q-1)\,d = \frac{1}{p(p+q)} \quad ...(2)$$

Solving (1), (2)

$$a = d = \frac{1}{pq(p+q)}$$

T_1 = Common difference

$T_{p+q} = a + (p+q-1)d$

$= (p+q)\,d$

$$= (p+q)\left(\frac{1}{pq(p+q)}\right)$$

$$T_{p+q} = \frac{1}{pq}$$

$T_{pq} = a + (pq-1)d$

$$= pq\left(\frac{1}{pq(p+q)}\right)$$

$$T_{pq} = \frac{1}{p+q}$$

27. All roots of equation $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ are in G.P. If the sum of their reciprocals is 10, then δ can be equal to :

समीकरण $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$ के सभी मूल गुणौतर श्रेणी में है। यदि उनके व्युत्क्रमों का योग 10 है, तो δ का मान हो सकता है :

(A) 32 (B) –32 (C) $\frac{1}{32}$ (D) $-\frac{1}{32}$

Ans. AB

Sol. $x^5 - 40x^4 + \alpha x^3 + \beta x^2 + \gamma x + \delta = 0$

Let $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$ are rootos of polynomial

$$\left[\frac{a}{r^2} + \frac{a}{r} + a + ar + ar^2\right] = 40$$

$$\left(\frac{1}{r^2} + r^2\right) + \left(\frac{1}{r} + r\right) + 1 = \frac{40}{a} \quad ...(1)$$

$$\frac{r^2}{a}+\frac{r}{a}+\frac{1}{a}+\frac{1}{ar}+\frac{1}{ar^2}=10$$

$$\left(\frac{1}{r^2}+r^2\right)+\left(r+\frac{1}{r}\right)+1=10a \qquad ...(2)$$

Solving (1), (2)

$$10a=\frac{40}{a}$$

$a^2 = 4$

$a = \pm 2$

$$\delta=-\left(\frac{a}{r^2}\times\frac{a}{r}\times a\times ar\times ar^2\right)$$

$\delta = -a^5$

at $a = 2$ $\qquad$ $a = -2$

$\delta = -32$ $\qquad$ $\delta = 32$

28. For the A.P. given by $a_1, a_2, \ldots\ldots\ldots, a_n, \ldots\ldots\ldots$, the equations satisfied are :

समान्तर श्रेढ़ी $a_1, a_2, \ldots\ldots\ldots, a_n, \ldots\ldots\ldots$, के लिए निम्न में से कौनसा सम्बन्ध सत्य हैं :

(A) $a_1 + 2a_2 + a_3 = 0$ $\qquad$ (B) $a_1 - 2a_2 + a_3 = 0$

(C) $a_1 + 3a_2 - 3a_3 - a_4 = 0$ $\qquad$ (D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5 = 0$

Ans. BD

Sol. $a_1, a_2, a_3, \ldots\ldots\ldots, a_n, \ldots\ldots\ldots$ are in A.P.

(A) $2a_2 = a_1 + a_3$

$\Rightarrow a_1 + 2a_2 + a_3 \neq 0$

(B) $a_1 - 2a_2 + a_3 = 0$

(C) $a_1 + 3a_2 - 3a_3 - a_4 = a_1 + 3(a_1 + d) - (a_1 + 2d) - (a_1 + 3d) \neq 0$

(D) $a_1 - 4a_2 + 6a_3 - 4a_4 + a_5$

$= a_1 - 4(a_1 + d) + 6(a_1 + 2d) - 4(a + 3d) + (a + 4d) = 0$

29. If sum of the infinite G.P., p, 1, $\frac{1}{p}, \frac{1}{p^2}, \frac{1}{p^3}, \ldots\ldots$ is $\frac{9}{2}$, the value of p is :

यदि अनन्त गु.श्रे. p, 1, $\frac{1}{p}, \frac{1}{p^2}, \frac{1}{p^3}, \ldots\ldots$ का योगफल $\frac{9}{2}$ हो, तो p का मान है :

(A) 3 $\qquad$ (B) $\frac{2}{3}$ $\qquad$ (C) $\frac{3}{2}$ $\qquad$ (D) $\frac{1}{3}$

Ans. AC

Sol. $\dfrac{p}{1-\dfrac{1}{p}}=\dfrac{9}{2}$

$\dfrac{p^2}{p-1}=\dfrac{9}{2}$

$2p^2 = 9p - 9$

$2p^2 - 9q + 9 = 0$

$2p^2 - 6p - 3p + p = 0$

$2p(p - 3) - 3(p - 3) = 0$

$p=\dfrac{3}{2}, p = 3$

30. If the arithmetic mean of two positive numbers a & b (a > b) is twice their geometric mean, then a: b is :

यदि दो धनात्मक संख्याओं a एवं b (a > b) का समान्तर माध्य उनके गुणोत्तर माध्य का दुगुना हो, तो a: b है :

(A) $2+\sqrt{3}:2-\sqrt{3}$ (B) $7+4\sqrt{3}:1$ (C) $1:7-4\sqrt{3}$ (D) $2:\sqrt{3}$

Ans. ABC

Sol. $\dfrac{a+b}{2}=2\sqrt{ab}$

$(a+b)=4\sqrt{ab}$

$(a + b)^2 = 16ab$

$a^2 + b^2 - 14\ ab = 0$

$a^2 - 14ab + b^2 = 0$

$$a=\frac{14b\pm\sqrt{(-14b)^2-4\times b^2}}{2}$$

$$a=\frac{14b\pm\sqrt{192b^2}}{2}$$

$$a=\frac{14b\pm8\sqrt{3}b}{2}$$

$$\frac{a}{b}=7\pm4\sqrt{3}$$

$$\frac{a}{b}=\frac{7+4\sqrt{3}}{1} \quad \text{or} \quad \frac{a}{b}=\frac{7-4\sqrt{3}}{1}$$

$$\frac{a}{b}=\frac{7+4\sqrt{3}}{1}\times\frac{7-4\sqrt{3}}{7-4\sqrt{3}}=\frac{1}{7-4\sqrt{3}}$$

$$\frac{a}{b}=\frac{2+\sqrt{3}}{2\sqrt{3}}\times\frac{2+\sqrt{3}}{2+\sqrt{3}}=\frac{7+4\sqrt{3}}{1}$$

31. If $\sum_{r=1}^{n} r(r+1)\ (2r+3) = an^4 + bn^3 + cn^2 + dn + e$, then :

(A) $a + c = b + d$ (B) $e = 0$

(C) $a, b - 2/3, c - 1$ are in A.P. (D) c/a is an integer

यदि $\sum_{r=1}^{n} r(r+1)\ (2r+3) = an^4 + bn^3 + cn^2 + dn + e$ हो, तो :

(A) $a + c = b + d$ (B) $e = 0$

(C) $a, b - 2/3, c - 1$ स.श्रे. में है (D) c/a एक पूर्णांक है

Ans. ABCD

Sol. $\sum_{r=1}^{n}(r+1)(2r+1)=2\sum_{r=1}^{n}r^3+5\sum_{r=1}^{n}r^2+3\sum_{r=1}^{n}r$

$$an^4+bn^3+cn^2+dn+e=2\left[\frac{n(n+1)}{2}\right]^2+5\left[\frac{n(n+1)(2n+1)}{6}\right]+3\left[\frac{n(n+1)}{2}\right]$$

$$=\frac{n(n+1)}{2}\left[n(n+1)+\frac{5(2n+1)}{3}+3\right]$$

$$=\left(\frac{n^2+n}{2}\right)\left[\frac{3n^2+3n+10n+5+9}{3}\right]$$

$$=\left(\frac{n^2+n}{6}\right)\left(3n^2+13n+14\right)$$

$$an^4+bn^3+n^2+dn+e=\frac{1}{6}\left(3n^4+16n^3+27n^2+14n\right)$$

$$e=0, d=\frac{14}{6}, c=\frac{27}{6}, b=\frac{16}{6}, a=\frac{3}{6}$$

$$a+c=\frac{3}{6}+\frac{27}{6}=5=b+d$$

(C) $a, b-\frac{2}{3}, c-1 \Rightarrow \frac{3}{6}, \frac{16}{6}-\frac{2}{3}, \frac{27}{6-1}$

$\Rightarrow \frac{1}{2}, 2, \frac{7}{2}$

$2 \times 2 = \frac{1}{2} + \frac{7}{2} = 4$

a, $b - \frac{2}{3}$, c – 1 are in A.P.

(D) $\frac{c}{a} = \frac{\left(\frac{27}{6}\right)}{\left(\frac{3}{6}\right)} = 9$

32. If the first two terms of a progression are 8 and 4 respectively, then correct statement(s) is/are:

(A) if third term is 2, then the term are in G. P.

(B) if third term is 0, then the terms are in A. P.

(C) if third term is $\frac{8}{3}$, then the terms are in H. P.

(D) if the third term is 3, then terms are in A. P.

यदि अनुक्रम के पहले दो पद क्रमशः 8 तथा 4 है, तो कौनसा/कौनसे कथन सत्य होंगे :

(A) यदि तीसरा पद 2 है, तो पद गुणौतर श्रेणी में होंगे

(B) यदि तीसरा पद 0 है, तो पद समान्तर श्रेणी में होंगे

(C) यदि तीसरा पद $\frac{8}{3}$ है, तो पद हरात्मक श्रेणी में होंगे

(D) यदि तीसरा पद 3 है, तो पद समान्तर श्रेणी में होंगे

Ans. ABC

Sol. $T_1 = 8, T_2 = 4$

(A) $T_3 = 2$

8,4,2,.....

$\frac{T_3}{T_2} = \frac{2}{4} = \frac{1}{2} = \frac{T_2}{T_1}$

$\Rightarrow$ Geometric progression

(B) $8, 4, \frac{8}{3}, ...$

$\frac{2 \times (8)\left(\frac{8}{3}\right)}{8 + \frac{8}{3}} = \frac{128}{32} = 4$

$\Rightarrow$ H.P.
(B) 8,4,0,.....
$2 \times 4 = 8 + 0$
$\Rightarrow$ A.P.
(D) 8,4,3,.....
$2 \times 4 \neq 8 + 3$
Not in A.P.

33. If $40 + \frac{155}{4} + \frac{150}{4} + \frac{145}{4} + \ldots + n$ terms $= 625$ then n can be :

(A) 24 (B) 25 (C) 40 (D) 37

यदि $40 + \frac{155}{4} + \frac{150}{4} + \frac{145}{4} + \ldots + n$ पद $= 625$ है, तो n का मान हो सकता है :

(A) 24 (B) 25 (C) 40 (D) 37

Ans. BC

Sol. $40 + \frac{155}{4} + \frac{100}{4} + \frac{145}{4} + \ldots\ldots + n \text{ terms} = 625$

$160 + 155 + 150 + \ldots\ldots\ldots + \ldots\ldots\ldots + n \text{ terms} = (625) \times 4$

$\frac{n}{2}\left[2 \times 160 + (n-1)(-5)\right] = 625 \times 4$

$\frac{n}{2}\left[325 - 5n\right] = 2500$

$325n - 5n^2 = 5000$

$5n^2 - 325n + 5000 = 0$

$n^2 - 65n + 1000 = 0$

$n = \frac{65 \pm 15}{2}$

$n = 40, n = 25$

MATCH THE COLUMN TYPE QUESTION(S)

34. Match the following columns :

Column – I	**Column – II**
(A) If $\log_5 2$, $\log_5(2^x - 5)$ and $\log_5(2^x - 7/2)$ are in A.P., then value of 2x is equal to	(P) 6
(B) Let S_n denote sum of first n terms of an A.P. If $S_{2n} = 3S_n$, then $\frac{S_{3n}}{S_n}$ is	(Q) 9
(C) Sum of infinite series $4 + \frac{8}{3} + \frac{12}{3^2} + \frac{16}{3^3} + \ldots\ldots$ is	(R) 3

(D) The length,breadth, height of a rectangular box are in G.P. The volume is 27, the total surface area is 78. Then the length is (S) 1

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) यदि $\log_5 2, \log_5(2^x - 5)$ तथा $\log_5(2^x - 7/2)$ स. श्रे. में है, तो 2x का मान होगा	(P) 6
(B) माना S_n एक स. श्रे. के प्रथम n पदों को निरूपित करता है यदि $S_{2n} = 3S_n$ तो, $\frac{S_{3n}}{S_n}$ का मान होगा	(Q) 9
(C) अनन्त श्रेणी $4 + \frac{8}{3} + \frac{12}{3^2} + \frac{16}{3^3} + \ldots\ldots$ का योग होगा	(R) 3
(D) एक आयताकार बक्से के लम्बाई, चौड़ाई तथा ऊँचाई गु. श्रे. में है इसका आयतन 27 है, कुल पृष्ठीय क्षेत्रफल 78 है, तो लम्बाई होगी	(S) 1

Ans. (A) P, (B) P, (C) q, (D) Q, S

Sol. (A) $\log_5 2, \log_5(2x-5), \log_5\left(2^x - \frac{7}{2}\right)$ are in A.P.

$$2\log_5(2^x - 5) = \log_5 + 2\log_5\left(2^x - \frac{7}{2}\right)$$

$$(2^x - 5)^2 = 2\left(2^x - \frac{7}{2}\right)$$

Let $t = 2^x$

$$(t-5)^2 = 2\left(t - \frac{7}{2}\right)$$

$t^2 + 25 - 10t = 2t - 7$
$t^2 - 12t + 32 = 0$
$(t - 8)(t - 4) = 0$
$t = 4, 8$
$2^x = 2$ or $2^x = 8$
$x = 1$ or $x = 3$
$2x = 2$ or $2x = 6$
But $2^x - 5 > 0$
$2^x > 5$
$\Rightarrow x > 2$ $\Rightarrow$ $x = 3$

(B) $S_n = \frac{n}{2}[2a + (n-1)d]$

$S_{2n} = 3S_n$

$$\frac{2n}{2}[2a + (2n-1)d] = 3 \times \frac{n}{2}[2a + (n-1)d]$$

$2\left[2a+(2n-1)d\right]=3\left[2a+(n-1)d\right]$

$4a + (4n - 2)\, d = 6a + (3n - 3)d$

$2a = (n + 1)d$

$$\frac{S_{3n}}{S_n}=\frac{\frac{3n}{2}\left[2a+(3n-1)d\right]}{\frac{n}{2}\left[2a+(n-1)d\right]}=\frac{3\left[(n+1)+(3n-1)\right]}{\left[(n+1)+(n-1)\right]}$$

$$=3\left[\frac{4n}{2n}\right]$$

$= 6$

(C) $S=4+\frac{8}{3}+\frac{12}{3^2}+\frac{16}{3^3}+.....$ ...(1)

$$\frac{S}{3}=\frac{4}{3}+\frac{8}{3^2}+\frac{12}{3^3}+.....$$ ...(2)

(1) – (2)

$$\frac{2S}{3}=4+\frac{4}{3}+\frac{4}{3^2}+\frac{4}{3^3}+.....$$

$$\frac{2S}{3}=4\left[\frac{1}{1-\frac{1}{3}}\right]$$

$$S=\frac{3}{2}\times 4\left(\frac{3}{2}\right)=9$$

(D) let length = a

breadth = b,

Height = c

a,b,c are in G.P. $\Rightarrow \frac{a}{r}$, a, ar

volume of ractangular box = abc = 27

$$\left(\frac{a}{r}\right)(a)(ar)=27$$

$a^3 = 3^3$

$a = 3$

Surface area = 2 (ab + bc + ca) = 78

$$2\left[\left(\frac{a}{r}\right)(a)+a(ar)+(ar)\left(\frac{a}{r}\right)\right]=78$$

$$a^2\left[\frac{r^2+r+1}{r}\right]=39$$

$$9\left[\frac{r^2+r+1}{r}\right]=39$$

$3r^2 + 3r + 3 = 13r$
$3r^2 - 10r + 3 = 0$
$3r^2 - 9r - r + 3 = 0$
$3r(r-3) - (r-3) = 0$
$(3r-1)(r-3) = 0$

$$r = 3, \frac{1}{3}$$

$$\text{length} = \frac{a}{r} = \frac{3}{3} \text{ or } \left(\frac{3}{\frac{1}{3}}\right)$$

= 1 or 9

35. Match the following columns :

Column – I	**Column – II**
(A) If $\log_x y$, $\log_z x$, $\log_y z$ are in G.P., $xyz = 64$ and x^3, y^3, z^3 are in A.P., then $\frac{3x}{y}$ =	(P) 2
(B) The value of $2^{\frac{1}{4}}.4^{\frac{1}{8}}.8^{\frac{1}{16}} \ldots\ldots \infty$ =	(Q) 1
(C) If x, y, z are in A.P., then $(x+2y-z)(2y+z-x)(z+x-y) = kxyz$, where $k \in N$, then k =	(R) 3
	(S) 4

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) यदि $\log_x y$, $\log_z x$, $\log_y z$ गु.श्रे. में हो, $xyz = 64$ तथा x^3, y^3, z^3 स.श्रे. में हो तो $\frac{3x}{y}$ =	(P) 2
(B) $2^{\frac{1}{4}}.4^{\frac{1}{8}}.8^{\frac{1}{16}} \ldots\ldots \infty$ =	(Q) 1
(C) यदि x, y, z स.श्रे. में हो तो $(x+2y-z)(2y+z-x)(z+x-y) = kxyz$, जहाँ $k \in N$, तो k =	(R) 3
	(S) 4

Ans. (A) R, (B) P, (C) S,

Sol. $\log_x y, \log_z x, \log_y z$ are in G.P.

$(\log_z x)^2 = (\log_x y)(\log_y z)$

$(\log_z x)^2 = (\log_x z)$

$(\log_z x)^3 = 1$

$\log_z x = 1 \quad \Rightarrow x = z$

$xyz = 64$

$x^2y = 64 \quad \Rightarrow y = \dfrac{64}{x^2}$

x^3, y^3, z^3 are in A.P.

$2y^3 = x^3 + z^3$

$y^3 = x^3$

$$\left(\frac{64}{x^2}\right)^3 = x^3$$

$(64)^3 = x^3 \cdot x^6$

$4^9 = x^9$

$x = 4 \Rightarrow y = 4 \Rightarrow z = 4$

$$\frac{3x}{y} = 3$$

(B) $S = 2^{\frac{1}{4}} \cdot 4^{\frac{1}{8}} \cdot 8^{\frac{1}{16}} \ldots\ldots\ldots\ldots = 2^{\frac{1}{4}} \cdot 2^{\frac{2}{8}} \cdot 2^{\frac{3}{16}}$

$$S = 2^{\left(\frac{1}{4}+\frac{2}{8}+\frac{3}{16}+\ldots\ldots\right)}$$

$$A = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \ldots\ldots \qquad \ldots(1)$$

$$\frac{A}{2} = 0 + \frac{1}{8} + \frac{2}{16} + \ldots\ldots\ldots \qquad \ldots(2)$$

(1) – (2)

$$\frac{A}{2} = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots\ldots\ldots$$

$$\frac{A}{2} = \frac{1}{4}\left[1 + \frac{1}{2} + \frac{1}{4} + \ldots\ldots\right] = \frac{1}{4}\left[\frac{1}{1-\frac{1}{2}}\right] = \frac{1}{2}$$

$A = 1$

$S = 2^1 = 2$

(C) $(x + 2y - z)(2y + z - x)(z + x - y) = k(xyz)$

x, y, z are in A.P.

$2y = x + z$

$(x + x + z - z)\,(x + z + z - x)\left(z + x - \left(\frac{x+z}{2}\right)\right) = k(xyz)$

$(2x)\,(2z)\left(\frac{x+z}{2}\right) = k(xyz)$

$4(xyz) = k(xyz)$

$k = 4$

36. Match the series in Column I with their sum in Column II :

Column I	**Column II**
(A) $\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)$ ..... to infinity equals	(P) $\frac{1}{4}$
(B) $\frac{2^3-1}{2^3+1}\times\frac{3^3-1}{3^3+1}\times\frac{4^3-1}{4^3+1}$ ....to inifinity equals	(Q) $\frac{1}{2}$
(C) $\frac{1}{1.2.3}+\frac{1}{2.3.4}+\frac{1}{3.4.5}$ +..... to inifinity equals	(R) $\frac{2}{3}$
(D) $\frac{1}{1.2.3.4}+\frac{1}{2.3.4.5}+\frac{1}{3.4.5.6}$ +.... to inifnity equals	(S) $\frac{1}{18}$

कॉलम I में दी गई श्रेणी का कॉलम II में दिये गये योग से मिलान कीजिए :

Column I	**Column II**
(A) $\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)$ ..... अनन्त तक	(P) $\frac{1}{4}$
(B) $\frac{2^3-1}{2^3+1}\times\frac{3^3-1}{3^3+1}\times\frac{4^3-1}{4^3+1}$ ....अनन्त तक	(Q) $\frac{1}{2}$
(C) $\frac{1}{1.2.3}+\frac{1}{2.3.4}+\frac{1}{3.4.5}$ +..... अनन्त तक	(R) $\frac{2}{3}$
(D) $\frac{1}{1.2.3.4}+\frac{1}{2.3.4.5}+\frac{1}{3.4.5.6}$ +.... अनन्त तक	(S) $\frac{1}{18}$

Ans. (A) Q, (B) R, (C) P, (D) S

Sol. (A) $S = \left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)$.....

$$T_r = \left[1 - \frac{1}{(r+1)^2}\right] = \frac{(r+1)^2 - 1}{(r+1)^2} = \frac{r(r+2)}{(r+1)^2}$$

$$S = \prod_{r=1}^{\infty} T_r = \left(\frac{1\cdot 3}{2^2}\right)\left(\frac{2\cdot 4}{3^2}\right)\left(\frac{3\cdot 5}{4^2}\right)\left(\frac{4\cdot 6}{5^2}\right).....$$

$$= \frac{1}{2}$$

(B) $$S = \frac{2^3 - 1}{2^3 + 1} \times \frac{3^3 - 1}{3^3 + 1} \times \frac{4^3 - 1}{4^3 + 1} \times .....$$

$$T_r = \frac{(r+1)^3 - 1}{(r+1)^3 + 1} = \frac{((r+1) - 1)}{((r+1) + 1)} \times \left[\frac{(r+1)^2 + 1 + (r+1)}{(r+1)^2 + 1 - (r+1)}\right]$$

$$= \left(\frac{r}{r+2}\right)\left[\frac{r^2 + 3r + 3}{r^2 + r + 1}\right]$$

$$S = \prod_{r=1}^{\infty} T_r$$

$$S = \prod_{r=1}^{\infty} \left(\frac{r}{r+2}\right)\left(\frac{r^2 + 3r + 3}{r^2 + r + 1}\right)$$

$$= \left(\frac{1}{3}\right)\left(\frac{7}{3}\right) \times \left(\frac{2}{4}\right)\left(\frac{13}{7}\right) \times \left(\frac{3}{5}\right)\left(\frac{21}{13}\right) \times .....$$

$$S = \frac{2}{3}$$

(C) $$S = \frac{1}{1\cdot 2\cdot 3} + \frac{1}{2\cdot 3\cdot 4} + \frac{1}{3\cdot 4\cdot 5} + ...$$

$$S = \sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)} = \sum_{n=1}^{\infty} \left[\frac{(n+2) - n}{n(n+1)(n+2)}\right]$$

$$S = \frac{1}{2} \sum_{n=1}^{\infty} \left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

$$S = \frac{1}{2}\left[\left(\frac{1}{1\cdot 2} - \frac{1}{2\cdot 3}\right) + \left(\frac{1}{2\cdot 3} - \frac{1}{3\cdot 4}\right) + ...\right]$$

$$S = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

(D) $$S = \frac{1}{1\cdot 2\cdot 3\cdot 4} + \frac{1}{2\cdot 3\cdot 4\cdot 5} + ...$$

$$S=\sum_{r=1}^{\infty}\frac{1}{n(n+1)(n+2)(n+3)}=\frac{1}{3}\sum_{n=1}^{\infty}\frac{(n+3)-n}{n(n+1)(n+2)(n+3)}$$

$$S=\frac{1}{3}\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)(n+2)}-\frac{1}{(n+1)(n+2)(n+3)}\right]$$

$$S=\frac{1}{6}\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)}-\frac{1}{(n+1)(n+2)}\right]-\frac{1}{6}\sum_{n=1}^{\infty}\frac{1}{(n+1)(n+2)}-\frac{1}{(n+2)(n+3)}$$

$$=\frac{1}{6}\sum_{n=1}^{\infty}\frac{1}{n(n+1)}-\frac{1}{3}\sum_{n=1}^{\infty}\frac{1}{(n+1)(n+2)}+\frac{1}{6}\sum_{n=1}^{\infty}\frac{1}{(n+2)(n+3)}$$

$$=\frac{1}{6}\left[\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+1}\right)\right]-\frac{1}{3}\left[\sum_{n=1}^{\infty}\left(\frac{1}{n+1}-\frac{1}{n+2}\right)\right]+\frac{1}{6}\left[\sum_{n=1}^{\infty}\left(\frac{1}{n+2}-\frac{1}{n+3}\right)\right]$$

$$=\frac{1}{6}-\frac{1}{6}+\frac{1}{6}\times\frac{1}{3}$$

$$=\frac{1}{18}$$

Exercise J-M

1. If sum of the first 21 terms of the series $\log_{9^{\frac{1}{2}}} x + \log_{9^{\frac{1}{3}}} x + \log_{9^{\frac{1}{4}}} x + \ldots.$, where $x > 0$ is 504, then x is equal to:

[JEE(Main)July2021]

यदि श्रेणी $\log_{9^{\frac{1}{2}}} x + \log_{9^{\frac{1}{3}}} x + \log_{9^{\frac{1}{4}}} x + \ldots.$, $x > 0$ के प्रथम 21 पदों का योग 504 है, तो x बराबर है :

[JEE(Main)July2021]

(A) 243 (B) 7 (C) 9 (D) 81

Ans. D

Sol. $2\log_9^x + 3\log_9^x + 4\log_9^x + \ldots\ldots\ldots\ldots$ 21 terms

$2\log_9^x (2+3+4+\ldots\ldots\ldots+22) = 504$

$252 \log_9^x = 504$

$\log_9^x = 2$

$x = 81$

2. Let S_n denote the sum of first n-terms of an arithmetic progression. If $S_{10} = 530$, $S_5 = 140$, then $S_{20} - S_6$ is equal to: **[JEE(Main)July2021]**

माना एक समान्तर श्रेढ़ी के प्रथम n पदों का योग S_n है। यदि $S_{10} = 530$, $S_5 = 140$ हैं, तो $S_{20} - S_6$ बराबर है :

[JEE(Main)July2021]

(A) 1862 (B) 1872 (C) 1852 (D) 1842

Ans. A

Sol. $S_{10} = 530$

$\frac{10}{2}(2a + 9d) = 530$

$2a + 9d = 106$ ....(1)

$S_5 = \frac{5}{2}(2a + 4d) = 140$

$a + 2d = 28$ ....(2)

By (1) and (2)

$d = 10, \quad a = 8$

Now $S_{20} - S_6 = \frac{20}{2}[2a + 19d] - \frac{6}{2}[2a + 5d]$

$= 1862$

3. Let S_n be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is :

[JEE(Main)July2021]

माना एक समान्तर श्रेढ़ी के प्रथम n पदों का योगफल S_n हैं। यदि $S_{3n} = 3S_{2n}$ है, तो $\frac{S_{4n}}{S_{2n}}$ बराबर है :

[JEE(Main)July2021]

(A) 8 (B) 4 (C) 2 (D) 6

Ans. D

Sol. $S_{3n} = 3S_{2n}$

$$\frac{3n}{2}[2a+(3n-1)d] = \frac{3.(2n)}{2}[2a+(2n-1)d]$$

$2a + (3n-1)d = 4a + (4n-2)d$

$(3n - 1 - 4n + 2)d = 2a$

$(1-n)\,d = 2a$

$$\frac{S_{4n}}{S_{2n}} = \frac{\frac{4n}{2}[2a+(4n-1)d]}{\frac{2n}{2}[2a+(2n-1)d]}$$

$$= 2\frac{[(1-n)d+(4n-1)d]}{[(1-n)d+(2n-1)d]}$$

$= 2\,(3) = 6$

4. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_1 = 1$, $a_2 = 1$ and $a_{n+2} = 2a_{n+1} + a_n$ for all $n \geq 1$. Then the value of 47 $\sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ is equal to _____. **[JEE(Main)July2021]**

माना $\{a_n\}_{n=1}^{\infty}$ एक अनुक्रम है, जिसके लिए $a_1 = 1$, $a_2 = 1$ तथा $a_{n+2} = 2a_{n+1} + a_n$ $\forall\, n \geq 1$ हैं। तो 47 $\sum_{n=1}^{\infty} \frac{a_n}{2^{3n}}$ बराबर है _____

[JEE(Main)July2021]

Ans. 7

Sol. $a_{n+2} = 2a_{n+1} + a_n$

$$\frac{a_{n+2}}{8^n} = 2\,\frac{a_{n+1}}{8^n} + \frac{a_n}{8^n}$$

$$64\sum_{n=1}^{\infty}\left(\frac{a_{n+2}}{8^{n+2}}\right) = 16\sum_{n=1}^{\infty}\left(\frac{a_{n+1}}{8^{n+1}}\right) + \sum_{n=1}^{\infty}\frac{a_n}{8^n}$$

Let $\sum_{n=1}^{\infty}\left(\frac{a_n}{8^n}\right)=p$

$64\left(p-\frac{a_1}{8}-\frac{a_2}{8^2}\right)=16\left(p-\frac{a_1}{8}\right)+p$

$p=\frac{7}{47}$

$47\,p=7$

5. If the value of $\left(1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\ldots\ldots\text{ upto}\infty\right)^{\log_{(0.25)}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\ldots\text{upto}\infty\right)}$ is l, then l^2 is equal to _____.

[JEE(Main)July2021]

यदि $\left(1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}+\ldots\ldots\ \infty\text{ तक}\right)^{\log_{(0.25)}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\ldots\infty\text{ तक}\right)}$ का मान l है, तो l^2 बराबर है _____

[JEE(Main)July2021]

Ans. 3

Sol. Let

$S_1=1+\frac{2}{3}+\frac{6}{3^2}+\frac{10}{3^3}\ldots\ldots$

$\frac{S_1}{3}=\frac{1}{3}+\frac{2}{3^2}+\frac{6}{3^3}\ldots\ldots$

$\frac{2S_1}{3}=1+\frac{1}{3}+\frac{4}{3^2}+\frac{4}{3^3}\ldots\ldots$

$\frac{2S_1}{3}=\frac{4}{3}+\frac{4}{3^2}+\frac{4}{3^3}\ldots\ldots$

$\frac{2S_1}{3}=\frac{4}{3}\left[\frac{1}{1-\frac{1}{3}}\right]$

$\frac{2S_1}{3}=\frac{4}{3}\times\frac{3}{2}$

$S_1=3$

$S_2=\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}\ldots\ldots$

$$S_2 = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{1}{2}$$

$$\ell = (S_1)^{\log_{0.25}(S_2)} = (3)^{\log_{0.25}(0.5)} = \sqrt{3}$$

$$\ell^2 = 3$$

6. If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to_____.

[JEE(Main)July2021]

यदि $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ एक समांतर श्रेढ़ी में है, तो x का मान बराबर है _____

[JEE(Main)July2021]

Ans. 3

Sol. $2\log_3(2^x - 5) = \log_3 2 + \log_3\left(2x - \frac{7}{2}\right)$

$$(2^x - 5)^2 = 2\left(2^x - \frac{7}{2}\right)$$

$(t-5)^2 = 2t - 7$

$\Rightarrow t^2 - 12t + 32 = 0$

$2^x = 4, 8$

$x = 2, 3$ (for $x = 2$, $2^x - 5 < 0$)

So $x = 3$

7. If the sum of an infinite GP a, ar, ar^2, ar^3, ... is 15 and the sum of the squares of its each term is 150, then the sum of ar^2, ar^4, ar^6, ... is : **[JEE(Main)Aug.2021]**

एक अनन्त GP a, ar, ar^2, ar^3, ... का योग 15 है तथा इसके प्रत्येक पद का वर्ग करने पर योग 150 है, तो ar^2, ar^4, ar^6, ... का योग है : **[JEE(Main)Aug.2021]**

(A) $\frac{25}{2}$ (B) $\frac{1}{2}$ (C) $\frac{5}{2}$ (D) $\frac{9}{2}$

Ans. B

Sol. $\frac{a}{1-r} = 15$ $\frac{a^2}{1-r^2} = 150$

$$\frac{(15)^2(1-r)^2}{1-r^2} = 150$$

$$\frac{1-r}{1+r}=\frac{2}{3}$$

$3 - 3r = 212r$

$$r=\frac{1}{5}$$

$\Rightarrow a = 15\,(1 - r) = 12$

$S = ar^2 + ar^4 + ar^6 + ...\infty$

$$= \frac{ar^2}{1-r^2} = \frac{12\left(\frac{1}{5}\right)^2}{1-\left(\frac{1}{5}\right)^2} = \frac{12}{24} = \frac{1}{2}$$

8. The sum of the series $\frac{1}{x+1}+\frac{2}{x^2+1}+\frac{2^2}{x^4+1}+.....+\frac{2^{100}}{x^{2^{100}}+1}$ when $x = 2$ is : **[JEE(Main)Aug.2021]**

श्रेणी $\frac{1}{x+1}+\frac{2}{x^2+1}+\frac{2^2}{x^4+1}+.....+\frac{2^{100}}{x^{2^{100}}+1}$ का योग, जब $x = 2$ है, है : **[JEE(Main)Aug.2021]**

(A) $1+\frac{2^{101}}{2^{2^{101}}-1}$ (B) $1-\frac{2^{101}}{2^{2^{101}}-1}$ (C) $1+\frac{2^{100}}{2^{2^{101}}-1}$ (D) $1-\frac{2^{100}}{2^{2^{101}}-1}$

Ans. B

Sol. $$S=\frac{1}{x+1}+\frac{2}{x^2+1}+\frac{2^2}{x^4+1}+.....$$

$$S-\frac{1}{x-1}=\frac{1}{x+1}-\frac{1}{x-1}+\frac{2}{x^2+1}+.....$$

$$S-\frac{1}{x-1}=\frac{2}{x^2+1}-\frac{2}{x^2-1}+\frac{2}{x^4+1}$$

$$S-\frac{1}{x-1}=-\frac{2^{101}}{x^{2^{101}}-1}$$

put $x = 2$

$$S=1-\frac{2^{101}}{2^{2^{101}}-1}$$ Bonus

9. If for $x, y \in \mathbf{R}, x > 0, y = \log_{10}x + \log_{10}x^{1/3} + \log_{10}x^{1/9} + ...$upto ∞ terms and $\frac{2+4+6+...+2y}{3+6+9+...+3y} = \frac{4}{\log_{10} x}$, then the ordered pair (x, y) is equal to : **[JEE(Main)Aug.2021]**

यदि $x, y \in \mathbf{R}, x > 0$ के लिए $y = \log_{10}x + \log_{10}x^{1/3} + \log_{10}x^{1/9} + ...$अनन्त पदों तक तथा $\frac{2+4+6+...+2y}{3+6+9+...+3y} = \frac{4}{\log_{10} x}$, है, तो क्रमित युग्म (x, y) बराबर है : **[JEE(Main)Aug.2021]**

(A) $(10^6, 6)$ (B) $(10^4, 6)$ (C) $(10^2, 3)$ (D) $(10^6, 9)$

Ans. D

Sol. $y = \log_{10} x + \log_{10} x^{1/3} + \log_{10} x^{1/9} + ...\infty$

$$y = \left(1 + \frac{1}{3} + \frac{1}{9} + ...\infty\right)\log_{10} x$$

$$y = \frac{3}{2}\log_{10} x$$

$$\frac{2+4+6+.....+2y}{3+6+9+.....+3y} = \frac{4}{\log_{10} x}$$

$$\Rightarrow \frac{2}{3} = \frac{4}{\log_{10} x} \Rightarrow \log_{10} x = 6$$

$$x = 10^6$$

$$y = \frac{3}{2}\log_{10} 10^6$$

$$y = 9$$

$$(x, y) = (10^6, 9)$$

10. The sum of 10 terms of the series $\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + ...$ is : **[JEE(Main)Aug.2021]**

श्रेणी $\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + ...$ के 10 पदों का योग है : **[JEE(Main)Aug.2021]**

(A) $\frac{143}{144}$ (B) $\frac{120}{121}$ (C) 1 (D) $\frac{99}{100}$

Ans. B

Sol. $S = \sum_{r=1}^{10} \frac{2r+1}{r^2(r+1)^2} = \sum_{r=1}^{10} \frac{(r+1)^2 - r^2}{r^2(r+1)^2}$

$$S=\sum_{r=1}^{10}\frac{1}{r^2}-\frac{1}{(r+1)^2}=\left(\left(\frac{1}{1^2}-\frac{1}{2^2}\right)+\left(\frac{1}{2^2}-\frac{1}{3^2}\right)+.....+\left(\frac{1}{10^2}-\frac{1}{11^2}\right)\right)$$

$$S=1-\frac{1}{121}=\frac{120}{121}$$

11. Three numbers are in an increasing geometric progression with common ratio r. If the middle number is doubled, then the new numbers are in an arithmetic progression with common difference d. If the fourth term of GP is $3r^2$, then $r^2 - d$ is equal to : **[JEE(Main)Aug.2021]**

तीन संख्याऐं एक वर्धमान गुणोत्तर श्रेढ़ी, जिसका सार्व अनुपात r है, में हैं। यदि बीच की संख्या को दुगुना कर दिया जाये, तो नयी संख्याऐं एक समान्तर श्रेढ़ी, जिसका सार्वअंतर d हैं, में हैं। यदि गुणोत्तर श्रेढ़ी का चौथा पद $3r^2$ हैं, तो $r^2 - d$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $7-\sqrt{3}$ (B) $7-7\sqrt{3}$ (C) $7+3\sqrt{3}$ (D) $7+\sqrt{3}$

Ans. D

Sol. $\frac{3}{r}, 3, 3r, 3r^2, ...$

$\frac{3}{r}, 6, 3r$ are in A.P.

$$12=\frac{3}{r}+3r$$

$$\Rightarrow r^2-4r+1=0 \quad (r>0)$$

$$r=2+\sqrt{3}$$

$$d=3r-6=3\sqrt{3}$$

$$r^2-d=(2+\sqrt{3})^2-(3\sqrt{3})$$

$$=7+\sqrt{3}$$

12. Let $a_1, a_2, a_3, ...$ be an A.P. If $\frac{a_1+a_2+...+a_{10}}{a_1+a_2+...+a_p}=\frac{100}{p^2}$, $p \neq 10$, then $\frac{a_{11}}{a_{10}}$ is equal to : **[JEE(Main)Aug.2021]**

माना $a_1, a_2, a_3, ...$ एक A.P. है। यदि $\frac{a_1+a_2+...+a_{10}}{a_1+a_2+...+a_p}=\frac{100}{p^2}$, $p \neq 10$ है, तो $\frac{a_{11}}{a_{10}}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) $\frac{121}{100}$ (B) $\frac{21}{19}$ (C) $\frac{100}{121}$ (D) $\frac{19}{21}$

Ans. B

Sol. $$\frac{\frac{10}{2}(2a+9d)}{\frac{P}{2}(2a+(P-1)d)}=\frac{100}{P^2}$$

$$\frac{2a+9d}{2a+(P-1)d}=\frac{10}{P}$$

$$2aP+9dP=20a+10(P-1)d$$

$$2aP=20a=10(P-1)d-9Pd$$

$$a(2P-20)=d(10P-10-9P)$$

$$2a(P-10)=d(P-10)$$

$$2a=d$$

$$\frac{a_{11}}{a_{10}}=\frac{a+10d}{a+9d}=\frac{a+10(2a)}{a+9(2a)}=\frac{21a}{19a}=\frac{21}{19}$$

13. Let $a_1, a_2, ..., a_{21}$ be an AP such that $\sum_{n=1}^{20}\frac{1}{a_n a_{n+1}}=\frac{4}{9}$. In the sum of this A.P. is 189, then $a_6 a_{16}$ is equal to :

[JEE(Main)Aug.2021]

माना $a_1, a_2, ..., a_{21}$ समांतर श्रेढ़ी में इस प्रकार हैं कि $\sum_{n=1}^{20}\frac{1}{a_n a_{n+1}}=\frac{4}{9}$ है। यदि इस समांतर श्रेढ़ी का योगफल 189 है, तब $a_6 a_{16}$ बराबर है :

[JEE(Main)Aug.2021]

(A) 36 (B) 57 (C) 72 (D) 48

Ans. C

Sol. $$\frac{1}{a_1.a_2}+\frac{1}{a_2a_3}+\frac{1}{a_3a_4}+.....+\frac{1}{a_{20}.a_{21}}$$

$\therefore$ $a_1, a_2, a_3, ..., a_{21}$ be an A.P.

$\Rightarrow$ $a_2-a_1=a_3-a_2=.....=a_{21}-a_{20}=d$

$$=\frac{1}{d}\left(\frac{a_2-a_1}{a_1a_2}+\frac{a_3-a_3}{a_2a_3}+.....+\frac{a_{21}+a_{20}}{a_{20}.a_{21}}\right)$$

$$=\frac{1}{d}\left[\frac{1}{a_1}-\frac{1}{a_2}+\frac{1}{a_2}-\frac{1}{a_3}+.....+\frac{a_{21}-a_{20}}{a_{20}-a_{21}}\right]$$

$$= \frac{1}{d}\left[\frac{1}{a_1} - \frac{1}{a_{21}}\right]$$

$$= \frac{a_{21} - a_1}{d.a_1a_{21}} = \frac{a_1 + (21-1)d - a_1}{d.(a_1a_{21})}$$

$$= \frac{20}{a_1a_{21}}$$

$$\therefore \frac{20}{a_1a_{21}} = \frac{4}{9}$$

$\Rightarrow a_1\, a_{21} = 45$ .....(1)

Sum = 189

$$\Rightarrow \frac{21}{2}[a_1 + a_{21}] = 189$$

$\therefore a_1 + a_{21} = 18$ .....(2)

using (1) & (2)

$a_1 = 3$ and $a_{21} = 15$

$\therefore 15 = 3 + 20d$

$$\therefore d = \frac{12}{20}$$

$$d = \frac{3}{5}$$

$a_6 = a_1 + 5d = 3 + 3 = 6$

$a_{16} = a_1 + 15d = 3 + 9 = 12$

$a_6 . a_{16} = 72$

14. The number of 4-digit numbers which are neither multiple of 7 nor multiple of 3 is_____.

[JEE(Main)Aug.2021]

4-अंको की संख्याएँ, जो न तो 7 की गुणज हैं न ही 3 की गुणज हैं, की संख्या है_____ **[JEE(Main)Aug.2021]**

Ans. 5143

Sol. 4 digit numbers divisible by 3

1002, 1005,.....9999

9999 = 1002 + (n –) 3

$$\frac{9999 - 1002}{3} + 1 = n$$

$$\frac{8997}{3} + 1 = n$$

2999 + 1 = n

n = 3000

4-digit numbers divisible by 7

1001, 1008,.....9996

9996 = 1001 + (n – 1)7

$$\frac{9996-1001}{7}+1=n$$

$$\frac{8995}{7}+1=n$$

1285 + 1 = n

n = 1286

4-digit numbers divisible by both 3 and 7

1008, 1029.....9996

9996 = 1008 + (n – 1)21

$$\frac{9996-1008}{21}+1=n$$

$$\frac{8988}{21}+1=n$$

428 + 1 = n

n = 429

No. divisible by either 3 or 7

= 3000 + 1286 – 429

= 3857

No. not divisible by 3 and 7

= 9000 – 3857

= 5143

15. If $S=\frac{7}{5}+\frac{9}{5^2}+\frac{13}{5^3}+\frac{19}{5^4}+...$, then 160 S is equal to_____. **[JEE(Main)Aug.2021]**

यदि $S=\frac{7}{5}+\frac{9}{5^2}+\frac{13}{5^3}+\frac{19}{5^4}+...$ है, तो 160 S बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 305

Sol. $S=\frac{7}{5}+\frac{9}{5^2}+\frac{13}{5^3}+\frac{19}{5^4}+.....$

$$\frac{S}{5}=\frac{7}{5^2}+\frac{9}{5^3}+\frac{13}{5^4}+....$$

$$\frac{4S}{5}=\frac{7}{5}+\frac{2}{5^2}+\frac{4}{5^3}+\frac{6}{5^9}+.....$$

$$S_0 = \frac{2}{5^2} + \frac{4}{5^3} + \frac{6}{5^4} + ...$$

$$\frac{S_0}{5} = \quad \frac{2}{5^3} \quad + \quad \frac{4}{5^4} + ...$$

$$\frac{4S_0}{5} = \frac{2}{5^2} + \frac{2}{5^3} + \frac{2}{5^4} + ...$$

$$2S_0 = \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + ...$$

$$2S_0 = \frac{\frac{1}{5}}{1-\frac{1}{5}} = \frac{\frac{1}{5}}{\frac{4}{5}} = \frac{1}{4}$$

$$S_0 = \frac{1}{8}$$

$$\frac{4S}{5} - \frac{7}{5} = \frac{1}{8}$$

$$\frac{4S}{5} = \frac{1}{8} + \frac{7}{5} = \frac{61}{40}$$

$$S = \frac{5}{4} \times \frac{61}{40}$$

$$160S = 160 \times \frac{5}{4} \times \frac{61}{40} = 61 \times 5 = 305$$

16. Let $a_1, a_2, ...a_{10}$ be an AP with common difference –3 and $b_1, b_2, ...b_{10}$ be a GP with common ratio 2. Let $c_k = a_k + b_k$, k = 1, 2,....., 10. If $c_2 = 12$ and $c_3 = 13$, then $\sum_{k=1}^{10} c_k$ is equal to ____. **[JEE(Main)Aug.2021]**

माना $a_1, a_2, ...a_{10}$ एक AP है, जिसका सार्वअंतर –3 है तथा $b_1, b_2, ...b_{10}$ एक GP है, जिसका सार्व अनुपात 2 है। माना $c_k = a_k + b_k$, k = 1, 2,....., 10 है। यदि $c_2 = 12$ तथा $c_3 = 13$ है, तो $\sum_{k=1}^{10} c_k$ बराबर है____ **[JEE(Main)Aug.2021]**

Ans. 2021

Sol. $c_2 = a_2 + b_2 = (a_1 - 3) + 2b_1 = 12 \Rightarrow a_1 = 11$
$c_3 = a_3 + b_3 = (a_1 - 6) + 4b_1 = 13 \Rightarrow b_1 = 2$
$c_k = a_k + b_k = (a_1 - 3(k-1)) + (b_1 . 2^{k-1})$
$= (11 - 3k + 3) + (2^k) = 14 - 3k + 2^k$

$$\sum_{k=1}^{10} c_k = \sum_{k=1}^{10}\left(2^k - 3k + 14\right)$$

$$= \sum_{k=1}^{10} 2^k - 3\sum_{k=1}^{10} k + \sum_{k=1}^{10} 14$$

$$= 2\left(2^{10} - 1\right) - 3\cdot\frac{10\cdot 11}{2} + 140$$

$$= 2021$$

17. If $\{a_i\}_{i=1}^{n}$, where n is an even integer, is an arithmetic progression with common difference 1, and $\sum_{i=1}^{n} a_i = 192, \sum_{i=1}^{\frac{n}{2}} a_{2i} = 120$, then n is equal to : **[JEE(Main)June2022]**

यदि $\{a_i\}_{i=1}^{n}$, जहाँ n एक सम पूर्णांक है, एक समांतर श्रेढ़ी है, जिसका सार्व अंतर 1 है, तथा $\sum_{i=1}^{n} a_i = 192, \sum_{i=1}^{\frac{n}{2}} a_{2i} = 120$ हैं, तो n बराबर है :

(1) 48 (2) 96 (3) 92 (4) 104

Ans. Official Answer NTA (2)

Sol. $a_1 + a_2 + a_3 + ... + a_n = 192$ ——(1)

$a_2 + a_4 + a_6 + ... + a_n = 120$ ——(2)

(1) – (2)

$a_1 + a_3 + a_5 + ... + a_{n-1} = 72$ ——(3)

(2) – (3)

$1 + 1 + ... + 1 = 48$, Total terms = n, number of odd terms = n/2

$$\frac{n}{2} = 48$$

$$n = 96$$

18. Let x, y > 0. If $x^3 y^2 = 2^{15}$, then the least values of 3x + 2y is : **[JEE(Main)June2022]**

माना x, y > 0 हैं। यदि $x^3 y^2 = 2^{15}$ है, तो 3x + 2y का न्यूनतम मान है : **[JEE(Main)June2022]**

(1) 30 (2) 32 (3) 36 (4) 40

Ans. Official Answer NTA (4)

Sol. $x^3y^2 = 2^{15}$

Let consider numbers x, x, x, y, y and apply AM & GM

$$\frac{x + x + x + y + y}{5} \geq \left(x^3.y^2\right)^{\frac{1}{5}}$$

$3x + 2y \geq 5\left(2^{15}\right)^{\frac{1}{5}}$

$3x + 2y \geq 40$

Then least value of $3x + 2y$ is 40.

19. If $\frac{1}{2.3^{10}} + \frac{1}{2^2.3^9} + ... + \frac{1}{2^{10}.3} = \frac{K}{2^{10}.3^{10}}$, then the remainder when K is divided by 6 is :

[JEE(Main)June2022]

यदि $\frac{1}{2.3^{10}} + \frac{1}{2^2.3^9} + ... + \frac{1}{2^{10}.3} = \frac{K}{2^{10}.3^{10}}$ है, तो K को 6 से विभाजित करने पर शेषफल है :

[JEE(Main)June2022]

(1) 1 (2) 2 (3) 3 (4) 5

Ans. Official Answer NTA (4)

Sol. $\frac{1}{2 \cdot 3^{10}} + \frac{1}{2^2 \cdot 3^9} + \frac{1}{2^3 \cdot 3^8} + ... + \frac{1}{2^{10} \cdot 3} = \frac{K}{2^{10} \cdot 3^{10}}$

$K = 2^9 + 2^8 \cdot 3 + 2^7 \cdot 3^2 + ... + 3^9$

$$= \frac{2^9\left(\left(\frac{3}{2}\right)^{10} - 1\right)}{\frac{3}{2} - 1} = 3^{10} - 2^{10}$$

Now, $3^{10} - 2^{10} = (3^5 - 2^5)(3^5 + 2^5)$

$= (211)(275)$

$= (35 \times 6 + 1)(45 \times 6 + 5)$

$= 6l + 5$

Remainder is 5.

20. The sum $1 + 2 \cdot 3 + 3 \cdot 3^2 + ... + 10 \cdot 3^9$ is equal to : **[JEE(Main)June2022]**

योगफल $1 + 2 \cdot 3 + 3 \cdot 3^2 + ... + 10 \cdot 3^9$ बराबर है : **[JEE(Main)June2022]**

(1) $\frac{2 \cdot 3^{12} + 10}{4}$ (2) $\frac{19 \cdot 3^{10} + 1}{4}$ (3) $5 \cdot 3^{10} - 2$ (4) $\frac{9 \cdot 3^{10} + 1}{2}$

Ans. Official Answer NTA (2)

Sol. $S = 1 + 2.3 + 3.3^2 + ... + 10.3^9$

$3S = 1.3 + 2.3^2 + ... + 9.3^9 + 10.3^{10}$

$-2S = 1 + 3 + 3^2 + ...+ 3^9 - 10.3^{10}$

$-2S = \frac{1.\left(3^{10} - 1\right)}{3 - 1} - 10.3^{10}$

$S = \frac{19.3^{10} + 1}{4}$

21. If $A=\sum_{n=1}^{\infty}\frac{1}{\left(3+(-1)^n\right)^n}$ and $B=\sum_{n=1}^{\infty}\frac{(-1)^n}{\left(3+(-1)^n\right)^n}$, then $\frac{A}{B}$ is equal to : **[JEE(Main)June2022]**

यदि $A=\sum_{n=1}^{\infty}\frac{1}{\left(3+(-1)^n\right)^n}$ तथा $B=\sum_{n=1}^{\infty}\frac{(-1)^n}{\left(3+(-1)^n\right)^n}$ हैं, तो $\frac{A}{B}$ बराबर है : **[JEE(Main)June2022]**

(1) $\frac{11}{9}$ (2) 1 (3) $-\frac{11}{9}$ (4) $-\frac{11}{3}$

Ans. Official Answer NTA (3)

Sol. $A=\sum_{n=1}^{\infty}\frac{1}{\left(3+(-1)^n\right)^n}$ $B=\sum_{n=1}^{\infty}\frac{(-1)^n}{\left(3+(-1)^n\right)^n}$

$A=\frac{1}{2}+\frac{1}{4^2}+\frac{1}{2^3}+\frac{1}{4^4}+.....$

$A=\left(\frac{1}{2}+\frac{1}{2^3}+\frac{1}{2^5}+.....\right)+\left(\frac{1}{4^2}+\frac{1}{4^4}+.....\right)$

$A=\frac{\frac{1}{2}}{1-\frac{1}{2^2}}+\frac{\frac{1}{4^2}}{1-\frac{1}{4^2}}$

$A=\frac{\frac{1}{2}}{\frac{3}{4}}+\frac{\frac{1}{16}}{\frac{15}{16}}$

$A=\frac{2}{3}+\frac{1}{15}$

$A=\frac{10+1}{15}=\frac{11}{15}$

$B=\sum_{n=1}^{\infty}\frac{(-1)^n}{\left(3+(-1)^n\right)^n}$

$B=\frac{-1}{2}+\frac{1}{4^2}-\frac{1}{2^3}+\frac{1}{4^4}+.....$

$B=-\left(\frac{1}{2}+\frac{1}{2^3}+\frac{1}{2^5}.....\right)+\left(\frac{1}{4^2}+\frac{1}{4^4}+.....\right)$

$$B = -\left(\frac{\frac{1}{2}}{1-\frac{1}{2^2}}\right)+\left(\frac{\frac{1}{4^2}}{1-\frac{1}{4^2}}\right)$$

$$B = -\left(\frac{\frac{1}{2}}{\frac{3}{4}}\right)+\frac{\frac{1}{16}}{\frac{15}{16}}$$

$$B = \frac{-2}{3}+\frac{1}{15} \qquad = \frac{-10+1}{15} = \frac{-9}{15}$$

$$\frac{A}{B} = \frac{\frac{11}{15}}{\frac{-9}{15}} \qquad = \frac{-11}{9}$$

22. If $x = \sum_{n=0}^{\infty} a,^{n}\ y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=0}^{\infty} c^n$, where a, b, c are in A.P. and $|a| < 1$, $|b| < 1$, $|c| < 1$, $abc \neq 0$, then :

[JEE(Main)June2022]

(1) x, y, z are in A.P

(2) x, y, z are in G.P.

(3) $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ are in A.P.

(4) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z} = 1 - (a + b + c)$

यदि $x = \sum_{n=0}^{\infty} a^n y = \sum_{n=0}^{\infty} b^n, z = \sum_{n=c}^{\infty} c^n$ हैं, जहाँ a, b, c एक A.P. में हैं तथा $|a| < 1$, $|b| < 1$, $|c| < 1$, $abc \neq 0$ है, तो :

[JEE(Main)June2022]

(1) x, y, z एक A.P. में हैं

(2) x, y, z एक G.P. में हैं

(3) $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ एक A.P. में हैं

(4) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z} = 1 - (a + b + c)$

Ans. Official Answer NTA (3)

Sol. $x = \sum_{n=0}^{\infty} a^n = 1 + a + a^2 + a^3 .....\infty = \frac{1}{1-a}$

$$y = \sum_{n=0}^{\infty} b^n = 1 + b + b^2 + b^3 .....\infty = \frac{1}{1-b}$$

$$z = \sum_{n=0}^{\infty} c^n = 1 + c + c^2 .....\infty = \frac{1}{1-c}$$

$a, b, c \rightarrow$ A.P.

$1 - a, 1 - b, 1 - c \rightarrow$ A.P.

$\frac{1}{x}, \frac{1}{y}, \frac{1}{z} \rightarrow$ A.P.

23. Let $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4}$ +..... Then 4S is equal to : **[JEE(Main)June2022]**

माना $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4}$ +..... है। तो 4S बराबर है : **[JEE(Main)June2022]**

(1) $\left(\frac{7}{3}\right)^2$ (2) $\frac{7^3}{3^2}$ (3) $\left(\frac{7}{3}\right)^3$ (4) $\frac{7^2}{3^3}$

Ans. Official Answer NTA (3)

Sol. $S = 2 + \frac{6}{7} + \frac{12}{7^2} + \frac{20}{7^3} + \frac{30}{7^4} + ...$

$\frac{S}{7} = \frac{2}{7} + \frac{6}{7^2} + \frac{12}{7^3} + \frac{20}{7^4} + ...$

$\frac{6S}{7} = \frac{2}{1} + \frac{4}{7} + \frac{6}{7^2} + \frac{8}{7^3} + \frac{10}{7^4} + ...$

$\frac{6S}{7} = \frac{2}{1 - \frac{1}{7}} + \frac{(2)\left(\frac{1}{7}\right)}{\left(1 - \frac{1}{7}\right)^2}$

$\frac{6S}{7} = \frac{14}{6} + \frac{14}{36}$

$\frac{6S}{7} = \frac{14}{36} \times 7$

$S = 2 \times \frac{7^3}{6^3}$

$4S = \left(\frac{7}{3}\right)^3$

24. If a_1, a_2, a_3.....and b_1, b_2, b_3.....are A.P., and $a_1 = 2$, $a_{10} = 3$, $a_1 b_1 = 1 = a_{10} b_{10}$, then $a_4 b_4$ is equal to : **[JEE(Main)June2022]**

यदि a_1, a_2, a_3..... तथा b_1, b_2, b_3..... समांतर श्रेढ़ीं (A.P.) हैं तथा $a_1 = 2$, $a_{10} = 3$, $a_1 b_1 = 1 = a_{10} b_{10}$ हैं, तो $a_4 b_4$ बराबर है : **[JEE(Main)June2022]**

(1) $\frac{35}{27}$ (2) 1 (3) $\frac{27}{28}$ (4) $\frac{28}{27}$

Ans. Official Answer NTA (4)

Sol. $a_{10} - a_1 = 1$ $\quad b_1 = \frac{1}{2}$ $\quad b_{10} = \frac{1}{3}$

$d = \frac{1}{9}$ $\quad 9D = \frac{1}{3} - \frac{1}{2}$ $\quad \Rightarrow D = \frac{-1}{54}$

$a_4 = a_1 + 3d$

$= 2 + \frac{1}{3}$

$= \frac{7}{3}$

$b_4 = b_1 + 3D$

$b_4 = \frac{1}{2} - \frac{1}{18}$

$b_4 = \frac{4}{9}$

$a_4 b_4 = \frac{28}{27}$

25. Let $A_1, A_2, A_3, \ldots\ldots$ be an increasing geometric progression of positive real numbers. If $A_1A_3A_5A_7 = \frac{1}{1296}$ and $A_2 + A_4 = \frac{7}{36}$, then, the value of $A_6 + A_8 + A_{10}$ is equal to : **[JEE(Main)June2022]**

माना $A_1, A_2, A_3, \ldots\ldots$धनात्मक वास्तविक संख्याओं की वर्धमान गुणोत्तर श्रेढ़ी है। यदि $A_1A_3A_5A_7 = \frac{1}{1296}$ तथा $A_2 + A_4 = \frac{7}{36}$ हैं, तो $A_6 + A_8 + A_{10}$ का मान बराबर है : **[JEE(Main)June2022]**

(1) 33 (2) 37 (3) 43 (4) 47

Ans. Official Answer NTA (3)

Sol. Let $A_4 = a$ and common ratio = r

$\frac{a}{r^3} \cdot \frac{a}{r^2} \cdot a^r \cdot ar^3 = \frac{1}{1296} \Rightarrow a^4 = \frac{1}{64} \Rightarrow a = \frac{1}{6}$

$A_2 + A_4 = \frac{7}{36} \Rightarrow \frac{a}{r^2} + a = \frac{7}{36} \Rightarrow \frac{1}{r^2} + 1 = \frac{7}{6} \Rightarrow r^2 = 6$

$A_6 + A_8 + A_{10} = a(r^2 + r^4 + r^6) = 1/6\ (6 + 6^2 + 6^3) = 1 + 6 + 36 = 43$

26. If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last mean is 1 : 7 and $a + n = 33$, then the value of n is : **[JEE(Main)June2022]**

यदि a तथा 100 के बीच n समान्तर माध्य इस प्रकार डाले गए हैं कि पहले माध्य का अंतिम माध्य से अनुपात 1 : 7 है तथा $a + n = 33$ है, तो n का मान है : **[JEE(Main)June2022]**

(1) 21 (2) 22 (3) 23 (4) 24

Ans. Official Answer NTA (3)

Sol. $a, A_1, A_2, \ldots\ldots\ldots A_n, 100$

Let the common difference of above A.P. is d then $\frac{a+d}{100-d}=\frac{1}{7}$

$7a + 8d = 100$ .....(1)

and $a + n = 33$ .....(2)

also, $100 = a + (n + 1)d$

$\Rightarrow 100 = a+(34-a)\times\left(\frac{100-7a}{8}\right)$

$800 = 8a + 7a^2 - 338a + 3400$

$7a^2 - 330a + 2600 = 0$

$7a^2 - 260a - 70a + 2600 = 0$

$a = 10, \frac{260}{7}$ but $a \neq \frac{260}{7}$

$\therefore$ from eqn (2)

$10 + n = 33$

$\Rightarrow n = 23$

27. Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and $a_{n+2} = 2a_{n+1} - a_n + 1$ for all $n \geq 0$. Then, $\sum_{n=2}^{\infty}\frac{a_n}{7^n}$ is equal to :

[JEE(Main)June2022]

माना $\{a_n\}_{n=0}^{\infty}$ एक अनुक्रम है जिसमें $a_0 = a_1 = 0$ हैं तथा सभी $n \geq 0$ के लिए $a_{n+2} = 2a_{n+1} - a_n + 1$ हैं। तो $\sum_{n=2}^{\infty}\frac{a_n}{7^n}$ बराबर हैं :

[JEE(Main)June2022]

(1) $\frac{6}{343}$ (2) $\frac{7}{216}$ (3) $\frac{8}{343}$ (4) $\frac{49}{216}$

Ans. Official Answer NTA (2)

Sol. $a_0 = a_1 = 0$

$a_{n+2} = 2a_{n+1} - a_n + 1$

$a_2 = 2a_1 - a_0 + 1 = 1$

$a_3 = 2a_2 - a_1 + 1 = 2 + 1 = 3$

$a_4 = 2a_3 - a_2 + 1 = 6$

$a_5 = 2a_4 - a_3 + 1 = 10$

$S_\infty = \sum_{n=2}^{\infty}\frac{a_n}{7^n}$

$S_\infty = \frac{1}{7^2}+\frac{3}{7^3}+\frac{6}{7^4}+\frac{10}{7^5}+$ .....(1)

$\frac{1}{7}S_\infty = \frac{1}{7^3}+\frac{3}{7^4}+\frac{4}{7^5}+$ .....(2)

(1)-(2)

$$\frac{6}{7}S_\infty = \frac{1}{7^2}+\frac{2}{7^3}+\frac{3}{7^4}+\frac{4}{7^5}+ \quad \text{.....(3)}$$

$$\frac{6}{7^2}S_\infty = \frac{1}{7^3}+\frac{2}{7^4}+\frac{3}{7^5}+ \quad \text{.....(4)}$$

(3)-(4)

$$\frac{6}{7}S_\infty\left(1-\frac{1}{7}\right) = \frac{1}{7^2}+\frac{1}{7^3}+\frac{1}{7^4}+\text{.....}$$

$$36S_\infty = 1+\frac{1}{7}+\frac{1}{7^2}+\text{.....}$$

$$S_\infty = \frac{1}{1-\frac{1}{7}}.\frac{1}{36} = \frac{7}{216} \text{ Ans.}$$

28. The sum of the infinite series $1+\frac{5}{6}+\frac{12}{6^2}+\frac{22}{6^3}+\frac{35}{6^4}+\frac{51}{6^5}+\frac{70}{6^6}+\text{.....}$ is equal to : **[JEE(Main)June2022]**

अपरिमित श्रेणी $1+\frac{5}{6}+\frac{12}{6^2}+\frac{22}{6^3}+\frac{35}{6^4}+\frac{51}{6^5}+\frac{70}{6^6}+\text{.....}$ का योगफल बराबर है : **[JEE(Main)June2022]**

(1) $\frac{425}{216}$ (2) $\frac{429}{216}$ (3) $\frac{288}{125}$ (4) $\frac{280}{125}$

Ans. Official Answer NTA (3)

Sol. $S = 1+\frac{5}{6}+\frac{12}{6^2}+\frac{22}{6^3}+\text{.....}$

$$\frac{1}{6}S = \frac{1}{6}+\frac{5}{6^2}+\frac{12}{6^3}+\frac{22}{6^4}+\frac{35}{6^5}+\frac{51}{6^6}+\text{.....}$$

$$\frac{5}{6}S = 1+\frac{4}{6}+\frac{7}{6^2}+\frac{10}{6^3}+\frac{13}{6^4}+\frac{16}{6^5}+\frac{19}{6^6}+\text{.....}$$

$$\frac{5}{36}S = \frac{1}{6}+\frac{4}{6^2}+\frac{7}{6^3}+\frac{10}{6^4}+\frac{13}{6^5}+\frac{16}{6^6}+\text{.....}$$

$$\frac{25}{36}S = 1+\frac{3}{6}+\frac{3}{6^2}+\frac{3}{6^3}+\frac{3}{6^4}+\frac{3}{6^5}+\frac{3}{6^6}+\text{.....}$$

$$\frac{25}{36}S = 1+\frac{\frac{3}{6}}{1-\frac{1}{6}} = 1+\frac{3}{5} = \frac{8}{5}$$

$$S = \frac{288}{125}$$

29. The sum of all the elements of the set $\{\alpha \in \{1,2,.....100\} : \text{HCF}(\alpha, 24) = 1\}$ is _____.

[JEE(Main)June2022]

समुच्चय $\{\alpha \in \{1,2,.....100\} : \text{HCF}(\alpha, 24) = 1 \text{ है}\}$ के सभी अवयवों का योगफल है _____.

[JEE(Main)June2022]

Ans. Official Answer NTA (1633)

Sol. from set $\alpha \in \{1, 2, ...100\}$ numbers which are divisible by '2' or '3' cannot have HCF '1' with 24 number divisible by 2 = {2, 4, 6..... 100} = 50 numbers.

$$\text{sum} = \frac{50}{2}(2+100) = 50 \times 51$$

numbers divisible by 3 = {3, 6, ...99} = 33 numbers

$$\text{sum} = \frac{33}{2}(3+99) = 33 \times 51$$

numbers divisible by 6 = {6, 12, ...96} = 16 numbers

$$\text{sum} = \frac{16}{2}(6+96) = 16 \times 51$$

sum of all number = 1 + 2 + ... + 100 = 50 × 101
hence sum of number divisible by '2' or '3'
can be given as = 50 × 51 + 33 × 51 – 16 × 51
= 67 × 51
sum of required numbers = 50 × 101 – 67 × 51

30. The greatest integer less than or equal to the sum of first 100 terms of the sequence $\frac{1}{3}, \frac{5}{9}, \frac{19}{27}, \frac{65}{81}, \ldots\ldots$ is equal to _____.

[JEE(Main)June2022]

महत्तम पूर्णांक, जो अनुक्रम $\frac{1}{3}, \frac{5}{9}, \frac{19}{27}, \frac{65}{81}, \ldots\ldots$ के प्रथम 100 पदों के योगफल के बराबर या उससे कम है, है _____

[JEE(Main)June2022]

Question ID : 101490

Ans. Official Answer NTA (98)

Sol. $\frac{1}{3}+\frac{5}{9}+\frac{19}{27}+\frac{65}{81}+\ldots$

$\left(1-\frac{2}{3}\right)+\left(1-\frac{4}{9}\right)+\left(1-\frac{8}{27}\right)+\left(1-\frac{16}{81}\right)\ldots 100 \text{ terms}$

$100-\left[\frac{2}{3}+\left(\frac{2}{3}\right)^2+\ldots\right]$

$$100-\frac{\frac{2}{3}\left(1-\left(\frac{2}{3}\right)^{100}\right)}{1-\frac{2}{3}}$$

$$100-2\left(1-\left(\frac{2}{3}\right)^{100}\right)$$

$$S=98+2\left(\frac{2}{3}\right)^{100}$$

$$\Rightarrow [S]=98$$

31. Let $A=\sum_{i=1}^{10}\sum_{j=1}^{10}\min\{i,j\}$ and $B=\sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$. Then $A+B$ is equal to _____.

[JEE(Main)June2022]

माना $A=\sum_{i=1}^{10}\sum_{j=1}^{10}\min\{i,j\}$ तथा $B=\sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$ हैं। तो $A+B$ बराबर है _____

[JEE(Main)June2022]

Ans. Official Answer NTA (1100)

Sol. $$A=\sum_{i=1}^{10}\sum_{j=1}^{10}\min\{i,j\}$$

$$B=\sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$$

$$A=\sum_{i=1}^{10}\min(i,1)+\min(i,2)+....\min(i,10)$$

$$=\underbrace{(1+1+1+...+1)}_{19\text{ times}}+\underbrace{(2+2+2+...+2)}_{17\text{ times}}+\underbrace{(3+3+3+...+3)}_{15\text{ times}}+....(1)\ \ 1\text{ times}$$

$$B=\sum_{i=1}^{10}\max(i,1)+\max(i,2)+.....\max(i,10)$$

$$=\underbrace{(10+10+...+10)}_{19\text{ times}}+\underbrace{(9+9+...+9)}_{17\text{ times}}+...+1\ \ 1\text{ times}$$

$$A+B=20(1+2+3+....+10)$$

$$=20\times\frac{10\times11}{2}=10\times110=1100$$

32. If a_1 (> 0), a_2, a_3, a_4, a_5 are in G.P., $a_2 + a_4 = 2a_3 + 1$ and $3a_2 + a_3 = 2a_4$, then $a_2 + a_4 + 2a_5$ is equal to _____.

[JEE(Main)June2022]

यदि a_1 (> 0), a_2, a_3, a_4, a_5 एक G.P. में हैं, $a_2 + a_4 = 2a_3 + 1$ तथा $3a_2 + a_3 = 2a_4$ हैं, तो $a_2 + a_4 + 2a_5$ बराबर है _____

[JEE(Main)June2022]

Ans. Official Answer NTA (40)

Sol. $a_2 + a_4 = 2a_3 + 1$

$ar + ar^3 = 2ar^2 + 1$ .....(1)

$3a_2 + a_3 = 2a_4$

$3ar + ar^2 = 2ar^3$

$3r + r^2 = 2r^3$

$r(3 + r) = 2r^3$

$2r^2 - r - 3 = 0$

$r = \frac{3}{2}, r = -1$

when $r = \frac{3}{2}$

$\frac{3a}{2} + \frac{27a}{8} = 2a\frac{9}{4} + 1$

$a = \frac{r}{3}$

for $r = -1$, $a < 0$

$\therefore a_2 + a_4 + 2a_5 = ar + ar^3 + 2ar^4$

$= a[r + r^3 + 2r^4]$

$= 40$

33. If the sum of the first ten terms of the series $\frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501} + ...$ is $\frac{m}{n}$, where m and n are co-prime numbers, then m + n is equal to _____. **[JEE(Main)June2022]**

यदि श्रेणी $\frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501} + ...$ के प्रथम दस पदों का योगफल $\frac{m}{n}$ है, जहाँ m तथा n सहअभाज्य (co-prime) संख्याएँ हैं, तो m + n बराबर है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (276)

Sol. $\frac{1}{5} + \frac{2}{65} + \frac{3}{325} + \frac{4}{1025} + \frac{5}{2501}$

$T_r = \frac{r}{4r^4 + 1}$

$$S_{10}=\sum_{r=1}^{10}\frac{r}{4r^4+1}=\frac{1}{4}\sum_{r=1}^{10}\left(\frac{1}{2r^2-2r+1}-\frac{1}{2r^2+2r+1}\right)$$

$$S_{10}=\frac{1}{4}\left(1-\frac{1}{221}\right)$$

$$S_{10}\Rightarrow\frac{220}{4\times221}=\frac{55}{221}=\frac{m}{n}$$

$m+n=276$

34. Let for $n=1, 2, ...,50$, S_n be the sum of the infinite geometric progression whose first term is n^2 and whose common ratio is $\frac{1}{(n+1)^2}$. Then the value of $\frac{1}{26}+\sum_{n=1}^{50}\left(S_n+\frac{2}{n+1}-n-1\right)$ is equal to _____.

[JEE(Main)June2022]

माना $n=1, 2, ...,50$ के लिए अनंत गुणोत्तर श्रेढ़ी जिसका पहला पद n^2 तथा सार्व अनुपात $\frac{1}{(n+1)^2}$ है का योग S_n है।

तो $\frac{1}{26}+\sum_{n=1}^{50}\left(S_n+\frac{2}{n+1}-n-1\right)$ का मान बराबर है _____ **[JEE(Main)June2022]**

Ans. Official Answer NTA (41651)

Sol. Infinite series whose $a=n^2$.

$$r=\frac{1}{(n+1)^2}$$

$$S_n=S_\infty=\frac{n^2}{1-\frac{1}{(n+1)^2}}$$

$$S_n=\frac{(n+1)^2}{(n+2)}=\frac{n(n+1)^2}{(n+2)}$$

$$S_n=(n^2+1)-\frac{2}{n+2}\text{ [add and sub 2]}$$

Now,

$$\Rightarrow\frac{1}{26}+\sum_{n=1}^{50}\left[(n^2+1)-\frac{2}{n+2}+\frac{2}{n+1}-n-1\right]$$

$$\Rightarrow\frac{1}{26}+\sum_{n=1}^{50}\left[(n^2-n)+2\left[\frac{1}{n+1}-\frac{1}{n+2}\right]\right]$$

$$\Rightarrow\frac{1}{26}+\sum_{n=1}^{50}\left[(n^2-n)\right]+2.\sum_{n=1}^{50}\left[\frac{1}{n+1}-\frac{1}{n+2}\right]$$

$$\Rightarrow \frac{1}{26}+\frac{50\times51\times101}{6}-\frac{50\times51}{2}+2.\left[\frac{1}{2}-\frac{1}{52}\right]$$

$$=\frac{1}{26}+\frac{57550-7650}{6}+1-\frac{1}{26}$$

$$= 41650 + 1 = 41651$$

35. Let 3, 6, 9, 12, ... upto 78 terms and 5, 9, 13, 17, ... upto 59 terms be two series. Then, the sum of the terms common to both the series is equal to _____. **[JEE(Main)June2022]**

माना दो श्रेणियाँ 3, 6, 9, 12, ... (78 पदों तक) तथा 5, 9, 13, 17, ... (59 पदों तक) हैं। तो दोनों श्रेणियों के उभयनिष्ठ पदों का योगफल है_____. **[JEE(Main)June2022]**

Ans. Official Answer NTA (2223)

Sol. $S_1 = 3, 6, 9, 12$ .....78 - terms

$S_2 = 5, 9, 13, 17$ .....59 - terms

Common terms are 9, 21, ...

Now T_{78} of $S_1 = 3 + (78 - 1)\ 3 = 234$

T_{59} of $S_1 = 5 + (59 - 1)\ 4 = 237$

So, common terms ≤ 234

$9 + (n-1)\ 21 \leq 234$

$$n < \frac{225}{12}+1$$

$n = 19$

So, S_{19} of common terms

$$5_{19} = \frac{19}{2}\left(2\times9+(19-1)\times12\right)=19\left(9+108\right)$$

$= 2223$

36. The sum $\sum_{n=1}^{21}\frac{3}{(4n-1)(4n+3)}$ is equal to : **[JEE(Main)July2022]**

योगफल $\sum_{n=1}^{21}\frac{3}{(4n-1)(4n+3)}$ बराबर है : **[JEE(Main)July2022]**

(1) $\frac{7}{87}$ (2) $\frac{7}{29}$ (3) $\frac{14}{87}$ (4) $\frac{21}{29}$

Ans. Official Answer NTA (2)

Sol. $$\sum_{n=1}^{21}\frac{3}{(4n-1)(4n+3)}=\frac{3}{4}\sum_{n=1}^{21}\frac{1}{4n-1}-\frac{1}{4n+3}$$

$$=\frac{3}{4}\left[\left(\frac{1}{3}-\frac{1}{7}\right)+\left(\frac{1}{7}-\frac{1}{11}\right)+.....+\left(\frac{1}{83}-\frac{1}{87}\right)\right]$$

$$= \frac{3}{4}\left[\frac{1}{3} - \frac{1}{87}\right] = \frac{3}{4}\frac{84}{3.87} = \frac{7}{29}$$

37. Consider two G.Ps. $2, 2^2, 2^3, ...$ and $4, 4^2, 4^3, ...$ of 60 and n terms respectively. If the geometric mean of all the $60 + n$ terms is $(2)^{\frac{225}{8}}$, then $\sum_{k=1}^{n} k(n-k)$ is equal to : **[JEE(Main)July2022]**

60 तथा n पदों की दो G.P. क्रमशः $2, 2^2, 2^3, ...$ तथा $4, 4^2, 4^3, ...$ हैं। यदि सभी $60 + n$ पदों गुणोत्तर माध्य $(2)^{\frac{225}{8}}$ है, तो $\sum_{k=1}^{n} k(n-k)$ बराबर है : **[JEE(Main)July2022]**

(1) 560 (2) 1540 (3) 1330 (4) 2600

Ans. Official Answer NTA (3)

Sol. $\left(\left(2^1 2^2 \ldots . 2^{60}\right)\left(4^1 \cdot 4^2 \ldots \ldots 4^n\right)\right)^{\frac{1}{60+n}} = 2^{\frac{225}{8}}$

$$\left(2^{30\times 61} 4^{\frac{n(n+1)}{2}}\right)\frac{1}{60+n} = 2^{\frac{225}{8}}$$

$$2^{1830+n^2+n} = 2^{\frac{(225)(60+n)}{8}}$$

$$= 8n^2 - 217n + 1140 = 0$$

$$n = 20, \frac{57}{8}$$

$$\sum_{k=1}^{n} nk - k^2 = \frac{n^2(n+1)}{2} - \frac{n(n+1)(2n+1)}{6}$$

$$= 1330$$

38. Suppose $a_1, a_2, ...a_n, ...$ be an arithmetic progression of natural numbers. If the ratio of the sum of first five terms to the sum of first nine terms of the progression is 5 : 17 and $110 < a_{15} < 120$, then the sum of the first terms of the progression is equal to : **[JEE(Main)July2022]**

माना $a_1, a_2, ...a_n, ...$ वास्तविक संख्याओं की एक समांतर श्रेढ़ी है। यदि इस श्रेढ़ी के प्रथम पाँच पदों का योग का, नौ पदों के योग से अनुपात 5 : 17 है तथा $110 < a_{15} < 120$ है, तो इस श्रेढ़ी के प्रथम दस पदों का योग है : **[JEE(Main)July2022]**

(1) 290 (2) 380 (3) 460 (4) 510

Ans. Official Answer NTA (2)

Sol. $\frac{S_5}{S_9}=\frac{5}{17}\Rightarrow\frac{\frac{5}{2}(2a+4d)}{\frac{9}{2}(2a+8d)}=\frac{5}{17}$

$\Rightarrow d = 4a$

$a_{15} = a + 14d = 57\,a$

Now , $110 < a_{15} < 120$

$\Rightarrow 110 < 57a < 120$

$\Rightarrow a = 2\ \therefore\ d = 8$

$S_{10}=\frac{10}{2}(2\times2+9\times8)=380$

39. Let the sum of an infinite G.P., whose first term is a and the common ratio is r, be 5. Let the sum of its first five terms be $\frac{98}{25}$. Then the sum of the first 21 terms of an AP, whose first term is 10ar, n^{th} term is a_n and the common difference is $10ar^2$, is equal to : **[JEE(Main)July2022]**

माना एक अपरिमित G.P., जिसका पहला पद a है तथा सार्व अनुपात r है, का योग 5 है। माना इसके प्रथम पाँच पदों का योग $\frac{98}{25}$ है। तो उस AP, जिसका पहला पद 10ar है, n^{th} पद a_n है तथा सार्व अंतर $10ar^2$ है, के प्रथम 21 पदों का योग है : **[JEE(Main)July2022]**

(1) 21 a_{11} (2) 22 a_{11} (3) 15 a_{16} (4) 14 a_{16}

Ans. Official Answer NTA (1)

Sol. $S_{21}=\frac{21}{2}[20ar+20.10ar^2]$

$= 21\,[10\,ar + 100\,ar^2]$

$= 21.\ a_{11}$

40. If the minimum value of $f(x)=\frac{5x^2}{2}+\frac{\alpha}{x^5}$, $x>0$, is 14, then the value of α is equal to : **[JEE(Main)July2022]**

यदि $f(x)=\frac{5x^2}{2}+\frac{\alpha}{x^5}$, $x>0$, का निम्नतम मान 14 है, तो α का मान बराबर है : **[JEE(Main)July2022]**

(1) 32 (2) 64 (3) 128 (4) 256

Ans. Official Answer NTA (3)

Sol. $\frac{x^2}{2}+\frac{x^2}{2}+\frac{x^2}{2}+\frac{x^2}{2}+\frac{x^2}{2}+\frac{\alpha}{2x^5}+\frac{\alpha}{2x^5}$

$$\geq 7\left(\frac{\alpha^2}{2^7}\right)^{\frac{1}{7}}$$

$$\frac{7.(\alpha)^{2/7}}{2}=14$$

$(\alpha^2)^{1/7} = 2^2$

$\alpha = (2^2)^{7/2} = 2^7$

$\alpha = 128$

41. If $\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+....+\frac{1}{(180-a)(60-a)}=\frac{1}{256}$, then the maximum value of a is :

[JEE(Main)July2022]

यदि $\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+....+\frac{1}{(180-a)(60-a)}=\frac{1}{256}$ है, तो a का अधिकतम मान है :

[JEE(Main)July2022]

(1) 198 (2) 202 (3) 212 (4) 218

Ans. Official Answer NTA (3)

Sol. $$\frac{1}{20}\left[\left(\frac{1}{20-a}-\frac{1}{40-a}\right)+\left(\frac{1}{40-a}-\frac{1}{60-a}\right)+....+\left(\frac{1}{180-a}-\frac{1}{200-a}\right)\right]$$

$$\Rightarrow \frac{1}{20}\left(\frac{1}{20-a}-\frac{1}{200-a}\right)=\frac{1}{256}$$

$(20 - a)(200 - a) = 256 \times 9$

$a^2 - 220a + 1696 = 0$

$a = 8, 212$

Hence maximum value of a is 212.

42. Let a, b be two non-zero real numbers. If p and r are the roots of the equation $x^2 - 8ax + 2a = 0$ and q and s are the roots of the equation $x^2 + 12bx + 6b = 0$, such that $\frac{1}{p},\frac{1}{q},\frac{1}{r},\frac{1}{s}$ are in A.P., then $a^{-1} - b^{-1}$ is equal to _____.

[JEE(Main)July2022]

माना a, b दो शून्येत्तर वास्तविक संख्याएँ हैं। यदि समीकरण $x^2 - 8ax + 2a = 0$ के मूल p मूल r हैं और समीकरण $x^2 + 12bx + 6b = 0$, के मूल है q तथा s हैं, जबकि $\frac{1}{p},\frac{1}{q},\frac{1}{r},\frac{1}{s}$ A.P. में हैं, तो $a^{-1} - b^{-1}$ बराबर है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (38)

Sol. $x^2 - 8ax + 2a = 0$ $x^2 + 12bx + 6b = 0$

$p + r = 8a$ $\quad\quad$ $q + s = -12b$

$pr = 2a$ $\quad\quad$ $qs = 6b$

$\frac{1}{p}+\frac{1}{r}=4$ $\quad\quad$ $\frac{1}{q}+\frac{1}{s}=-2$

$\frac{2}{q}=4$ $\quad\quad$ $\frac{2}{r}=-2$

$q=\frac{1}{2}$ $\quad\quad$ $r=-1$

$p=\frac{1}{5}$ $\quad\quad$ $s=\frac{-1}{4}$

Now, $\frac{1}{a}-\frac{1}{b}=\frac{2}{pr}-\frac{6}{qs}=38$

43. Let $a_1 = b_1 = 1$, $a_n = a_{n-1} + 2$ and $b_n = a_n + b_{n-1}$ for every natural number $n \geq 2$. Then $\sum_{n=1}^{15} a_n.b_n$ is equal to _____.

[JEE(Main)July2022]

माना $a_1 = b_1 = 1$ हैं तथा प्रत्येक $n \geq 2$ के लिए $a_n = a_{n-1} + 2$ और $b_n = a_n + b_{n-1}$ हैं। तो $\sum_{n=1}^{15} a_n.b_n$ बराबर है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (27560)

Sol. $a_1 = b_1 = 1$

$a_2 = a_1 + 2 = 3$

$a_3 = a_2 + 2 = 5$

$a_4 = a_2 + 2 = 7$

$\Rightarrow an = 2n - 1$

$b_2 = a_1 + b_1 = 4$

$b_3 = a_3 + b_2 = 9$

$b_4 = a_4 + b_3 = 16$

$b_n = n^2$

$\sum_{n=1}^{15} a_n b_n$

$\sum_{n=1}^{15} (2n-1)n^2$

$\sum_{n=1}^{15} (2n^3 - n^2)$

$= 2\frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6}$

Put n = 15

$$= \frac{2\times 225\times 16\times 16}{4} - \frac{15\times 16\times 31}{6}$$

$= 27560$

44. The series of positive multiples of 3 is divided into sets " {3}, {6, 9, 12}, {15, 18, 21, 24, 27},..... Then the sum of the elements in the 11th set is equal to _____. **[JEE(Main)July2022]**

3 के धनात्मक गुणजों की श्रेणी को समुच्चयों {3}, {6, 9, 12}, {15, 18, 21, 24, 27},.....में विभाजित किया गया है। तो 11 वें समुच्चय में अवयवों का योग बराबर है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (6993)

Sol. $S_{11} = 3[101+102 + ... +121]$

$$= \frac{3}{2}(222)\times 21 = 6993$$

45. If $\sum_{k=1}^{10} \frac{k}{k^4+k^2+1} = \frac{m}{n}$, where m and n are co-prime, then m + n is equal to _____.

[JEE(Main)July2022]

यदि $\sum_{k=1}^{10} \frac{k}{k^4+k^2+1} = \frac{m}{n}$ है, जहाँ m तथा n असहभाज्य हैं, तो m + n बराबर है _____.

[JEE(Main)July2022]

Ans. Official Answer NTA (166)

Sol. $\sum_{k=1}^{10} \frac{k}{k^4+k^2+1} = \frac{m}{n}$

$$\Rightarrow \frac{1}{2}\sum_{k=1}^{10} \frac{\left(k^2+k+1\right)-\left(k^2-k+1\right)}{\left(k^2+k+1\right)\left(k^2-k+1\right)}$$

$$\Rightarrow \frac{1}{2}\left(\sum_{k=1}^{10}\left(\frac{1}{\left(k^2-k+1\right)} - \frac{1}{k^2+k+1}\right)\right)$$

$$\Rightarrow \frac{55}{111} = \frac{m}{n}$$

$m + n = 166$

46. Different A.P.'s are constructed with the first term 100, the last term 199, and integral common differences. The sum of the common differences of all such A.P.'s having at least 3 terms and at most 33 terms is _____.

[JEE(Main)July2022]

भिन्न A.P. बनाई गई हैं, जिनके प्रथम पद 100, अंतिम पद 199 तथा सार्व अंतर पूर्णांक हैं। इस प्रकार की सभी A.P. जिनमें कम से कम 3 पद तथा अधिक से अधिक 33 पद हैं, के सार्व अंतरों का योगफल हैं _____

[JEE(Main)July2022]

Ans. Official Answer NTA (53)

Sol. 1st term = 100 = a

Last term = 199 = ℓ

If 3 term

a, a + d, a + 2d

$a_n = \ell = a + (n-1)d$

$d_i = \frac{\ell - a}{n-1}$

$n \rightarrow$ number of terms

$n = 3,\ d_1 = \frac{199-100}{2}$

$= \frac{99}{2} \notin I$

$n = 4,\ d_2 = \frac{99}{3} = 33 \in I$

$n = 10,\ d_3 = \frac{99}{9} = 11 \in I$

$n = 12,\ d_4 = \frac{99}{11} = 9 \in I$

$\therefore\ \Sigma d_1 = 33 + 11 + 9 = 53$

47. Let $f(x) = 2x^2 - x - 1$ and $S = \{n \in Z : |f(n)| \le 800\}$. Then, the value of $\sum_{n \in S} f(n)$ is equal to _____.

[JEE(Main)July2022]

माना $f(x) = 2x^2 - x - 1$ तथा $S = \{n \in Z :\ |f(n)| \le 800\}$ हैं। तो $\sum_{n \in S} f(n)$ का मान है _____.

[JEE(Main)July2022]

Ans. Official Answer NTA (10620)

Sol. $f(x) = 2x^2 - x - 1$

$|f(x)| \le 800$

$2n^2 - n - 801 \le 0$

$n^2 - \frac{1}{n}n - \frac{801}{2} \le 0$

$$\left(n-\frac{1}{4}\right)^2-\frac{801}{2}-\frac{1}{16}\le 0$$

$$\left(n-\frac{1}{4}\right)^2-\frac{6409}{16}\le 0$$

$$\left(n-\frac{1}{4}-\frac{\sqrt{6409}}{4}\right)\left(n-\frac{1}{4}+\frac{\sqrt{6409}}{16}\right)\le 0$$

$$\frac{1-\sqrt{6409}}{4}\le n\le\frac{1+\sqrt{6409}}{4}$$

$$n=\{-19,-18-17,.....0,1,2,.....,20\}$$

$$\sum_{n\in s} f(x)=\Sigma\left(2x^2-x-1\right)$$

$= 2\ [19^2 + 18^2 +.....+ 1^2 + 1^2 + 2^2 +....+19^2 + 20^2]$

$= 4\ [1^2 + 2^2 +.....+19^2] +2\ [20^2] - 20 - 40$

$$=\frac{4\times19\times20\times(2\times19+1)}{6}+2\times400-60$$

$$=\frac{4\times19\times20\times39}{6}+800-60-9880+800-60$$

$= 10620$

48. $\frac{2^3-1^3}{1\times7}+\frac{4^3-3^3+2^2-1^3}{2\times11}+\frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times15}+.....+\frac{30^3-29^3+28^3-27^3+....+2^3-1^3}{15\times63}$ is equal to _____.

[JEE(Main)July2022]

$\frac{2^3-1^3}{1\times7}+\frac{4^3-3^3+2^2-1^3}{2\times11}+\frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times15}+.....+\frac{30^3-29^3+28^3-27^3+....+2^3-1^3}{15\times63}$ बराबर है_____

[JEE(Main)July2022]

Ans. Official Answer NTA (120)

Sol. $$T_n=\frac{2\sum_{r=1}^{n}(2r)^3-\left(\sum_{r=1}^{2n}r^3\right)}{n(4n+3)}$$

$$\Rightarrow T_n=n$$

So, $\sum_{n=1}^{15} T_n=120$

49. Let the coefficient of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}}+\beta x\right)^4$, $(1-3\beta x)^2$ and $\left(1-\frac{\beta}{2}x\right)^6$, $\beta > 0$, respectively form the first three terms of an A.P. If is the common difference of this A.P., then $50-\frac{2d}{\beta^2}$ is equal to _____. **[JEE(Main)July2022]**

माना $\left(\frac{1}{\sqrt{6}}+\beta x\right)^4$, $(1-3\beta x)^2$ तथा $\left(1-\frac{\beta}{2}x\right)^6$, $\beta > 0$ के प्रसार में मध्य पदों के गुणांक क्रमशः एक A.P. के पहले तीन पद हैं। यदि इस A.P. का सार्व अंतर d है, तो $50-\frac{2d}{\beta^2}$ बराबर है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (57)

Sol. ${}^4C_2\times\frac{\beta^2}{6}, -6\beta, -{}^6C_3\times\frac{\beta^3}{8}$ are in A.P

$$\beta^2-\frac{5}{2}\beta^3=-12\beta$$

$$\beta=\frac{12}{5} \text{ or } \beta=-2 \therefore \beta=\frac{12}{5}$$

$$d=-\frac{72}{5}-\frac{144}{25}=-\frac{504}{25}$$

$$\therefore 50-\frac{2d}{\beta^2}=57$$

50. If $\frac{6}{3^{12}}+\frac{10}{3^{11}}+\frac{20}{3^{10}}+\frac{40}{3^9}+.....+\frac{10240}{3}=2^n.m$, where m is odd, then m . n is equal to _____.

[JEE(Main)July2022]

यदि $\frac{6}{3^{12}}+\frac{10}{3^{11}}+\frac{20}{3^{10}}+\frac{40}{3^9}+.....+\frac{10240}{3}=2^n.m$ है, जहाँ m एक विषम संख्या है, तो m . n बराबर है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (12)

Sol. $$\frac{6}{3^{12}}+10\left(\frac{1}{3^{11}}+\frac{2}{3^{10}}+\frac{2^2}{3^9}+\frac{2^3}{3^8}+.....+\frac{2^{10}}{3}\right)$$

$$\frac{6}{3^{12}}+\frac{10}{3^{11}}\left(\frac{6^{11}-1}{6-1}\right)$$

$= 2^{12}$. 1; m.n = 12

51. Let $a_1, a_2, a_3, ...$ be and A.P. If $\sum_{r=1}^{\infty} \frac{a_r}{2^r} = 4$, then $4a_2$ is equal to _____ **[JEE(Main)July2022]**

माना $a_1, a_2, a_3, ...$ एक A.P. है। यदि $\sum_{r=1}^{\infty} \frac{a_r}{2^r} = 4$ है, तो $4a_2$ बराबर है _____ **[JEE(Main)July2022]**

Ans. Official Answer NTA (16)

Sol. $S = \frac{a_1}{2} + \frac{a_2}{2^2} + \frac{a_3}{2^3} + \ldots$

$\frac{S}{2} = \frac{a_1}{2^2} + \frac{a_2}{2^3} + \ldots$

$\frac{S}{2} = \frac{a_1}{2} + d\left(\frac{1}{2^2} + \frac{1}{2^3} + \ldots\right)$

$\frac{S}{2} = \frac{a_1}{2} + d\left(\frac{\frac{1}{4}}{1-\frac{1}{2}}\right)$

$\therefore S = a_1 + d = a_2 = 4$

Or $4a_2 = 16$

52. If $\frac{1}{2\times3\times4} + \frac{1}{3\times4\times5} + \frac{1}{4\times5\times6} + ... + \frac{1}{100\times101\times102} = \frac{k}{101}$, then 34 k is equal to _____

[JEE(Main)July2022]

यदि $\frac{1}{2\times3\times4} + \frac{1}{3\times4\times5} + \frac{1}{4\times5\times6} + ... + \frac{1}{100\times101\times102} = \frac{k}{101}$, है, तो 34 k बराबर है _____

[JEE(Main)July2022]

Ans. Official Answer NTA (286)

Sol. $S = \frac{a_1}{2} + \frac{a_2}{2^2} + \frac{a_3}{2^3} + \ldots$

$\frac{S}{2} = \frac{a_1}{2^2} + \frac{a_2}{2^3} + \ldots$

$\frac{S}{2} = \frac{a_1}{2} + d\left(\frac{1}{2^2} + \frac{1}{2^3} + \ldots\right)$

$\frac{S}{2} = \frac{a_1}{2} + d\left(\frac{\frac{1}{4}}{1-\frac{1}{2}}\right)$

$\therefore S = a_1 + d = a_2 = 4$

Or $4a_2 = 16$

Exercise J - A

1. Let $a_1, a_2, a_3, \ldots$ be in harmonic progression with $a_1 = 5$ and $a_{20} = 25$. The least positive integer n for which $a_n < 0$ is : **[JEE-2012]**

   माना $a_1, a_2, a_3, \ldots$ ह.श्रे. बनाते है जहाँ $a_1 = 5$ तथा $a_{20} = 25$ तो n का न्यूनतम धनात्मक पूर्णांक मान ज्ञात कीजिये जिसके लिये $a_n < 0$ है : **[JEE-2012]**

   (A) 22 (B) 23 (C) 24 (D) 25

Ans. D

Sol. Corresponding A.P.

$$\frac{1}{5}, \ldots\ldots\ldots\ldots \frac{1}{25} \ (20^{th} \text{ term})$$

$$\frac{1}{25} = \frac{1}{5} + 19\,d \qquad \Rightarrow \qquad d = \frac{1}{19}\left(\frac{-4}{25}\right) = -\frac{4}{19\times 25}$$

$a_n < 0$

$$\frac{1}{5} - \frac{4}{19\times 25}\times(n-1) < 0 \qquad \Rightarrow \qquad \frac{19\times 5}{4} < n-1 \qquad \Rightarrow \qquad n > 24.75$$

Hindi. संगत समान्तर श्रेढ़ी है

$$\frac{1}{5}, \ldots\ldots\ldots\ldots \frac{1}{25} \ (20 \text{ वाँ पद})$$

$$\frac{1}{25} = \frac{1}{5} + 19\,d \qquad \Rightarrow \qquad d = \frac{1}{19}\left(\frac{-4}{25}\right) = -\frac{4}{19\times 25}$$

$a_n < 0$

$$\frac{1}{5} - \frac{4}{19\times 25}\times(n-1) < 0 \qquad \Rightarrow \qquad \frac{19\times 5}{4} < n-1 \qquad \Rightarrow \qquad n > 24.75$$

2. A pack contains n cards numbered from 1 to n. Two consecutive numbered cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is k, then k – 20 = **[JEE(Adv.)2013]**

   एक गडडी में n कार्ड हैं जो संख्याओं 1 से n द्वारा चिन्हित है। दो क्रमागत संख्याओं वाले कार्ड गडडी से निकाल दिये जाते है और शेष कार्डो की संख्याओं का योग 1224 है। यदि निकाले गए कार्डो की चिन्हित संख्याओं में से लघुतर संख्या k है, तब k – 20 = **[JEE(Adv.)2013]**

Ans. 5

Sol. Numbers removed are k and k + 1. now $\frac{n(n+1)}{2} - k - (k+1) = 1224 \qquad \Rightarrow \qquad n^2 + n - 4k = 2450$

$$\Rightarrow \qquad n^2 + n - 2450 = 4k \qquad \Rightarrow \qquad (n+50)(n-49) = 4k \qquad \Rightarrow \qquad n > 49$$

Alternative

$\therefore$ To satisfy this equation n should be of the form of $(4p + 1)$ or $(4p + 2)$ taking $n = 50$

$\Rightarrow$ $4k = 100$ $\Rightarrow$ $k = 25$ $\therefore$ $k - 20 = 5$

Now if we take $n = 53$ $\Rightarrow$ $k = 103$ $\Rightarrow$ $n < k$

so not possible. Hence $n \geq 53$ will not be possible.

Hindi. k तथा k + 1 संख्याएं हटाने पर

अब $\frac{n(n+1)}{2} - k - (k + 1) = 1224$ $\Rightarrow$ $n^2 + n - 4k = 2450$ $\Rightarrow$ $n^2 + n - 2450 = 4k$

$\Rightarrow$ $(n + 50)(n - 49) = 4k$ $\Rightarrow$ $n > 49$

Alternative

$\therefore$ इस समीकरण को संतुष्ठ करने के लिए $(4p + 1)$ या $(4p + 2)$ रूप होगा $n = 50$

$\Rightarrow$ $4k = 100$ $\Rightarrow$ $k = 25$

$\therefore$ $k - 20 = 5$ अब यदि $n = 53$ लेते है

$k = 103$ $\Rightarrow$ $n < k$ इसलिए संभव नही अतः $n \geq 53$ भी संभव नही होगा।

3. Let $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$. Then S_n can take value(s) : **[JEE(Adv.)2013]**

माना कि $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2$, तब S_n निम्न मान ले सकता है : **[JEE(Adv.)2013]**

(A) 1056 (B) 1088 (C) 1120 (D) 1332

Ans. AD

Sol. $S_n = \sum_{k=1}^{4n} (-1)^{\frac{k(k+1)}{2}} k^2 = -1^2 - 2^2 + 3^2 + 4^2 - 5^2 - 6^2 + 7^2 + 8^2 + ...$

$= (3^2 - 1) + (4^2 - 2^2) + (7^2 - 5^2) + (8^2 - 6^2)..... = 2[\underbrace{4 + 6 + 12 + 14 + 20 + 22 + ...}_{2n \text{ terms}}]$

$= 2[(4 + 12 + 20 .....) + (6 + 14 + 22 .....)]$

n terms n terms

$= 2\left[\frac{n}{2}(4 \times 2 + (n-1)8)\frac{n}{2}(2 \times 6 + (n+1)8)\right] = 2[n(4 + 4n - 4) + n(6 + 4n - 4)]$

$= 2(4n^2 + (4n + 2)n) = 2(8n^2 + 2n) = 4n(4n + 1)$

(A) $1056 = 32 \times 33$ $n = 8$

(B) $1088 = 32 \times 34$

(C) $1120 = 32 \times 35$

(D) $1332 = 36 \times 37$ $n = 9$

4. Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a,b, c are in geometric progression and the arithmetic mean of a, b, c is b+2, then the value of $\frac{a^2+a-14}{a+1}$ is. **[JEE(Adv.)2014]**

माना कि a, b, c धनात्मक पूर्णांक हैं तथा $\frac{b}{a}$ एक पूर्णांक है। यदि a, b, c गुणोत्तर श्रेणी में हैं तथा a, b, c का समान्तर माध्य b + 2 है, तो $\frac{a^2+a-14}{a+1}$ का मान है। **[JEE(Adv.)2014]**

Ans. 4

Sol. Let $b = ar$, $c = ar^2 \Rightarrow$ r is Integers. Also $\frac{a+ar+ar^2}{3} = ar + 2 \Rightarrow a + ar^2 = 2ar + 6$

$\Rightarrow$ $a(r-1)^2 = 6$ $\Rightarrow$ r must be 2 and a = 6. Thus $\frac{a^2+a-14}{a+1} = \frac{36+6-14}{7} = 4$ Ans.

Hindi. माना $b = ar$, $c = ar^2 \Rightarrow$ r पूर्णांक है। तथा $\frac{a+ar+ar^2}{3} = ar + 2 \Rightarrow a + ar^2 = 2ar + 6$

$\Rightarrow$ $a(r-1)^2 = 6$ $\Rightarrow$ r, 2 होगा और a = 6. अतः $\frac{a^2+a-14}{a+1} = \frac{36+6-14}{7} = 4$ Ans.

5. Suppose that all the terms of an arithmetic progression (A.P.) are positive integers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is. **[JEE(Adv.)2015]**

माना कि एक समान्तर श्रेणी (arithmetic progression (A.P.)) के सभी पद धन पूर्णांक हैं। इस समान्तर श्रेणी में यदि पहले सात (7) पदों के योग और पहले ग्यारह (11) पदों के योग का अनुपात 6 : 11 है तथा सातवाँ पद 130 और 140 के बीच में स्थित है, तब इस समान्तर श्रेणी के सार्व अन्तर (common difference) का मान है। **[JEE(Adv.)2015]**

Ans. 9

Sol. $\frac{S_7}{S_{11}} = \frac{6}{11}$

$$\frac{\frac{7}{2}[2a+6d]}{\frac{11}{2}[2a+10d]}=\frac{6}{11}$$ Given $130 < a + 6d < 140$

$$\frac{7(a+3d)}{11(a+5d)}=\frac{6}{11}$$

$7a + 21d = 6a + 30d \Rightarrow 130 < 15d < 140$

$a = 9d$ Hence $d = 9$

$a = 81$

Hence $d = 9$

Alternative :

Let the AP be a, a + d, a + 2d, ...

where $a, d \in N$

Given $\frac{S_7}{S_{11}}=\frac{6}{11}$ and $130 < a + 6d < 140$ .....(2)

$$\Rightarrow \quad \frac{\frac{7}{2}\{2a+6d\}}{\frac{11}{2}\{2a+10d\}}=\frac{6}{11}$$

$$\Rightarrow \quad \frac{14a+42d}{22a+110d}=\frac{6}{11}$$

$\Rightarrow \quad 154a + 462d = 132a + 660d$

$\Rightarrow \quad 22a = 198d$

$$\Rightarrow \quad a=\frac{99d}{11}=9d$$

(2) $\Rightarrow \quad \therefore \quad 130 < 9d + 6d < 140$

$\Rightarrow \quad 8.6 < d < 9.3$

$\therefore \quad d = 9$

6. The least value of $\alpha \in \mathbb{R}$ for which $4\alpha x^2+\frac{1}{x}\geq 1$, for all $x > 0$, is : **[JEE(Adv.)2016]**

यदि $\alpha \in \mathbb{R}$ और सभी $x > 0$ है, तब $4\alpha x^2+\frac{1}{x}\geq 1$ के लिए α का न्यूनतम मान क्या होगा : **[JEE(Adv.)2016]**

(A) $\frac{1}{64}$ (B) $\frac{1}{32}$ (C) $\frac{1}{27}$ (D) $\frac{1}{25}$

Ans. C

Sol. $4\alpha x^2 + \frac{1}{2x} + \frac{1}{2x} \geq 1$

By A.M. $\geq$ G.M.

$$\frac{4\alpha x^2 + \frac{1}{2x} + \frac{1}{2x}}{3} \geq \left(4\alpha x^2 . \frac{1}{2x} . \frac{1}{2x}\right)^{\frac{1}{3}}$$

$$4\alpha x^2 + \frac{1}{2x} + \frac{1}{2x} \geq 3(\alpha)^{\frac{1}{3}}$$

For the least value of α

$$3(\alpha)^{\frac{1}{3}} = 1 \Rightarrow \alpha = \frac{1}{27}$$

7. Let $b_i > 1$ for i = 1, 2, ..., 101. Suppose $\log_e b_1, \log_e b_2, ..., \log_e b_{101}$ are in Arithmetic Progression (A.P.) with the common difference $\log_e 2$. Suppose $a_1, a_2, ... a_{101}$ are in A.P. such that $a_1 = b_1$ and $a_{51} = b_{51}$. If $t = b_1 + b_2 + ... + b_{51}$ and $s = a_1 + a_2 + ... + a_{51}$, then : **[JEE(Adv.)2016]**

माना कि i = 1, 2, ..., 101 के लिए $b_i > 1$ है। मान लीजिए कि $\log_e b_1, \log_e b_2, ..., \log_e b_{101}$ सार्वअंतर $\log_e 2$ वाली समांतर श्रेणी में है। मान लीजिए कि $a_1, a_2, ... a_{101}$ समांतर श्रेणी में इस प्रकार है कि $a_1 = b_1$ तथा $a_{51} = b_{51}$. यदि $t = b_1 + b_2 + ... + b_{51}$ तथा $s = a_1 + a_2 + ... + a_{51}$ हैं, तो : **[JEE(Adv.)2016]**

(A) $s > t$ & $a_{101} > b_{101}$ (B) $s > t$ & $a_{101} < b_{101}$ (C) $s < t$ & $a_{101} > b_{101}$ (D) $s < t$ & $a_{101} < b_{101}$

Ans. B

Sol. If $\log_e b_1, \log_e b_2 ..... \log_e b_{101}$ $\rightarrow$ A.P.

then $b_1, b_2, ... b_{101}$ $\rightarrow$ G.P.

Given $a_1 = b_1$ and $a_{51} = b_{51}$

$a_1, a_2, a_3 ..... a_{50}, a_{51}$ are in Arithmetic progression.

Here, We can say $a_2, a_3, ... a_{50}$ are Arithmetic means between a_1 and a_{50}.

By A.M. > G.M.

$a_2 > b_2, a_3 > b_3, ...$

$\Rightarrow s > t$

Also a_1, a_{51}, a_{101} are in Arithmetic Progression and b_1, b_{51}, b_{101} are in Geometric Progression.

$\Rightarrow b_{101} > a_{101}$ ($\because a_1 = b_1, a_{51} = b_{51}$)

8. Let X be the set consisting of the first 2018 terms of the arithmetic progression 1,6,11,... , and Y be the set consisting of the first 2018 terms of the arithmetic progression 9,16,23, ... Then, the number of elements in the set $X \cup Y$ is _____. **[JEE(Adv.)2018]**

माना कि X समान्तर श्रेणी (arithmetic progression) 1,6,11,... के प्रथम 2018 पदों का समुच्चय है, और Y समान्तर श्रेणी 9,16,23, ... के प्रथम 2018 पदों का समुच्चय है। तब समुच्चय $X \cup Y$ में अवयवों की संख्या है _____

[JEE(Adv.)2018]

Ans. 3748

Sol. $X = \{1, 6, 11, ...\}$

$Y = \{9, 16, 23, ...\}$

Common terms : 16, 51, 86, ...

$t_p = 16 + (p-1)35 = 35p - 19 \le 10086$

$\Rightarrow p \le 288.7$

$\therefore\ n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$

$= 2018 + 2018 - 288$

$= 3748$

9. Let α and β be the roots of $x^2 - x - 1 = 0$ with $\alpha > \beta$. For all positive integers n, define $a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}$, $n \ge 1$, $b_1 = 1$ and $b_n = a_{n-1} + a_{n+1}$, $n \ge 2$ Then which of the following options is/are correct : **[JEE(Adv.)2019]**

(A) $b_n = \alpha^n + \beta^n$ for all $n \ge 1$

(B) $a_1 + a_2 + a_3 + ... + a_n = a_{n+2} - 1$ for all $n \ge 1$

(C) $\sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{10}{89}$

(D) $\sum_{n=1}^{\infty} \frac{b_n}{10^n} = \frac{8}{89}$

माना कि $x^2 - x - 1 = 0$ के मूल α और β हैं जहा $\alpha > \beta$ है। सभी धनात्मक पूर्णांकों n के लिए निम्न को परिभाषित किया गया है।

$a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}$, $n \ge 1$, $b_1 = 1$ तथा $b_n = a_{n-1} + a_{n+1}$, $n \ge 2$ तब निम्न मे से कौन सा (से) विकल्प सही है :

[JEE(Adv.)2019]

(A) प्रत्येक $n \ge 1$ के लिए, $b_n = \alpha^n + \beta^n$

(B) प्रत्येक $n \ge 1$ के लिए, $a_1 + a_2 + a_3 + ... + a_n = a_{n+2} - 1$

(C) $\sum_{n=1}^{\infty} \frac{a_n}{10^n} = \frac{10}{89}$

(D) $\sum_{n=1}^{\infty} \frac{b_n}{10^n} = \frac{8}{89}$

Ans. ABC

Sol. $x^2 - x - 1 = 0$ roots are α and β $\quad a_n = \frac{\alpha^n - \beta^n}{\alpha - \beta}$, $\alpha = \frac{1+\sqrt{5}}{2}$, $\beta = \frac{1-\sqrt{5}}{2}$

$\alpha^2 - \alpha - 1 = 0$ and $\beta^2 - \beta - 1 = 0$

Option (1) $b_n = a_{n-1} + a_{n+1}$

$= \frac{\left(\alpha^{n-1} - \beta^{n-1}\right) + \left(\alpha^{n+1} - \beta^{n-1}\right)}{\alpha - \beta}$

$$= \frac{\alpha^{n-1}\left(1+\alpha^2\right)-\beta^{n-1}\left(1+\beta^2\right)}{\alpha-\beta}$$

$$= \frac{\alpha^{n-1}\left(\alpha+2\right)-\beta^{n-1}\left(\beta+2\right)}{\alpha-\beta}$$

$$= \frac{\alpha^{n-1}\left(\frac{5+\sqrt{5}}{2}\right)-\beta^{n-1}\left(\frac{5-\sqrt{5}}{2}\right)}{\alpha-\beta}$$

$$= \frac{\sqrt{5}\left(\alpha^{n-1}\left(\frac{\sqrt{5}+1}{2}\right)-\beta^{n-1}\left(\frac{\sqrt{5}-1}{2}\right)\right)}{\alpha-\beta}$$

$$= \sqrt{5}\left(\frac{\alpha^{n-1}\left(\alpha\right)+\beta^{n-1}\left(\beta\right)}{\alpha-\beta}\right)$$

$$b_n = \frac{\left(\alpha^n+\beta^n\right)\sqrt{5}}{\sqrt{5}} = \alpha^n+\beta^n$$

Option (2) $a_n = \frac{\alpha^n-\beta^n}{\alpha-\beta}$

$$= \frac{\alpha^{n-2}\left(\alpha^2\right)-\beta^{n-2}\left(\beta^2\right)}{\alpha-\beta}$$

$$= \frac{\alpha^{n-2}\left(\alpha+1\right)-\beta^{n-2}\left(\beta+1\right)}{\alpha-\beta}$$

$$= \frac{\alpha^{n-1}-\beta^{n-1}}{\alpha-1}+\frac{\alpha^{n-2}-\beta^{n-2}}{\alpha-\beta}$$

$$a_n = a_{n-1}+a_{n-2}$$

$$a_{n+2} = a_{n+1}+a_n$$

$$a_{n-1} = a_n+a_{n-1} \qquad \Big| \qquad a_2 = \frac{\alpha^2-\beta^2}{\alpha-\beta} = \alpha+\beta = 1$$

$$a_n = a_{n-1}+a_{n-2}$$

|
|
|

$a_3 = a_2 + a_1$

Add

$$a_{n+2} = (a_n + a_{n-1} + a_{n-2} + ... + a_2 + a_3) + a_2$$

$$a_{n+2} - a_2 = a_n + a_{n-1} + ... + a_1$$

$$a_{n+2} - 1 = a_n + a_{n-1} + ... + a_1$$

Option (3) $\sum \frac{\alpha^n - \beta^n}{(\alpha-\beta)10^n} = \frac{1}{\alpha-\beta}\left(\sum \frac{\alpha^n}{10^n} - \sum \frac{\beta^n}{10^n}\right)$

$$= \frac{1}{\alpha-\beta}\left(\frac{\frac{\alpha}{10}}{1-\frac{\alpha}{10}} - \frac{\frac{\beta}{10}}{1-\frac{\beta}{10}}\right)$$

$$= \frac{1}{\alpha-\beta}\left(\frac{\alpha}{10-\alpha} - \frac{\beta}{10-\beta}\right)$$

$$= \frac{1}{\alpha-\beta}\left(\frac{(10-\beta)\alpha - \beta(10-\alpha)}{(10-\alpha)(10-\beta)}\right)$$

$$= \frac{(10\alpha - 10\beta)}{(\alpha-\beta)(100-(\alpha+\beta)10+\alpha\beta)}$$

$$= \frac{10}{100-10+1} = \frac{10}{89}$$

Option (4) $\sum\left(\frac{\alpha^n + \beta^n}{10^n}\right) = \sum \frac{\alpha^n}{10^n} + \sum \frac{\beta^n}{10^n}$

$$= \frac{\frac{\alpha}{10}}{1-\frac{\alpha}{10}} + \frac{\frac{\beta}{10}}{1-\frac{\beta}{10}}$$

$$= \frac{\alpha}{10-\alpha} + \frac{\beta}{10-\beta}$$

$$= \frac{\alpha(10-\beta) + \beta(10-\alpha)}{(10-\alpha)(10-\beta)}$$

$$= \frac{10(\alpha-\beta)-2\alpha\beta}{100-(\alpha+\beta)+\alpha\beta}$$

$$= \frac{10+2}{100-10-1} = \frac{12}{89}$$

10. Let AP (a;d) denote the set of all the terms of an infinite arithmeitc progression with first term a and common difference d > 0. If AP(1;3) ∩ AP (2;5)∩ AP (3;7) = AP(a;d) then a + d equals. **[JEE(Adv.)2019]**

माना कि AP (a;d)एक अनंत समान्तर श्रेण के पदों का समूच्चय है जिसका प्रथम पद a तथा सार्वन्तर d > 0 है। यदि AP(1;3) ∩ AP (2;5)∩ AP (3;7) = AP(a;d) है तब a + d बराबर होगा। **[JEE(Adv.)2019]**

Ans. 157

Sol. AP (1,3) ∩ AP(2,5) ∩ AP(3,7)
I AP 1,4,7,10,.....
II AP 2,7,12,17,.....
III AP 3,10,17,24,.....
first time 52 will be present in all three AP
d = LCM (3,5,7) = 105
a = 52
d = 105
a + d = 157

11. Let m be the minimum possible value of $\log_3(3^{y_1}+3^{y_2}+3^{y_3})$, where y_1, y_2, y_3 are real numbers for which $y_1+y_2+y_3=9$. Let M be the maximum possible value of $(\log_3 x_1+\log_3 x_2+\log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1+x_2+x_3=9$. Then the value of $\log_2(m^3)+\log_3(M^2)$ is _____. **[JEE(Adv.)2020]**

माना कि $\log_3(3^{y_1}+3^{y_2}+3^{y_3})$ का न्यूनतम संभावित मान m है, जहाँ y_1, y_2, y_3 वास्तविक संख्याएँ हैं जिनके लिये $y_1+y_2+y_3=9$ है। माना कि $(\log_3 x_1+\log_3 x_2+\log_3 x_3)$ का अधिकतम संभावित मान M है, जहाँ x_1, x_2, x_3 धनात्मक वास्तविक संख्याएँ है जिनके लिये $x_1+x_2+x_3=9$ है। तब $\log_2(m^3)+\log_3(M^2)$ का मान है _____ **[JEE(Adv.)2020]**

Ans. 8

Sol. $$\frac{3^{y_1}+3^{y_2}+3^{y_3}}{3} \geq \left(3^{y_1}3^{y_2}3^{y_3}\right)^{\frac{1}{3}}$$

$$3^{y_1}+3^{y_2}+3^{y_3} \geq 3\left(3^{\frac{y_1+y_2+y_3}{3}}\right)$$

$3^{y_1} + 3^{y_2} + 3^{y_3} \geq 3^4$

$\log_3(3^{y_1} + 3^{y_2} + 3^{y_3}) \geq 4$

$\Rightarrow m = 4$

$$\frac{x_1 + x_2 + x_3}{3} \geq \left(x_1 x_2 x_3\right)^{\frac{1}{3}}$$

$$3 \geq \left(x_1 x_2 x_3\right)^{\frac{1}{3}}$$

$27 \geq x_1 x_2 x_3$

$\log_3(27) \geq \log_3(x_1 x_2 x_3)$

$\Rightarrow M = 3$

$\log_2(m^3) + \log_3(M^2) = \log_2(64) + \log_3(9)$

$= 6 + 2 = 8$

12. Let a_1, a_2, a_3,... be a sequence of positive integers in arithmetic progression with common difference 2. Also, let b_1, b_2, b_3,... be a sequence of positive integers in geometric progression with common ratio 2. If $a_1 = b_1 = c$, then the number of all possible values of c, for which the equality $2(a_1 + a_2 + ...+ a_n) = b_1 + b_2 + ...+ b_n$ holds for some positive integer n, is _____. **[JEE(Adv.)2020]**

माना कि धनात्मक पूर्णांकों का एक अनुक्रम a_1, a_2, a_3,...समान्तर श्रेढी में है जिसका सार्व अन्तर 2 है। तथा माना कि धनात्मक पूर्णांकों का एक अनुक्रम b_1, b_2, b_3,... गुणोत्तर श्रेढ़ी में है जिसका सार्व अनुपात 2 है। यदि $a_1 = b_1 = c$ है, तब c के सभी संभावित मानों कि संख्या, जिनके लिये समीका $2(a_1 + a_2 + ...+ a_n) = b_1 + b_2 + ...+ b_n$ किसी धनात्मक पूर्णांक n के लिए सही हो, है_____

[JEE(Adv.)2020]

Ans. 1

Sol. 2 (sum of first n–terms of AP) = (sum of first n–terms of GP)

$$\Rightarrow 2\left(\frac{n}{2}[2c + (n-1)2]\right) = c \cdot \left(\frac{2^n - 1}{2 - 1}\right)$$

$\Rightarrow 2nc + 2n^2 - 2n = c2^n - c$

$$c = \frac{2n^2 - 2n}{2^n - 2n - 1} = \text{Positive Integer (since } a_i \text{ \& } b_i \text{ are sequences of positive integers)}$$

Now for $n = 1$; $c = 0$

$n = 2$; $c = -4$

$n = 3$; $c = 12$

$$n = 4; c = \frac{24}{7}$$

$$n = 5; c = \frac{40}{21}$$

$n = 6;\ c = \frac{60}{51}$

And for all $n \geq 7$; $c < 1$

So no integral value of c can be obtained further.

So only possible value of $c = 12$.

13. Let $l_1, l_2 \ldots, l_{100}$ be consecutive terms of an arithmetic progression with common difference d_1, and let $w_1, w_2, \ldots, w_{100}$ be consecutive terms of another arithmetic progression with common difference d_2, where $d_1 d_2 = 10$. For each $i = 1, 2, \ldots, 100$, let R_i be a rectangle with length l_i, width w_i and area A_i. If $A_{51} - A_{50} = 1000$, then the value of $A_{100} - A_{90}$ is _____. **[JEE(Adv.)2022]**

माना कि $l_1, l_2 \ldots, l_{100}$ सार्वअंतर (common difference) d_1 वाली एक समांतर श्रेढ़ी (aritmhetic progression) के क्रमागत पद (consecutive terms) है, एवं माना कि $w_1, w_2, \ldots, w_{100}$ सार्वअंतर (common difference) d_2 वाली एक दूसरी समांतर श्रेढ़ी (arithmetic progression) के क्रमागत पद है, जहाँ $d_1 d_2 = 10$ है। प्रत्येक $i = 1, 2, \ldots, 100$ के लिए, माना कि R_i एक आयत (rectangle) है जिसकी लम्बाई l_i, चौड़ाई w_i एवं क्षेत्रफल A_i है। यदि $A_{51} - A_{50} = 1000$ है, त $A_{100} - A_{90}$ का मान _____ है।

Ans. 18900

Sol. $A_{51} - A_{50} = 1000$

$\ell_{51} w_{51} - \ell_{50} w_{50} = 1000$

$(\ell_1 + 50d_1)(w_1 + 50d_2) - (\ell_1 + 49d_1)(w_1 + 49d_2) = 1000$

$(50 - 49)\ell_1 d_2 + (50 - 49) w_1 d_1 + (50^2 - 49^2) d_1 d_2 = 1000$

$\ell_1 d_2 + w_1 d_1 + 99 d_1 d_2 = 1000$

$\ell_1 d_2 + w_1 d_1 = 10$

Now $A_{100} - A_{90}$

$= \ell_{100} w_{100} - \ell_{90} w_{90}$

$= (\ell_1 + 99d_1)(w_1 + 99d_2) - (\ell_1 + 89d_1)(w_1 + 89d_2)$

$= (99 - 89)\ell_1 d_2 + (99 - 89) W_1 d_1 + (99^2 - 89^2) d_1 d_2$

$= 10\ell_1 d_2 + 10 w_1 d_1 + (10)(188) d_1 d_2$

$= 10 \times 10 + 100 \times 188$

$= 100\ (189)$

$= 18900$

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14. Let $a_1, a_2, a_3, \ldots$ be an arithmetic progression with $a_1 = 7$ and common difference 8. Let $T_1, T_2, T_3, \ldots$ be such that $T_1 = 3$ and $T_{n+1} - T_n = a_n$ for $n \geq 1$. Then, which of the following is/are TRUE? **[JEE(Adv.)2022]**

माना कि a_1, a_2, a_3, ... एक सार्वअंतर (common difference) 8 वाली समांतर श्रेढ़ी (arithmetic progression) है, जहाँ $a_1 = 7$ है। माना कि $T_1, T_2, T_3, \ldots$ इस प्रकार से है कि $T_1 = 3$ एवं सभी $n \geq 1$ के लिए $T_{n+1} - T_n = a_n$ है। तब निम्न में से कौनसा सत्य हैं? **[JEE(Adv.)2022]**

(A) $T_{20} = 1604$ (B) $\sum_{k=1}^{20} T_k = 10510$ (C) $T_{30} = 3454$ (D) $\sum_{k=1}^{30} T_k = 35610$

Ans. BC

Sol. $a_1 = 7, d = 8$

$T_{n+1} - T_n = a_n \forall n \geq 1$

$S_n = T_1 + T_2 + T_3 + \ldots + T_{n-1} + T_n$

$S_n = \quad T_1 + T_2 + T_3 + \ldots + T_{n-1} + T_n$

on subtraction

$T_n = T_1 + a_1 + a_2 + \ldots + a_{n-1}$

$T_n = 3 + (n-1)(4n-1)$

$T_n = 4n^2 - 5n + 4$

$\sum_{k=1}^{n} T_k = 4\sum n^2 - 5\sum n + 4n$

$T_{20} = 1504$

$T_{30} = 3454$

$\sum_{k=1}^{30} T_k = 35615$

$\sum_{k=1}^{20} T_k = 10510$

15. Let $7\overbrace{5\cdots5}^{r}7$ denote the (r + 2) digit number where the first and the last digits are 7 and the remaining r digits are 5. Consider the sum $S = 77 + 757 + 7557 + \cdots + 7\overbrace{5\cdots5}^{98}7$. If $S = \frac{7\overbrace{5\cdots5}^{99}7 + m}{n}$, where m and n are natural numbers less than 3000 , then the value of m + n is _____. **[JEE(Adv.)2023]**

माना कि $7\overbrace{5\cdots5}^{r}7$ वह (r + 2) अंको वाली संख्या है जिसका पहला एवं अंतिम अंक 7 है तथा बाकी के r अंक 5 हैं। योगफल $S = 77 + 757 + 7557 + \cdots + 7\overbrace{5\cdots5}^{98}7$ पर विचार कीजिए। यदि $S = \frac{7\overbrace{5\cdots5}^{99}7 + m}{n}$, जहाँ m तथा n, 3000 से छोटी प्राकृत संख्याएँ (natural numbers) हैं, तब m + n का मान है _____ **[JEE(Adv.)2023]**

Ans. 1219

Sol. $S = 77 + 757 + 7557 + ... + 7\overbrace{5.....5}^{98}7$

$$= 7\left(10+10^2+...+10^{99}\right)+50\left(1+11+...+\overbrace{111...1}^{98}\right)+7\times 99$$

$$= 70\left(\frac{10^{99}-1}{9}\right)+\frac{50}{9}\left[(10-1)+\left(10^2-1\right)+...+\left(10^{98}-1\right)\right]+7\times 99$$

$$= 70\left(\frac{10^{99}-1}{9}\right)+\frac{50}{9}\left[10\left(\frac{10^{98}-1}{9}\right)-98\right]+7\times 99$$

$$= \frac{7\times 10^{100}}{9}-\frac{70}{9}+\frac{50}{9}\left[\frac{10^{99}-1-9}{9}-98\right]+7\times 99$$

$$= \frac{7\times 10^{100}}{9}-\frac{70}{9}+\frac{50}{9}\left[\overbrace{111....1}^{99}-99\right]+7\times 99$$

$$= \frac{7\times 10^{100}-70+\overbrace{555...50}^{99}}{9}-550+693$$

$$= \frac{7\overbrace{555...5}^{99}-70+143\times 9}{9}$$

$$= \frac{7\overbrace{55...57}^{99}+1210}{9}$$

$m + n = 1219$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CHOICE TYPE QUESTIONS

1. If a, a_1, a_2, a_3,..., a_{2n}, b are in A.P. and a, g_1, g_2, g_3,.....g_{2n}, b are in G.P. and h is the harmonic mean of a and b, then $\frac{a_1+a_{2n}}{g_1g_{2n}}+\frac{a_2+a_{2n-1}}{g_2g_{2n-1}}+.....+\frac{a_n+a_{n+1}}{g_ng_{n+1}}$ is equal to :

यदि a, a_1, a_2, a_3,..., a_{2n}, b स.श्रे. में है और a, g_1, g_2, g_3,.....g_{2n}, b गु.श्रे. में हैं तथा a एवं b का हरात्मक माध्य h हो, तो $\frac{a_1+a_{2n}}{g_1g_{2n}}+\frac{a_2+a_{2n-1}}{g_2g_{2n-1}}+.....+\frac{a_n+a_{n+1}}{g_ng_{n+1}}$ का मान है :

(A) $\frac{2n}{h}$ (B) 2nh (C) nh (D) $\frac{n}{h}$

Ans. A

Sol. a, a_1, a_2 ....., a_{2n}, b are in AP and a, g_1, g_2, ...g_{2n}, b are in GP and $h=\frac{2ab}{a+b}$

$$\therefore \frac{a_1+a_{2n}}{g_1\ g_{2n}}+\frac{a_2+a_{2n-1}}{g_2\ g_{2n-1}}+.....+\frac{a_n+a_{n+1}}{g_n\ g_{n+1}}=\frac{a+b}{ab}+\frac{a+b}{ab}+.....+\frac{a+b}{ab}=2n\,.\left(\frac{a+b}{2ab}\right)=\frac{2n}{h}$$

Hindi a, a_1, a_2 ....., a_{2n}, b समान्तर श्रेढी में है और a, g_1, g_2, ...g_{2n}, b गुणोत्तर श्रेढी में है और $h=\frac{2ab}{a+b}$

$$\therefore \frac{a_1+a_{2n}}{g_1\ g_{2n}}+\frac{a_2+a_{2n-1}}{g_2\ g_{2n-1}}+.....+\frac{a_n+a_{n+1}}{g_n\ g_{n+1}}=\frac{a+b}{ab}+\frac{a+b}{ab}+.....+\frac{a+b}{ab}=2n\,.\left(\frac{a+b}{2ab}\right)=\frac{2n}{h}$$

2. If $H_n=1+\frac{1}{2}+\frac{1}{3}+\ .....+\frac{1}{n}$, then value of $1+\frac{3}{2}+\frac{5}{3}+.....+\frac{2n-1}{n}$ is :

यदि $H_n=1+\frac{1}{2}+\frac{1}{3}+\ .....+\frac{1}{n}$ हो, तो $1+\frac{3}{2}+\frac{5}{3}+.....+\frac{2n-1}{n}$ का मान हैं :

(A) $2n-H_n$ (B) $2n+H_n$ (C) H_n-2n (D) H_n+n

Ans. A

Sol. $H_n=1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n}$

$$1+\frac{3}{2}+\frac{5}{3}+\ \ldots\ldots\ +\frac{2n-1}{n}\ =(2-1)+\left(2-\frac{1}{2}\right)+\left(2-\frac{1}{3}\right)+\ldots\ldots+\left(2-\frac{1}{n}\right)=2n-H_n$$

3. If S_1, S_2, S_3 are the sums of first n natural numbers, their squares, their cubes respectively, then $\frac{S_3(1+8S_1)}{S_2^2}$ is equal to :

यदि S_1, S_2, S_3 क्रमशः प्रथम n प्राकृत संख्याओं का, उनके वर्गो का और उनके घनों का योगफल हो, तो $\frac{S_3(1+8S_1)}{S_2^2}$ का मान हैं :

(A) 1 (B) 3 (C) 9 (D) 10

Ans. C

Sol. $\frac{S_3(1+8S_1)}{S_2^2}=\frac{\left[\frac{n(n+1)}{2}\right]^2\left[1+\frac{8n(n+1)}{2}\right]}{\left[\frac{n(n+1)(2n+1)}{6}\right]^2}\ =\frac{\left[1+4n(n+1)\right]9}{(2n+1)^2}=9$ Ans

4. Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. if $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is :

माना a, b, c स.श्रे. में हैं और a^2, b^2, c^2 गु.श्रे. में है। यदि $a < b < c$ और $a + b + c = \frac{3}{2}$ हो, तो a का मान है :

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2\sqrt{3}}$ (C) $\frac{1}{2}-\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2}-\frac{1}{\sqrt{2}}$

Ans. D

Sol. $2b = a + c$ and $b^2 = \pm ac$

case-I

if $b^2 = ac$ and $a + c + b = \frac{3}{2} \Rightarrow b = \frac{1}{2}$

$a + c = 1 \Rightarrow ac = \frac{1}{4} \Rightarrow (1 - c)\,c = \frac{1}{4}$

$c^2 - c + \frac{1}{4} = 0 \Rightarrow c = \frac{1}{2} \Rightarrow a = \frac{1}{2}$

$a = b = c$ so not valid

case-II

$b^2 = -ac$ and $b = \frac{1}{2}$; $a + c = 1$ $\Rightarrow$ $ac = -\frac{1}{4}$

$(1-c)\,c = -\frac{1}{4}$ $\Rightarrow$ $c^2 - c - \frac{1}{4} = 0$ $\Rightarrow$ $c = \frac{1 \pm \sqrt{1+1}}{2} = \frac{1 \pm \sqrt{2}}{2}$

$c = \frac{1+\sqrt{2}}{2}$ $\Rightarrow$ $a = \frac{1-\sqrt{2}}{2}$

Hindi. $2b = a + c$ और $b^2 = \pm ac$

स्थिति-I

यदि $b^2 = ac$ और $a + c + b = \frac{3}{2}$ $\Rightarrow$ $b = \frac{1}{2}$

$a + c = 1$ $\Rightarrow$ $ac = \frac{1}{4}$ $\Rightarrow$ $(1-c)\,c = \frac{1}{4}$

$c^2 - c + \frac{1}{4} = 0$ $\Rightarrow$ $c = \frac{1}{2}$ $\Rightarrow$ $a = \frac{1}{2}$

$a = b = c$ अतः सत्य नहीं

स्थिति-II

$b^2 = -ac$ और $b = \frac{1}{2}$; $a + c = 1$ $\Rightarrow$ $ac = -\frac{1}{4}$

$(1-c)\,c = -\frac{1}{4}$ $\Rightarrow$ $c^2 - c - \frac{1}{4} = 0$; $c = \frac{1 \pm \sqrt{1+1}}{2} = \frac{1 \pm \sqrt{2}}{2}$

$c = \frac{1+\sqrt{2}}{2}$ $\Rightarrow$ $a = \frac{1-\sqrt{2}}{2}$

5. Let the positive numbers a, b, c, d be in A.P. Then abc, abd, acd, bcd are :

(A) not in A.P./G.P./H.P. (B) in A.P.

(C) in G.P. (D) in H.P.

यदि धनात्मक संख्याएँ a, b, c, d स.श्रे. में हो, तो abc, abd, acd, bcd किस श्रेढ़ी में हैं :

(A) स.श्रे./गु.श्रे./ह.श्रे. में नहीं है (B) स.श्रे. में है

(C) गु.श्रे. में है (D) ह.श्रे. में है

Ans. D

Sol. a, b, c, d $\rightarrow$ समान्तर श्रेढी; $\frac{a}{abcd}, \frac{b}{abcd}, \frac{c}{abcd}, \frac{d}{abcd}$ $\rightarrow$ समान्तर श्रेढी

$\frac{1}{bcd}, \frac{1}{acd}, \frac{1}{abd}, \frac{1}{abc} \rightarrow$ समान्तर श्रेढी ; $\frac{1}{abc}, \frac{1}{abd}, \frac{1}{acd}, \frac{1}{bcd} \rightarrow$ समान्तर श्रेढी

abc, abd, acd, bcd $\rightarrow$ हरात्मक श्रेढी

6. The first term of an infinite geometric progression is x and its sum is 5. Then :
एक अनन्त गुणोतर श्रेणी का प्रथम पद x है तथा योग 5 है तो :
(A) $0 \le x \le 10$ (B) $0 < x < 10$ (C) $-10 < x < 0$ (D) $x > 10$

Ans. B

Sol. $\frac{x}{1-r} = 5 \Rightarrow 5 - 5r = x \Rightarrow r = 1 - \frac{x}{5}$

As $|r| < 1$ i.e. $\left|1-\frac{x}{5}\right| < 1; -1 < 1 - \frac{x}{5} < 1$

$-5 < 5 - x < 5 = -10 < -x < 0 = 10 > x > 0$

i.e. $0 < x < 10$

7. In the quadratic equation $ax^2 + bx + c = 0$, if $\Delta = b^2 - 4ac$ and $\alpha + \beta$, $\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$ are in G.P. where α, β are the roots of $ax^2 + bx + c = 0$, then :
एक द्विघात समीकरण यदि $ax^2 + bx + c = 0$ में यदि $\Delta = b^2 - 4ac$ एवं $\alpha + \beta$, $\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$ गु.श्रे. में है जहाँ α, β सभी $ax^2 + bx + c = 0$ के मूल है तो :
(A) $\Delta \neq 0$ (B) $b\Delta = 0$ (C) $c\,\Delta = 0$ (D) $\Delta = 0$

Ans. C

Sol. $\alpha + \beta = \frac{-b}{a}$, $\alpha\,\beta = \frac{c}{a}$ $\Rightarrow$ $(\alpha + \beta), \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in G.P

then $(\alpha^2 + \beta^2)^2 = (\alpha + \beta)(\alpha^3 + \beta^3)$ $\Rightarrow$ $2\alpha^2\beta^2 = \alpha\beta^3 + \beta\alpha^3$

$\Rightarrow$ $\alpha\,\beta\,[\alpha^2 + \beta^2 - 2\alpha\,\beta] = 0$ so $\alpha\,\beta\,(\alpha - \beta)^2 = 0$

$\Rightarrow$ $\frac{c}{a} \cdot \frac{b^2 - 4ac}{a^2} = 0,$ $a \neq 0 \Rightarrow c \cdot \Delta = 0$

8. If $(m + 1)^{th}$, $(n + 1)^{th}$ and $(r + 1)^{th}$ terms of a non-constant A.P. are in G.P. and m, n, r are in H.P., then the ratio of first term of the A.P. to its common difference is :
यदि $(m + 1)^{th}$, $(n + 1)^{th}$ तथा $(r + 1)^{th}$ समान्तर श्रेणी के पद गुणोतर श्रेणी में है तथा m, n, r हरात्मक श्रेणी में है तो समान्तर श्रेणी के प्रथम तथा सार्वअन्तर का अनुपात होगा :

(A) $-\frac{n}{2}$ (B) $-n$ (C) $-2n$ (D) $+n$

Ans. A

Sol. $T_{m+1} = a + md = A$ ...(1)
$T_{n+1} = a + nd = AR$ ...(2)
$T_{r+1} = a + rd = AR^2$ ...(3)
m, n, r are in H.P.

So, $\frac{2}{n} = \frac{m+r}{mr} \Rightarrow m + r = \frac{2mr}{n}$

$T_{m+1}, T_{n+1}, T_{r+1}$ are in G.P.
$(a + nd)^2 = (a + md)(a + rd)$

$$\left(\frac{a}{d} + n\right)^2 = \left(\frac{a}{d} + m\right)\left(\frac{a}{d} + r\right)$$

let $\frac{a}{d} = x$

$(x + n)^2 = (x + m)(x + r)$
$x^2 + n^2 + 2nx = x^2 + (m + r)x + mr$
$x[m + r - 2n] + mr - n^2 = 0$

$$x = \frac{n^2 - mr}{m + r - 2n}$$

$$x = \frac{n^2 - mr}{\frac{2mr}{n} - 2n} = \frac{n\left(n^2 - mr\right)}{2\left(mr - n^2\right)} = -\frac{n}{2}$$

$$x = -\frac{n}{2} = \frac{a}{d}$$

9. If α, β be roots of the equation $375x^2 - 25x - 2 = 0$ and $s_n = \alpha^n + \beta^n$, then $\lim_{n\to\infty}\left(\sum_{r=1}^{n} S_r\right) = ...$

यदि α, β समीकरण $375x^2 - 25x - 2 = 0$ के मूल है तथा $s_n = \alpha^n + \beta^n$ है, तो $\lim_{n\to\infty}\left(\sum_{r=1}^{n} S_r\right) = ...$

(A) $\frac{1}{12}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) 1

Ans. A

Sol. $375x^2 - 25x - 2 = 0$ (α, β)
$S_n = \alpha^n + \beta^n$

$$\sum_{r=1}^{n} S_r = \sum_{r=1}^{n} \alpha^r + \sum_{r=1}^{n} \beta^r = (\alpha + \alpha^2 + ... + \alpha^n) + (\beta + \beta^2 + ... + \beta^n)$$

$$\sum_{r=1}^{n} S_r = \frac{\alpha\left(\alpha^n - 1\right)}{(\alpha - 1)} + \frac{\beta\left(\beta^n - 1\right)}{(\beta - 1)}$$

$375 - x^2 - 25x - 2 = 0$

$$x = \frac{25 \pm \sqrt{(25)^2 - 4(-2)(375)}}{2 \times 375}$$

$$\alpha = \frac{25 + \sqrt{3625}}{750}, \beta = \frac{25 - \sqrt{3625}}{750}$$

$|\alpha| < 1, |\beta| < 1$

So,

$$\lim_{n \to \infty} \sum_{r=1}^{n} S_r = \lim_{n \to \infty} \left[\frac{\alpha(\alpha^n - 1)}{\alpha - 1} + \frac{\beta(\beta^n - 1)}{\beta - 1} \right]$$

$$= \left[\frac{-\alpha}{\alpha - 1} - \frac{\beta}{\beta - 1} \right]$$

$$= -\left[\frac{\alpha}{\alpha - 1} + \frac{\beta}{\beta - 1} \right]$$

$$= -\left[\frac{2\alpha\beta - (\alpha + \beta)}{(\alpha - 1)(\beta - 1)} \right]$$

$$= -\left[\frac{2\left(\frac{-2}{375}\right) - \left(\frac{+25}{375}\right)}{\frac{348}{375}} \right]$$

$$= \frac{1}{12}$$

10. The sum of the first n terms of the series $1^2 + 2.\ 2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ...$ is $\frac{n\ (n+1)^2}{2}$ when n is even. When n is odd the sum is :

अनुक्रम $1^2 + 2.\ 2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ...$ के प्रथम n पदों का योग, जब n सम संख्या है, $\frac{n\ (n+1)^2}{2}$ है। यदि n विषम संख्या है तो यह योग है :

(A) $\frac{3n\ (n+1)}{2}$ (B) $\frac{n^2\ (n+1)}{2}$ (C) $\frac{n\ (n+1)^2}{4}$ (D) $\left[\frac{n\ (n+1)}{2}\right]^2$

Ans. B

Sol. $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ...n \text{ terms} = \frac{n(n+1)^2}{2}$, when n is even

$1^2 + 2 \cdot 2^2 + 3^2 + ... 2 \cdot n^2 = n \frac{(n+1)^2}{2}$ $\Rightarrow$ when n is odd $n+1$ is even

$1^2 + 2 \cdot 2^2 + 3^2 + ... n^2 + 2 \cdot (n+1)^2 = (n+1) \frac{(n+2)^2}{2}$

$1^2 + 2 \cdot 2^2 + 3^2 + ... n^2 = (n+1) \left[\frac{(n+2)^2}{2} - 2(n+1) \right] = \frac{(n+1) n^2}{2}$

Hindi $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + ...n$ पदों तक $= \frac{n(n+1)^2}{2}$, जब n सम है

$1^2 + 2 \cdot 2^2 + 3^2 + ... 2 \cdot n^2 = n \frac{(n+1)^2}{2}$ जब n विषम है तब $n+1$ सम है

$1^2 + 2 \cdot 2^2 + 3^2 + ... n^2 + 2 \cdot (n+1)^2 = (n+1) \frac{(n+2)^2}{2}$

$1^2 + 2 \cdot 2^2 + 3^2 + ... n^2 = (n+1) \left[\frac{(n+2)^2}{2} - 2(n+1) \right] = \frac{(n+1) n^2}{2}$

11. Let a_n denotes the nth term of an A.P. If $a_2 \cdot a_{12} = 1$ & $a_4 \cdot a_{10} = b$, then all the terms of A.P. are distinct and real for the true set of values of 'b' given by :

माना a_n किसी स.श्रे. का n वां पद है तथा $a_2 \cdot a_{12} = 1$ & $a_4 \cdot a_{10} = b$ तो 'b' के मान होंगे (स.श्रे. की सभी पद भिन्न है) :

(A) $\left(\frac{9}{25}, \infty \right)$ (B) $(0, \infty)$ (C) $(1, \infty)$ (D) $(-\infty, \infty)$

Ans. C

Sol. $a_n = a + (n-1) d$

$a_2 a_{12} = (a + d)(a + 11d) = 1 \Rightarrow a^2 + 12ad + 11d^2 = 1$ ...(1)

$a_4 a_{10} = (a + 3d)(a + 9d) = b \Rightarrow a^2 + 12ad + 27d^2 = b$ ...(2)

(1) – (2) :

$-16d^2 = 1 - b$

$d^2 = \frac{b-1}{16}$

$d = \pm \sqrt{\frac{b-1}{16}}$

$\frac{b-1}{16} > 0$ ($\because$ All terms of A.P. are distinct)

$b > 1$

$b \in (1, \infty)$

12. An A.P. whose first terms is unity , and in which the sum of the first half of any even number of terms of that of the second half of the same number of terms is in constant ratio, then the common difference is d is given by :

एक स.श्रे. जिसका प्रथम पद एक है इस प्रकार कि किन्हीं सम संख्याओं में प्रथम आधी संख्याओं का योग तथा बाकी आधी संख्याओं का योग का अनुपात हमेशा नियत हो तो सार्वअन्तर होगा :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $a = 1$, common difference $= d$

let sequence has total 2n terms, then

$$\frac{a_1 + a_2 + ... + a_n}{a_{n+1} + a_{n+2} + ...a_{2n}} = k \text{ [constant]}$$

$$\frac{\frac{n}{2}\left[2 + (n-1)d\right]}{\frac{n}{2}\left[2(1+nd) + (n-1)d\right]} = k$$

$$\frac{2 + (n-1)d}{2 + 3nd - d} = k$$

$$\frac{(2-d) + nd}{(2-d) + 3nd} = k$$

ratio constant, so, $d = 2$

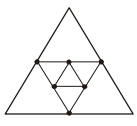
13. One side of an equilateral triangle is 24 cm. The mid–points of its sides are joined to form another triangle whose mid–points are in turn joined to form still another triangle. This process continues indefinitely. Then the sum of the perimeters of all the triangles is :

(A) 144 cm (B) 212 cm (C) 288 cm (D) None of these

एक समबाहु त्रिभुज की एक भुजा 24 सेमी. है। इसकी भुजाओं के मध्य बिन्दुओं को मिलाने से अन्य त्रिभुज बनता है जिसके मध्य बिन्दुओं को पुनः मिलाने पर किसी अन्य त्रिभुज का निर्माण होता है। यह क्रम अनन्त तक चलता हो, तो सभी त्रिभुजों के परिमापों का योगफल है :

(A) 144 सेमी. (B) 212 सेमी. (C) 288 सेमी. (D) इनमें से कोई नहीं

Ans. A

Sol.  $= 3[24 + 12 + 6 + ...\infty] = 3\ \frac{24}{1 - \frac{1}{2}} = 144$

14. If in a ΔABC, the altitudes from the vertices A, B, C on opposite sides are in H.P., then sin A, sin B, sin C are in :

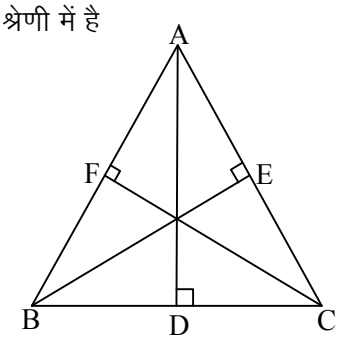
(A) G.P. (B) A.P.

(C) Arithmetic – Geometric progression (D) H.P.

यदि ΔABC में, शीर्षों A, B तथा C से सम्मुख भुजाओं पर डाले गए शीर्ष लम्ब हरात्मक श्रेणी में हैं, तो $\sin A$, $\sin B$ तथा $\sin C$:

(A) गुणोत्तर श्रेणी में है (B) समान्तर श्रेणी में है

(C) समान्तर गुणोत्तर श्रेणी में है (D) हरात्मक श्रेणी में है



Ans. B

Sol. AD, BE, CF are in H.P.

$$\frac{\sin A}{BC} = \frac{\sin B}{AC} = \frac{\sin c}{AB} = 2R$$

In ΔACF

$$\sin A = \frac{CF}{AC} \Rightarrow CF = AC\sin A$$

In ΔABD

$$\sin B = \frac{AD}{AB} \Rightarrow AD = AB\sin B$$

In ΔBCF

$$\sin C = \frac{BE}{BC} \Rightarrow BE = BC\sin C$$

$\frac{1}{AD}, \frac{1}{BE}, \frac{1}{CF}$ are in A.P.

$\frac{1}{AB\sin B}, \frac{1}{BC\sin C}, \frac{1}{AC\sin A}$ in A.P.

$\frac{1}{(AB)(2R\cdot AC)}, \frac{1}{(BC)(2R\cdot AB)}, \frac{1}{(AC)(2R\cdot BC)}$ are in A.P.

multiply by 2R(AB)(BC(AC)

BC, AC, AB are in A.P.

multiply by $2R \Rightarrow 2RBC, 2RAC, 2RAB$ are in A.P.

sin A, sin B, sin C are in A.P.

15. A new sequence is obtained from the sequence of positive integers [1, 2, 3, ...] by deleting all the perfect squares. Then find the 2009th term of the new sequence :

दी गयी धनात्मक पूर्णांक पदों की श्रेढ़ी [1, 2, 3, ...] से पूर्ण वर्ग संख्याओं को हटाकर एक नयी श्रेढ़ी प्राप्त की जाती है तब नयी श्रेढ़ी का 2009वां पद होगा :

(A) 2052 (B) 2053 (C) 2054 (D) 2055

Ans. C

Sol. $1,2,3,4,.....,2009 \Rightarrow$ total 2009 terms

$1^2, 2^2, 3^2, ...44^2, 45^2 = 2025$

total terms in sequence

$1,2,3,4,5,....., 2025 = 2025$

total terms in sequence

1,2,3,5,6,7,8,10,....., 2024 = 2025 – 45 = 1980
to get 2009th term of new sequence we add 29 in 2025
So, $T_{2009} = 2025 + 29 = 2054$

16. Consider $S_n = \frac{8}{5}+\frac{16}{65}+....+\frac{8r}{4r^4+1}$ Sum of infinite terms of above series will be :

(A) 0 (B) 1/2 (C) 2 (D) None of these

माना कि $S_n = \frac{8}{5}+\frac{16}{65}+....+\frac{8r}{4r^4+1}$ है तो दी गयी श्रेढ़ी के अनन्त पदों का योगफलन होगा :

(A) 0 (B) 1/2 (C) 2 (D) इनमें से कोई नहीं

Ans. C

Sol. $S_n = \frac{8}{5}+\frac{16}{65}+.....+\frac{8r}{4r^4+1}$

$$T_r = \frac{8r}{4r^4+1} = \frac{8r}{\left(2r^2+2r+1\right)\left(2r^2-2r+1\right)}$$

$$= 2\left[\frac{\left(2r^2+2r+1\right)-\left(2r^2-2r+1\right)}{\left(2r^2+2r+1\right)\left(2r^2-2r+1\right)}\right]$$

$$T_r = 2\left[\frac{1}{\left(2r^2-2r+1\right)}-\frac{1}{\left(2r^2+2r+1\right)}\right]$$

$$S_n = \sum_{r=1}^{n} T_r$$

$$S_\infty = 2\left[\left(\frac{1}{1}-\frac{1}{5}\right)+\left(\frac{1}{5}-\frac{1}{13}\right)+\left(\frac{1}{13}-\frac{1}{25}\right)+.....\right]$$

$S_\infty = 2$

17. If the sum of first n terms of an A.P. is cn^2, then the sum of squares of these n terms is :
यदि एक सामानान्तर श्रेणी के n पदों का योग cn^2 है तो उसके n पदों के वर्गो का योग होगा :

(A) $\frac{n(4n^2-1)c^2}{6}$ (B) $\frac{n(4n^2+1)c^2}{3}$ (C) $\frac{n(4n^2-1)c^2}{3}$ (D) $\frac{n(4n^2+1)c^2}{6}$

Ans. C

Sol. $S_n = cn^2$
$S_{n-1} = c(n-1)^2 = cn^2 - 2cn + c$

$T_n = S_n - S_{n-1} = 2cn - c$

$T_n^2 = (2cn - c)^2 = c^2 [4n^2 - 4n + 1]$

$$S = \sum_{r=1}^{n} T_r = c^2 \left[\sum_{r=1}^{n} 4r^2 - \sum_{r=1}^{n} 4r - \sum_{r=1}^{n} 1 \right]$$

$$= c^2 \left[4\left(\frac{n(n+1)(2n+1)}{6} \right) - \frac{4n(n+1)}{2} - n \right]$$

$$= \frac{nc^2\left(4n^2 - 1\right)}{3}$$

18. If x, y, z are in A.P., then $\frac{1}{\sqrt{y}+\sqrt{z}}, \frac{1}{\sqrt{z}+\sqrt{x}}, \frac{1}{\sqrt{x}+\sqrt{y}}$ will be in :

(A) A.P. (B) G.P. (D) H.P. (D) None

यदि x, y, z स.श्रे. में हो तो $\frac{1}{\sqrt{y}+\sqrt{z}}, \frac{1}{\sqrt{z}+\sqrt{x}}, \frac{1}{\sqrt{x}+\sqrt{y}}$ किसमें होंगे :

(A) A.P. (B) G.P. (D) H.P. (D) कोई नहीं

Ans. A

Sol. x, y, z are in A.P.

$$\frac{1}{\sqrt{y}+\sqrt{z}}, \frac{1}{\sqrt{z}+\sqrt{x}}, \frac{1}{\sqrt{x}+\sqrt{y}}$$

$$\frac{\sqrt{y}-\sqrt{z}}{y-z}, \frac{\sqrt{z}-\sqrt{x}}{z-x}, \frac{\sqrt{x}-\sqrt{y}}{x-y}$$

$x, y, z \Rightarrow x, x + d, x + 2d$

$y - z = -d$

$z - x = 2d$

$x - y = -d$

$$\frac{\sqrt{y}-\sqrt{z}}{-d}, \frac{\sqrt{z}-\sqrt{x}}{2d}, \frac{\sqrt{x}-\sqrt{y}}{-d}$$

$$\sqrt{z}-\sqrt{y}, \frac{\sqrt{z}-\sqrt{x}}{2}, \sqrt{y}-\sqrt{x}$$

$$\left(\sqrt{z}-\sqrt{y}\right)+\left(\sqrt{y}-\sqrt{x}\right) = \sqrt{z}-\sqrt{x} = 2\left(\frac{\sqrt{z}-\sqrt{x}}{2}\right)$$

$(2b = a + c)$

$\Rightarrow \frac{1}{\sqrt{y}+\sqrt{z}}, \frac{1}{\sqrt{z}+\sqrt{x}}, \frac{1}{\sqrt{x}+\sqrt{y}}$ are in A.P.

19. If $x \varepsilon R$, the numbers $5^{1+x}+5^{1-x}$, a/2, 25^x+25^{-x} form an A.P. then 'a' must lie in the interval :
यदि संख्याएँ $5^{1+x}+5^{1-x}$, a/2, 25^x+25^{-x} , $x \varepsilon R$ एक स.श्रे. बनाती हो, तो a किस अन्तराल में होना चाहिए :
(A) [1, 5] (B) [2, 5] (C) [5, 12] (D) $[12, \infty)$

Ans. D

Sol. $x \in R$
$5^{1+x}+5^{1-x}$, a/2 , $5^{2x}+5^{-2x}$ are in A.P समान्तर श्रेढ़ी में है।
$a=(5^{2x}+5^{-2x})+(5^{1+x}+5^{1-x}) \Rightarrow a=(5^{2x}+5^{-2x})+5(5^x+5^{-x}) = (5^x-5^{-x})^2+2+5(5^{x/2}-5^{-x/2})^2+10$
$a=12+(5^x-5^{-x})^2+5(5^{x/2}-5^{-x/2})^2 \Rightarrow a \geq 12$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

20. If positive numbers a, b, c are in A.P. and a^2, b^2, c^2 are in H.P., then :
(A) a = b = c (B) 2b = a + c (C) $b^2=\sqrt{\frac{ac}{8}}$ (D) None of these
यदि धनात्मक संख्याएँ a,b,c स.श्रे. में हो और a^2, b^2, c^2 ह.श्रे. में हो, तो :
(A) a = b = c (B) 2b = a + c (C) $b^2=\sqrt{\frac{ac}{8}}$ (D) इनमें से कोई नहीं

Ans. AB

Sol. a, b, c are in A.P. $\Rightarrow 2b = a + c$
a^2, b^2, c^2 are in H.P.

$$\Rightarrow \frac{1}{b^2}-\frac{1}{a^2}=\frac{1}{c^2}-\frac{1}{b^2}$$

$$\frac{(a-b)(a+b)}{a^2b^2}=\frac{(b-c)(c+b)}{b^2c^2}$$

$(a + b) c^2 = (b + c)a^2$
$ac^2 + bc^2 - a^2b - a^2c = 0$
$ac \cdot (c - a) + b (c - a) (c + a) = 0$
$(ac + bc + ab) (c - a) = 0$

$c - a = 0$	or	$ac + bc + ba = 0$
$c = a$	or	$(a + c)b + ac = 0$
$\Rightarrow a = b = c$		$2b^2 = - ac$

$$b^2=\frac{-ac}{2}$$

But a, b, c are positive numbers. So, this case is not possible.
$\Rightarrow a = b = c$

21. Let a and b be two positive real numbers. Suppose A_1, A_2 are two Arithmetic means, G_1, G_2 and H_1, H_2 are Geometric and Harmonic Means between a and b, then :

माना a तथा b दो धनात्मक वास्तविक संख्याऐं है। माना a तथा b के मध्य, A_1, A_2 दो समान्तर माध्य, G_1, G_2 दो गुणातर माध्य तथा H_1, H_2 दो हरात्मक माध्य है, तो :

(A) $\frac{G_1G_2}{H_1H_2} = \frac{A_1 + A_2}{H_1 + H_2}$

(B) $\frac{G_1G_2}{H_1H_2} - \frac{5}{9} = \frac{2}{9}\left(\frac{a}{b} + \frac{b}{a}\right)$

(C) $\frac{H_1 + H_2}{A_1 + A_2} = \frac{9ab}{(2a+b)(a+2b)}$

(D) $\frac{G_1G_2}{H_1H_2} = \frac{H_1 + H_2}{A_1 + A_2}$

Ans. ABC

Sol. a, A_1, A_2, b are in A.P.

$\Rightarrow A_1 = a + 1d$, $A_2 = a + 2d$ and $A_1 + A_2 = 2\left(\frac{a+b}{2}\right) = a + b$

a, G_1, G_2, b are in G.P.

$\Rightarrow G_1G_2 = \left[(ab)^{\frac{1}{2}}\right]^2 = ab$

a, H_1, H_2, b are in H.P. $\Rightarrow \frac{1}{H_1} + \frac{1}{H_2} = \frac{(a+b)}{ab}$

$\frac{H_1 + H_2}{H_1H_2} = \frac{(A_1 + A_2)}{G_1G_2}$

$\Rightarrow \frac{G_1G_2}{H_1H_2} = \frac{A_1 + A_2}{H_1 + H_2}$

$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{b}$ arre in A.P.

$\Rightarrow \frac{1}{b} = \frac{1}{a} + 3d \Rightarrow d = \frac{1}{3}\left(\frac{1}{b} - \frac{1}{a}\right)$

$\left(\frac{1}{H_1}\right)\left(\frac{1}{H_2}\right) = \left(\frac{2}{3a} + \frac{1}{3b}\right)\left(\frac{1}{3a} + \frac{2}{3b}\right) = \frac{(2a+b)(b+2a)}{9a^2b^2} = \frac{2a^2 + 2b^2 + 5ab}{9a^2b^2}$

$\frac{G_1G_2}{H_1H_2} - \frac{5}{9} = (ab)\left(\frac{2a^2 + 2b^2 + 5ab}{9a^2b^2}\right) - \frac{5}{9}$

$= \frac{2}{9}\left(\frac{a}{b} + \frac{b}{a}\right) + \frac{5}{9} - \frac{5}{9} = \frac{2}{9}\left(\frac{a}{b} + \frac{b}{a}\right)$

$$\frac{H_1+H_2}{A_1+A_2}=\frac{3ab\left[\frac{1}{b+2a}+\frac{1}{a+2b}\right]}{a+b}=\frac{9ab(a+b)}{(a+b)(a+2b)(b+2a)}$$

$$=\frac{9ab}{(a+2b)(b+2a)}$$

22. For ΔABC, if $81+144a^4+16b^4+9c^4=144abc$, (where notations have their usuall meaning), then :

(A) $a>b>c$ (B) $A<B<C$

(C) Area of $\Delta ABC=\frac{3\sqrt{3}}{8}$ (D) Triangle ABC is right angled

ΔABC के लिए, यदि $81+144a^4+16b^4+9c^4=144abc$ है, (जहां चिन्हों के अर्थ सामान्य है), तो :

(A) $a>b>c$ (B) $A<B<C$

(C) ΔABC का क्षेत्रफल $=\frac{3\sqrt{3}}{8}$ (D) ABC समकोण त्रिभुज है

Ans. BCD

Sol. $81+144a^4+16b^4+9c^4=144abc$

$(9)^2+(12a^2)^2+(4b^2)^2+(3c^2)^2=144abc$

Using A.M., G.M.

$$\frac{(9)^2+\left(12a^2\right)^2+\left(4b^2\right)^2+\left(3c^2\right)^2}{4}\geq\left[(9)^2\cdot\left(12a^2\right)^2\cdot\left(4b^2\right)^2\left(3c^2\right)^2\right]^{\frac{n}{4}}=36abc$$

$(9)^2+(12a^2)^2+(4b^2)^2+(3c^2)^2\geq 144abc$

But

$(9)^2+(12a^2)^2+(4b^2)^2+(3c^2)^2=144abc$

$\Rightarrow$ A.M. = G.M.

$\Rightarrow 81=144a^4=16b^4=9c^4$

$$a^4=\frac{81}{144},b^4=\frac{81}{16},c^4=\frac{81}{9}$$

$$a=\frac{\sqrt{3}}{2},b=\frac{3}{2},c=\sqrt{3}$$

$c>b>a$

Largest angle is opposite to largest side.

$\Rightarrow C>B>A$

$$\Rightarrow a^2+b^2=\frac{3}{4}+\frac{9}{4}=3=c^2$$

$a^2+b^2=c^2$ (right angle triangle)

Aea of $\Delta ABC = \frac{1}{2}\times\left(\frac{\sqrt{3}}{2}\right)\times\left(\frac{3}{2}\right)=\frac{3\sqrt{3}}{8}$

23. If $S_r=\sqrt{r+\sqrt{r+\sqrt{r+\sqrt{r+\sqrt{.....\infty}}}}}$, $r>0$, then which of the following is/are correct :

(A) S_2, S_6, S_{12}, S_{20} are in A.P. (B) S_4, S_9, S_{16} are irrational

(C) $(2S_3-1)^2, (2S_4-1)^2, (2S_5-1)^2$ are in A.P. (D) S_2, S_{12}, S_{56} are in G.P.

यदि $S_r=\sqrt{r+\sqrt{r+\sqrt{r+\sqrt{r+\sqrt{.....\infty}}}}}$, $r>0$ है, तो निम्न में से कौनसा/कौनसे कथन सत्य है :

(A) S_2, S_6, S_{12}, S_{20} समान्तर श्रेणी में है

(B) S_4, S_9, S_{16} अपरिमेय है

(C) $(2S_3-1)^2, (2S_4-1)^2, (2S_5-1)^2$ समान्तर श्रेणी में है

(D) S_2, S_{12}, S_{56} गुणोतर श्रेणी में है

Ans. ABCD

Sol. $S_r=\sqrt{r+\sqrt{r+\sqrt{r+.....\infty}}}$ $r>0$

$S_r=\sqrt{r+s_r}$

$S_r^2=r+S_r$

$S_r^2-S_r-r=0$

$S_r=\frac{1\pm\sqrt{1+4r}}{2}$

(A) $S_2=\frac{1\pm\sqrt{9}}{2}=2$, $S_2=\frac{1\pm\sqrt{9}}{2}=2,\underbrace{-1}_{\text{not possible}}$

$S_2=2$

$S_6=\frac{1\pm 25}{2}$

$S_6=3$

$S_{12}=\frac{1\pm\sqrt{49}}{2}$

$S_{12}=4$

$S_{20}=\frac{1\pm\sqrt{81}}{2}$

$S_{20}=5$ $(S_2, S_6. S_{12}, S_{20}=2,3,4,5)$ are in A.P.

$S_4=\frac{1\pm\sqrt{17}}{2}, S_9=\frac{1\pm\sqrt{37}}{2}, S_{16}=\frac{1\pm\sqrt{62}}{2}$ are irrational.

(C) $S_3 = \frac{1 \pm \sqrt{13}}{2} \Rightarrow 2S_3 = 1 + \sqrt{13} \Rightarrow 2S_3 - 1 = \sqrt{13}$

$S_4 = \frac{1 \pm \sqrt{17}}{2} \Rightarrow 2S_4 = 1 + \sqrt{17} \Rightarrow 2S_4 - 1 = \sqrt{17}$

$S_5 = \frac{1 \pm \sqrt{21}}{2} \Rightarrow 2S_5 = 1 + \sqrt{21} \Rightarrow 2S_5 - 1 = \sqrt{21}$

$(2S_3 - 1)^2, (2S_4 - 1)^2, (2S_5 - 1)^2 = 13, 17, 21$ (A.P.)

(D) $S_2 = 2, S_{12} = 4, S_{56} = 8$

$2, 4, 8 \rightarrow$ G.P.

24. If a, b, c are in H.P., then :

(A) $\frac{a}{b + c - a}, \frac{b}{c + a - b}, \frac{c}{a + b - c}$ are in H.P.

(B) $\frac{2}{b} = \frac{1}{b - a} + \frac{1}{b - c}$

(C) $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ are in G.P.

(D) $\frac{a}{b + c}, \frac{b}{c + a}, \frac{c}{a + b}$ are in H.P.

यदि a, b, c ह.श्रे. में हो तो :

(A) $\frac{a}{b + c - a}, \frac{b}{c + a - b}, \frac{c}{a + b - c}$ ह. श्रे. में है

(B) $\frac{2}{b} = \frac{1}{b - a} + \frac{1}{b - c}$

(C) $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ गु. श्रे. में है

(D) $\frac{a}{b + c}, \frac{b}{c + a}, \frac{c}{a + b}$ ह. श्रे. में है

Ans. ABCD

Sol. $a, b, c \rightarrow$ H.P.

$\Rightarrow b = \frac{2ac}{a + c}$

(A) $\frac{a}{b+c-a}, \frac{b}{c+a-b}, \frac{c}{a+b-c}$ are in H.P.

$\Rightarrow \frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P.

Add 2

$\frac{b+c+a}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$ are in A.P.

divide by $(a+b+c)$

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

$\Rightarrow$ a, b, c in H.P.

(B) $\frac{1}{b-a}+\frac{1}{b-c}=\frac{(b-c)+(b-a)}{(b-a)(b-c)}$

$=\frac{2b-(a+c)}{b^2-b(a+c)+ac}$

$=\frac{2b-\frac{2ac}{b}}{b^2-2ac+ac}$

$=\frac{\frac{2}{b}\left(b^2-ac\right)}{b^2-ac}$

$=\frac{2}{b}$

(C) $a-\frac{b}{2}, \frac{b}{2}, c-\frac{b}{2}$

$\left(c-\frac{b}{2}\right)\left(a-\frac{b}{2}\right)=ac+\frac{b^2}{4}-\frac{b}{2}(a+c)$

$=ac+\frac{b^2}{4}-ac$

$=\frac{b^2}{4}$

$\Rightarrow a-\frac{b}{2}, \frac{b}{2}, c-\frac{b}{2}$ are in G.P.

(D) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P.

$\Rightarrow \frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in A.P.

Add 1

$\Rightarrow \frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$ are in A.P.

Divide by (a + b + c)

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

$\Rightarrow$ a,b,c are in H.P.

25. A straight line through the vertex P of a triangle PQR intersects the side QR at the point S and the circumcircle of the triangle PQR at the point T. If S is not the centre of the circumcircle, then :

(A) $\frac{1}{PS}+\frac{1}{ST}<\frac{2}{\sqrt{QS \times SR}}$ (B) $\frac{1}{PS}+\frac{1}{ST}>\frac{2}{\sqrt{QS \times SR}}$

(C) $\frac{1}{PS}+\frac{1}{ST}<\frac{4}{QR}$ (D) $\frac{1}{PS}+\frac{1}{ST}>\frac{4}{QR}$

त्रिभुज PQR के शीर्ष P से गुजरने वाली एक सरल रेखा QR को एक बिन्दु S पर प्रतिच्छेद करती है तथा त्रिभुज PQR के परिवृत को बिन्दु T पर प्रतिच्छेद करती है। यदि S परिवृत का केन्द्र नही है तो :

(A) $\frac{1}{PS}+\frac{1}{ST}<\frac{2}{\sqrt{QS \times SR}}$ (B) $\frac{1}{PS}+\frac{1}{ST}>\frac{2}{\sqrt{QS \times SR}}$

(C) $\frac{1}{PS}+\frac{1}{ST}<\frac{4}{QR}$ (D) $\frac{1}{PS}+\frac{1}{ST}>\frac{4}{QR}$

Ans. BD

Sol. By intersecting chord theorem

(PS) (ST) = (QS) (SR)

using A.M.G.M.

$$\frac{1}{PS}+\frac{1}{PT}>\frac{2}{\sqrt{PS \cdot ST}}=\frac{2}{\sqrt{QS \cdot SR}} \Rightarrow \frac{1}{PS}+\frac{1}{PT}>\frac{2}{\sqrt{QS \cdot SR}}$$

$$\Rightarrow \sqrt{QS \cdot SR}>\frac{2}{\frac{1}{PS}+\frac{1}{PT}}$$

Using A.M. G.M.

$$\frac{QS+SR}{2}>\sqrt{QS \cdot SR}$$

$$\frac{QR}{2}>\frac{2}{\frac{1}{PS}+\frac{1}{PT}} \Rightarrow \frac{1}{PS}+\frac{1}{PT}>\frac{4}{QR}$$

Exercise C - II

1. If the first 3 consecutive terms of a geometrical progression are the real roots of the equation $2x^3 - 19x^2 + 57x - 54 = 0$ find the sum to infinite number of terms of G.P.

यदि एक गुणोतर श्रेणी के प्रथम तीन क्रमागत पद समीकरण $2x^3 - 19x^2 + 57x - 54 = 0$ के वास्तविक मूल है। तो गु. श्रे. के अनन्त पदों का योग निकालिये।

Ans. $\frac{27}{2}$

Sol. $2x^3 - 19x^2 + 57x - 54 = 0$ (α, β, γ are roots)

$\alpha+\beta+\gamma = \frac{19}{2}, \ \alpha\beta+\beta\gamma+\gamma\alpha = \frac{57}{2}, \alpha\beta\gamma = \frac{54}{2}$

α,β,γ are in G.P.

$\frac{a}{r}, a, ar = \alpha.\beta, \gamma$

$\alpha\beta\gamma = a^3 = 27 \Rightarrow a = 3$

$\frac{a}{r} + ar + a = \frac{19}{2}$

$3\left(\frac{1}{r}+1+r\right) = \frac{19}{2}$

$6[1+ r^2 + r] = 19r$

$6r^2 - 13r + 6 = 0$

$6r^2 - 9r - 4r + 6 = 0$

$3r(2r-3) - 2(2r-3) = 0$

$(3r-2)(2r-3) = 0$

$r = \frac{2}{3}, r = \frac{3}{2}$ not possible

$S_\infty = \frac{\frac{a}{r}}{1-r} = \frac{\frac{3}{\left(\frac{2}{3}\right)}}{1-\frac{2}{3}} = \frac{27}{2}$

2. Find the sum of the infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + ...\infty$.

अनन्त श्रेणी $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + ...\infty$ का योग निकालिए।

Ans. 23

Sol. $S = \frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} \ldots\ldots\ldots\infty$ ..... (i)

$S\left(\frac{1}{2}\right) = \frac{1.3}{2^2} + \frac{3.5}{2^3} + \frac{5.7}{2^4} + \ldots\ldots\infty$ ..... (ii)

(i) – (ii)

$\frac{S}{2} = \frac{1.3}{2} + \frac{3.4}{2^2} + \frac{4.5}{2^3} + \frac{4.7}{2^4} \ldots\ldots\ldots\infty$ .....

$\frac{S}{2} = \frac{1\cdot3}{2} + 4\left[\frac{3}{2^2} + \frac{5}{2^3} + \frac{7}{2^4} + \ldots\ldots\ldots\right]$

$\frac{S}{2} = \frac{1\cdot3}{2} + \left[3 + \frac{5}{2} + \frac{7}{2^2} + \ldots\ldots\right]$

Let $S' = 3 + \frac{5}{2} + \frac{7}{2^2} + \ldots\ldots\ldots$ ...(iii)

$\frac{S'}{2} = 0 + \frac{3}{2} + \frac{5}{2^2} + \ldots\ldots\ldots$ ...(iv)

(iii) – (iv)

$\frac{S'}{2} = 3 + \left[\frac{2}{2} + \frac{2}{2^2} + \ldots\ldots\right]$

$= 3 + \left[1 + \frac{1}{2} + \ldots\ldots\right]$

$\frac{S'}{2} = 5 \Rightarrow S' = 10$

By equation (ii)

$\frac{S}{2} = \frac{3}{2} + 10 = \frac{23}{2}$

$S = 23$

3. One of the roots of the equation $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ is of the form $\frac{m+\sqrt{n}}{r}$, where m is non zero integer and n and r are relatively prime natural numbers. Find the value of m + n + r.

समीकरण $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ का एक मूल $\frac{m+\sqrt{n}}{r}$ के रूप में है जहाँ m अशून्य पूर्णांक तथा n एवं r सापेक्षतः अभाज्य प्राकृतिक संख्यायें है। m + n + r का मान निकालिए।

Ans. 200

Sol. $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$

$$\alpha = \frac{m+\sqrt{n}}{r} \Rightarrow \beta = \frac{m-\sqrt{n}}{r}$$

$x + 10x^3 + 100x^5 = 2 - 2000x^6$

$$\frac{x\left(\left(10x^2\right)^3 - 1\right)}{10x^2 - 1} = 2\left(1 - \left(10x^2\right)^3\right)$$

$$\frac{x\left(1 - \left(10x^2\right)^3\right)}{1 - 10x^2} = 2\left(1 - \left(10x^2\right)^3\right)$$

$$\left[1 - \left(10x^2\right)^3\right]\left[\frac{x}{1-10x^2} - 2\right] = 0$$

$1000x^6 - 1 = 0$ or $x - 2 + 20x^2 = 0$

$x^6 = \frac{1}{1000}$ or $20x^2 + x - 2 = 0$

$$x = \frac{-1 \pm \sqrt{161}}{40}$$

$\Rightarrow m = -1, n = 161, = 40$

$m + n + r = 200$

4. Let $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$. Find [S]. Where [y] denotes largest integer less than or equal to y.

माना कि $S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$ तो [S] का मान बताओं। जहाँ [y] महत्तम पूर्णांक फलन है।

Ans. 49

Sol. $$S = \sum_{n=1}^{99} \frac{5^{100}}{(25)^n + 5^{100}}$$

$$S = \sum_{n=1}^{99} \frac{1}{1 + 5^{2n-100}}$$

$$= \frac{1}{1+5^{-98}} + \frac{1}{1+5^{-96}} + \frac{1}{1+5^{-94}} + \ldots\ldots\ldots + \frac{1}{1+5^{-2}} + \frac{1}{1+5^{0}} + \frac{1}{1+5^{2}} + \ldots\ldots + \frac{1}{1+5^{98}}$$

$$=\left(\frac{1}{1+5^{-98}}+\frac{1}{1+5^{98}}\right)+\left(\frac{1}{1+5^{-96}}+\frac{1}{1+5^{96}}\right)+.....+\left(\frac{1}{1+5^{-2}}+\frac{1}{1+5^{2}}\right)+\frac{1}{1+50}$$

$$S=\underbrace{1+1+.....+1}_{49\text{ times}}+\frac{1}{2}$$

$S = 49\cdot5$

$[S] = [49\cdot5] = 49$

5. If $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ form first three terms of an A.P., then find the sum $1 + \sin 2x + \sin^2 2x + ... \infty$.

यदि $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ किसी समान्तर श्रेणी के प्रथम तीन पद है तो $1 + \sin 2x + \sin^2 2x + ... \infty$ का मान बताओं।

Ans. 2

Sol. $3^{2\sin 2x-1}$, 14, $3^{4-2\sin 2x}$ are in A.P.

$2 \times 14 = 3^{2\sin 2x-1} + 3^{4-2\sin 2x}$

$$28=\frac{3^{2\sin 2x}}{3}+\frac{3^4}{3^{2\sin 2x}}$$

$$28=\frac{t}{3}+\frac{81}{t}$$

$$28=\frac{t^2+243}{3t}$$

$t^2 - 84t + 243 = 0$

$t^2 - 81t - 3t + 243 = 0$

$t(t - 81) - 3(t - 81) = 0$

$(t - 3)(t - 81) = 0$

$t = 3, t = 81$

$3^{2\sin 2x} = 3^1$	or	$3^{2\sin 2x} = 3^4$
$2\sin 2x = 1$	or	$2\sin 2x = 4$
$\sin 2x = \frac{1}{2}$		$\sin 2x = 2$
		Not possible

$S = 1 + \sin 2x + \sin^2 2x + ...$

$$S=1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....=\frac{1}{1-\frac{1}{2}}=2$$

6. Given that the cubic $ax^3 - ax^2 + 9bx - b = 0$ $(a \neq 0)$ has all three positive roots. Find the harmonic mean of the roots independent of a and b, hence deduce that the root are all equal. Find also the minimum value of (a + b) if a and $b \in N$.

घन समीकरण $ax^3 - ax^2 + 9bx - b = 0$ $(a \neq 0)$ के तीनों मूल धनात्मक है। a एवं b का हरात्मक माध्य निकालिए तदानुसार बताइए कि सभी मूल, समान है। (a + b) का न्युनतम मान भी निकालिए यदि $a, b \in N$ हो।

Ans. 28

Sol. $ax^3 - ax^2 + 9bx - b = 0$ $(a \infty 0)$
Let α, β, γ are roots.

$$H = \frac{3}{\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}} = \frac{3\alpha\beta\gamma}{\alpha\beta+\beta\gamma+\gamma\alpha} = \frac{3\left(\frac{b}{a}\right)}{\left(\frac{ab}{a}\right)} = \frac{1}{3}$$

Applying A.M., G.M., H.M.

$$\frac{\alpha+\beta+\gamma}{3} \geq (\alpha\beta\gamma)^{\frac{1}{3}} \geq \frac{3}{\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}}$$

$$\frac{1}{3} \geq (\alpha\beta\gamma)^{\frac{1}{3}} \geq \frac{1}{3}$$

$\Rightarrow$ A.M. = G.M. = H.M.
$\Rightarrow \alpha = \beta = \gamma$

$\because \alpha + \beta + \gamma = 1 \Rightarrow \alpha = \beta = \gamma = \frac{1}{3}$

$$\left(x - \frac{1}{3}\right)^3 = x^3 - x^2 + \frac{x}{3} - \frac{1}{27} = 0$$

$27x^3 - 27x^2 + 9x - 1 = 0$
compare with (1)
$\Rightarrow a = 27, b = 1 \quad \Rightarrow a + b = 28$

7. The sequence $a_1, a_2, a_3, \ldots\ldots a_{98}$ satisfies the relation $a_{n+1} = a_n + 1$ for $n = 1, 2, 3, \ldots\ldots 97$ and has the sum equal to 4949. Evaluate $\sum_{k=1}^{49} a_{2k}$.

श्रेणी $a_1, a_2, a_3, \ldots\ldots a_{98}$ सम्बन्ध $a_{n+1} = a_n + 1$ को $n = 1, 2, 3, \ldots\ldots 97$ के लिये संतुष्ट करती है एवं इसका योग 4949 है।

$\sum_{k=1}^{49} a_{2k}$ का मान निकालिए।

Ans. 2499

Sol. $S = a_1 + a_2 + a_3 + \ldots\ldots + a_{98} = 4949$
$a_{n+1} = a_n + 1 \Rightarrow d = a_{n+1} - a_n = 1$

$$S = \frac{98}{2}\left[2a_1 + (98-1)\cdot 1\right] = 4949$$

$\Rightarrow 2a + 97 = 101$

$2a = 4$

$\Rightarrow a = 2$

$$S' = \sum_{k=1}^{49} a_{2k} = \left[a_2 + a_4 + a_6 + ... + a_{98}\right]$$

$= [3 + 5 + 7 + ... + 99]$

$$= \frac{49}{2}\left[2\times 3 + (49-1)\times 2\right]$$

$= 2499$

8. The value of $x + y + z$ is 15 if a, x, y, z, b are in AP while the value of; $(1/x)+(1/y)+(1/z)$ is 5/3 if a, x, y, z, b are in HP. Find a and b.

$x + y + z$ का मान 15 है, यदि a, x, y, z, b AP में हों; जबकि $(1/x)+(1/y)+(1/z)$ का मान 5/3 है यदि a, x, y, z, b ह.श्रे. में हो तो a एवं b का मान निकालिये।

Ans. $a = 1, b = 9$ or $b = 1, a = 9$

Sol. $x + y + z = 15$ .....(i)

a, x, y, z, b are in AP

Suppose d is common difference $d = \frac{b-a}{4}$

$\therefore$ $x = a + \frac{b-a}{4} = \frac{b+3a}{4}$, $y = \frac{2b+2a}{4}$ and $z = \frac{3b+a}{4}$

on substituting the values of x, y and z in (i), we get

$\Rightarrow$ $\frac{6a+6b}{4} = 15$

$a + b = 10$ ..... (ii)

$\because$ $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{5}{3}$ .....(iii)

and a, x, y, z, b are in H.P. $\therefore$ $\frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b}$ are in A.P.

$\therefore$ $\frac{1}{x} = \frac{1}{a} + \frac{\left(\frac{1}{b} - \frac{1}{a}\right)}{4}$, $\frac{1}{y} = \frac{1}{a} + \frac{2}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$ and $\frac{1}{z} = \frac{1}{a} + \frac{3}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$

on substituting the value of $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ in (iii), we get

$$\frac{3}{a}+\frac{6}{4}\left(\frac{1}{b}-\frac{1}{a}\right)=\frac{5}{3} \quad \Rightarrow \quad \frac{3}{2a}+\frac{3}{2b}=\frac{5}{3} \Rightarrow \frac{1}{a}+\frac{1}{b}=\frac{10}{9} \quad \text{..... (iv)}$$

By equations (ii) and (iv), we get

$a = 9, b = 1$ or $a = 1, b = 9$

Hindi. $x + y + z = 15$ .....(i)

a, x, y, z, b समान्तर श्रेढ़ी में है माना d सार्वअन्तर है

$$d = \frac{b-a}{4} \quad \therefore \quad x = a + \frac{b-a}{4} = \frac{b+3a}{4}, \quad y = \frac{2b+2a}{4} \quad \text{and} \quad z = \frac{3b+a}{4}$$

समीकरण (i) में x, y और z का मान रखने पर

$$\Rightarrow \quad \frac{6a+6b}{4} = 15$$

$$\Rightarrow \quad a + b = 10 \quad \text{..... (ii)} \quad \because \quad \frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{5}{3} \quad \text{.....(iii)}$$

तथा a, x, y, z, b हरात्मक श्रेढ़ी में है

$$\therefore \quad \frac{1}{a},\frac{1}{x},\frac{1}{y},\frac{1}{z},\frac{1}{b} \text{ समान्तर श्रेढ़ी में है}$$

$$\therefore \quad \frac{1}{x}=\frac{1}{a}+\frac{\left(\frac{1}{b}-\frac{1}{a}\right)}{4}, \quad \frac{1}{y}=\frac{1}{a}+\frac{2}{4}\left(\frac{1}{b}-\frac{1}{a}\right) \text{ तथा } \frac{1}{z}=\frac{1}{a}+\frac{3}{4}\left(\frac{1}{b}-\frac{1}{a}\right)$$

समीकरण (iii) में $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ का मान रखने पर

$$\frac{3}{a}+\frac{6}{4}\left(\frac{1}{b}-\frac{1}{a}\right)=\frac{5}{3} \quad \Rightarrow \quad \frac{3}{2a}+\frac{3}{2b}=\frac{5}{3} \Rightarrow \frac{1}{a}+\frac{1}{b}=\frac{10}{9} \quad \text{..... (iv)}$$

समीकरण (ii) और (iv) से हमें प्राप्त होता है

$a = 9, b = 1$ या $a = 1, b = 9$

9. The values of xyz is 15/2 or 18/5 according as the series a, x, y, z, b is an AP or HP. Find the values of a and b assuming them to be positive integer.

xyz के मान 15/2 या 18/5 है जब a, x, y, z, b AP अथवा HP में क्रमशः है a, b को धनात्मक पूर्णांक मानते हुए इनका मान निकालिये।

Ans. $a = 1$; $b = 3$ or vice versa

Sol. If a, x, y, z, b are in A.P. then $xyz = \frac{15}{2}$

If a, x, y,z, b are in H.P. then $xyz = \frac{18}{5}$

$b = a + 4d \Rightarrow \left(\frac{b-a}{4}\right) = d$

$xyz = (a + d)(a + 2d)(a + 3d)$

$= \left(a + \frac{b-a}{4}\right)\left(a + \frac{2b-2a}{4}\right)\left(a + \frac{3b-3a}{4}\right)$

$xyz = \frac{(3a+b)(2a+2a)(a+3b)}{64} = \frac{15}{2}$ ...(1)

If a, x, y, z, b are in H.P.

$\frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b}$ are in A.P.

$\left(\frac{1}{b} - \frac{1}{a}\right) = 4d' \Rightarrow d' = \frac{1}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$

$\frac{1}{xyz} = \left(\frac{1}{a} + d'\right)\left(\frac{1}{a} + 2d'\right)\left(\frac{1}{a} + 3d'\right)$

$= \left[\frac{1}{a} + \frac{1}{4}\left(\frac{1}{b} - \frac{1}{a}\right)\right]\left[\frac{1}{a} + \frac{2}{4}\left(\frac{1}{b} - \frac{1}{a}\right)\right]\left[\frac{1}{a} + \frac{3}{4}\left(\frac{1}{b} - \frac{1}{a}\right)\right]$

$\frac{5}{18} = \left[\frac{3b+a}{4ab}\right]\left[\frac{2a+2b}{4ab}\right]\left[\frac{b+3a}{4ab}\right]$ ...(2)

$\frac{(1)}{(2)} : \frac{\left(\frac{15}{2}\right)}{\left(\frac{5}{18}\right)} = \frac{\left(\frac{(3a+b)(2a+2b)(a+3b)}{64}\right)}{\frac{(3a+b)(2a+2b)(a+3b)}{64(ab)^3}}$

$\frac{15}{2} \times \frac{18}{5} = (ab)^3 \Rightarrow ab = 3$

Since, a, b are positive integers.

So, a = 1, b = 3 or a = 3, b = 1

10. If the roots of $10x^3 - cx^2 - 54x - 27 = 0$ are in harmonic progression, then find *c* and all the roots.

समीकरण $10x^3 - cx^2 - 54x - 27 = 0$ के मूल यदि ह. श्रे. में है तो c का मान निकालिए एवं सभी मूल निकालिए।

Ans. C = 9 ;(3, –3/2 , -3/5)11

Sol. $10x^3 - cx^2 - 54x - 27 = 0$

let α, β, γ are roots

$$\alpha+\beta+\gamma = \frac{c}{10} \qquad \alpha\beta+\beta\gamma+\gamma\alpha = \frac{-54}{10} \qquad \alpha\beta\gamma = \frac{27}{10}$$

α, β, γ are in H.P.

$$\Rightarrow B = \frac{2\alpha\gamma}{\alpha+\gamma} \Rightarrow \alpha+\gamma = \frac{2\alpha\gamma}{B}$$

$$\alpha\beta+\beta\gamma+\gamma\alpha = \frac{-54}{10}$$

$$B(\alpha+\gamma)+\alpha\gamma = \frac{-54}{10}$$

$$B\left(\frac{2\gamma\alpha}{B}\right)+\gamma\alpha = \frac{-54}{10}$$

$$3\gamma\alpha = \frac{-54}{10} \Rightarrow \gamma\alpha = \frac{-18}{10} \Rightarrow B = \frac{-3}{2}$$

$$\alpha\beta+\beta\gamma+\gamma\alpha = \frac{-54}{10}$$

$$B(\alpha+\gamma)-\frac{18}{10} = \frac{-54}{10}$$

$$B(\alpha+\gamma) = \frac{-36}{10}$$

$$(\alpha+\gamma) = \frac{12}{5}$$

$x^2 - (\alpha + \gamma)\, x + \alpha\gamma = 0$

$10x^2 - 24x - 18 = 0 \Rightarrow (10x + 6)(x - 3) = 0$

$$x = \frac{-6}{10},\ x = 3$$

$$x = \frac{-3}{5},\ x = 3$$

let $\alpha = \frac{-3}{5}, \alpha = 3$

$$\alpha+\beta+\gamma = \frac{+c}{10}$$

$$\frac{-3}{5}+\left(\frac{-3}{2}\right)+3 = \frac{c}{10}$$

$$\frac{-6-15+30}{10} = \frac{c}{10}$$

$c = 9$

11. In a GP the ratio of the sum of the first eleven terms to the sum of the last eleven terms is 1/8 and the ratio of the sum of all the terms without the first nine to the sum of all the terms without the last nine is 2. Find the number of terms in the GP.

एक गु. श्रे. में प्रथम ग्यारह पदों के योग एवं अन्तिम ग्यारह पदों के योग का अनुपात 1/8 है तथा प्रथम नौ पदों को छोड़कर अन्य सभी पदों के योग एवं अन्तिम नौ पदों को छोड़कर अन्य सभी पदों के योग का अनुपात 2 है। गु. श्रे. में पदों की संख्या बताइए।

Ans. $n = 38$

Sol. $$\frac{a_1 + a_2 + ... + a_{11}}{a_{n-10} + a_{n-9} + ... + a_n} = \frac{1}{8}$$

$$\frac{a + ar + ... + ar^{10}}{ar^{n-11} + ar^{n-10} + ... + ar^{n-1}} = \frac{1}{8}$$

$$\frac{a\left(\frac{r^{11}-1}{r-1}\right)}{ar^{n-11}\left(\frac{r^{11}-1}{r-1}\right)} = \frac{1}{8}$$

$r^{n-11} = 2^3$ ...(1)

$$\frac{S_n - (a_1 + a_2 + ... + a_9)}{S_n - (a_{n-8} + a_{n-7} + ... + a_n)} = \frac{2}{1}$$

$$\frac{a\left(\frac{r^n-1}{r-1}\right) - a\left(\frac{r^9-1}{r-1}\right)}{a\left(\frac{r^n-1}{r-1}\right) - \left(\left(a\left(r^{n-9}\right)\right)\left(\frac{r^9-1}{r-1}\right)\right)} = \frac{2}{1}$$

$$\frac{r^n - r^9}{-1 + r^{n-9}} = \frac{2}{1}$$

$$r^9\left(\frac{r^{n-9}-1}{r^{n-9}-1}\right) = 2$$

$r^9 = 2$ ...(2)

By (1), (2)

$r^{n-11} = 2^3 = (r^9)^3$

$r^{n-11} = r^{27} \Rightarrow n = 27 + 11 = 38$

12. Let a_n be a sequence such that $a_1 = 3,\ a_{n+1} = 3a_n + 1$ $(n = 1, 2, 3, ...)$. If the value of $\sum_{n=1}^{\infty} \frac{a_n}{5^n} = \frac{p}{q}$. (where p and q are in their lowest form), then find the value of $(p + q)$.

यदि a_n श्रृंखला (sequence) इस प्रकार से है कि $a_1 = 3,\ a_{n+1} = 3a_n + 1$ $(n = 1, 2, 3, ...)$ यदि $\sum_{n=1}^{\infty} \frac{a_n}{5^n} = \frac{p}{q}$ तो $p + q$ का मान बताओं जहाँ p व q का महत्तम समापवर्तक 1 है।

Ans. 21

Sol. $a_1 = 3,\ a_{n+1} = 3a_{n+1}$

$a_1, a_2, a_3, a_4, ...$

$3, 3^2 + 1, 3^3 + 4, 3^4 + 13, ...$

$$\sum_{n=1}^{\infty} \frac{a_n}{5^n} = \frac{3}{5} + \frac{3^2+1}{5^2} + \frac{3^3+4}{5^3} + \frac{3^4+13}{5^4} + \frac{3^5+40}{5^5} + ...$$

$$= \underbrace{\left[\frac{3}{5} + \left(\frac{3}{5}\right)^2 + \left(\frac{3}{5}\right)^3 + \left(\frac{3}{5}\right)^4 + \left(\frac{3}{5}\right)^5 + ...\right]}_{S_1} \underbrace{\left[\frac{1}{5^2} + \frac{4}{5^3} + \frac{13}{5^4}\right]}_{S_2}$$

$$S_1 = \frac{\frac{3}{5}}{1 - \frac{3}{5}} = \frac{3}{2}$$

$$S_2 = \frac{1}{5^2} + \frac{4}{5^3} + \frac{13}{5^4} + \frac{40}{5^5} + ... \quad ...(1)$$

$$\frac{S_2}{5} = 0 + \frac{1}{5^3} + \frac{4}{5^4} + \frac{13}{5^5} + ... \quad ...(2)$$

$(1) - (2)$:

$$\frac{4S_2}{5} = \left(\frac{1}{5^2}\right) + \frac{3}{5^3} + \frac{9}{5^4} + \frac{27}{5^5} + ...$$

$$\frac{4S_2}{5} = \frac{1}{25} + \frac{1}{9}\left[\left(\frac{3}{5}\right)^3 + \left(\frac{3}{5}\right)^4 + \left(\frac{3}{5}\right)^5 + ...\right]$$

$$= \frac{1}{25} + \frac{1}{9}\left[\frac{\left(\frac{3}{5}\right)^3}{1 - \frac{3}{5}}\right]$$

$$\frac{4S_2}{5}=\frac{1}{25}+\frac{1}{9}\left[\frac{27\times5}{125\times2}\right]=\frac{1}{10}$$

$$S_2=\frac{1}{8}$$

$$S=\frac{1}{8}+\frac{3}{2}=\frac{13}{8}=\frac{p}{q}\Rightarrow p+q=21$$

13. Solve the following equations for x and y

$$\log_2 x+\log_4 x+\log_{16} x+\ldots\ldots\ldots\ldots=y$$

$$\frac{5+9+13+\ldots\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots\ldots\ldots+(2y-1)}=4\log_4 x$$

x तथा y के लिये निम्नलिखित समीकरणों को हल कीजिये।

$$\log_2 x+\log_4 x+\log_{16} x+\ldots\ldots\ldots\ldots=y$$

$$\frac{5+9+13+\ldots\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots\ldots+(2y-1)}=4\log_4 x$$

Ans. $x=2\sqrt{2}$ and $y=3$

Sol. $\log_2 x+\log_4 x+\log_{16} x+\ldots\ldots=y$

$$\log_2 x+1\log_2 x+\frac{1}{4}\log_2 x+\ldots\ldots=y$$

$$\log_2 x+\log_2 x^{\frac{1}{2}}+\log_2 x^{\frac{1}{4}}+\ldots\ldots=y$$

$$\log_2 x^{\left(1+\frac{1}{2}+\frac{1}{4}+\ldots\ldots\right)}=y$$

$$\left(\log_2 x\right)\left(\frac{1}{1-\frac{1}{2}}\right)=y$$

$$2\log_2 x=y \qquad \text{_____(1)}$$

$$\frac{5+9+13+\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots+(2y-1)}=4\log_4 x=2\log_2 x=y \qquad \text{(By (1))}$$

$$\frac{5+9+13+\ldots\ldots+(4y+1)}{1+3+5+\ldots\ldots+(2y-1)}=y \qquad \text{_____(2)}$$

$S = 5 + 9 + 13 + \ldots\ldots\ldots + (4y + 1) = \frac{y}{2}\left[2\times 5 + (y-1)\times 4\right] = y\,(2y + 3)$

$S' = 1 + 3 + 5 + \ldots\ldots\ldots + (2y - 1) = \frac{y}{2}\left[2\times 1 + (y-1)\times 2\right] = y^2$

By equation (2)

$$\frac{y(3+2y)}{y^2} = y$$

$y^2 - 2y - 3 = 0\ (y \neq 0)$

$$y = \frac{2 \pm \sqrt{4+12}}{2}$$

$$y = \frac{2 \pm 4}{2}$$

$y = 3, -1$ (reject)

$\Rightarrow y = 3$

$2\log_2 x = y = 3$

$$x = (2)^{\frac{3}{2}}$$

$$x = 2\sqrt{2}$$

14. Find the sum in the nth group of sequence :

निम्न अनुक्रमों के n वें समूह का योगफल ज्ञात कीजिए :

(i) 1, (2, 3); (4, 5, 6, 7); (8, 9,....., 15); ..... (ii) (1), (2, 3, 4), (5, 6, 7, 8, 9),.....

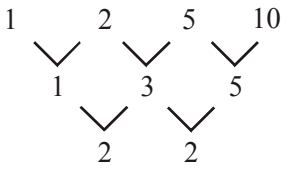
Ans. (i) $2^{n-2}\,(2^n + 2^{n-1} - 1)$ (ii) $(n-1)^3 + n^3$

Sol. (i) 1, (2, 3), (4, 5, 6, 7), (8, 9, ..., 15) ....

Number of terms in n^{th} group $= 2^{n-1}$; 1st term in n^{th} group $= 2^{n-1}$

So Sum of terms in n^{th} group $= \frac{2^{n-1}}{2}[2.2^{n-1} + (2^{n-1} - 1)\ 1] = 2^{n-2}\,[2^n + 2^{n-1} - 1]$

(ii) (1) , (2, 3, 4), (5, 6, 7, 8, 9) ; 1st term in n^{th} group let T_n



$T_n = a + bn + cn^2 \Rightarrow T_1 = a + b + c = 1$ .... (i)

$T_2 = a + 2b + 4c = 2$ .... (ii)

and $T_3 = a + 3b + 9c = 5$ .... (iii)

$a = 2, b = -2, c = 1$

On solving (i), (ii) and (iii), we get

So I^{st} term is $T_n = (2 - 2n + n^2)$. Number of terms in n^{th} group $= (2n - 1)$

$\therefore$ sum of terms in n^{th} group $= \frac{2n-1}{2}\,[2\,(2 - 2n + n^2) + 2n - 2]$

$= (2n - 1)\,[n^2 - n + 1] = 2n^3 - 2n^2 + 2n - n^2 + n - 1 = n^3 + (n - 1)^3$

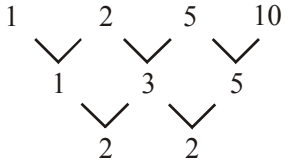
Hindi (i) 1, (2, 3), (4, 5, 6, 7), (8, 9, ..., 15) ....

n वें समूह में पदों की संख्या $= 2^{n-1}$

n वें समूह का प्रथम पद $= 2^{n-1}$

इसलिए n वें समूह के पदों का योग $= \frac{2^{n-1}}{2}\,[2.2^{n-1} + (2^{n-1} - 1)\,1] = 2^{n-2}\,[2^n + 2^{n-1} - 1]$

(ii) (1) , (2, 3, 4), (5, 6, 7, 8, 9) माना n वें समूह का प्रथम पद T_n है



$T_n = a + bn + cn^2 \Rightarrow T_1 = a + b + c = 1$ .... (i)

$T_2 = a + 2b + 4c = 2$ .... (ii)

और $T_3 = a + 3b + 9c = 5$ .... (iii)

$a = 2, b = -2, c = 1$

(i), (ii) और (iii) को हल करने पर

इसलिए प्रथम पद $T_n = (2 - 2n + n^2)$ $\Rightarrow$ n वें समूह में पदों की संख्या $= (2n - 1)$

$\therefore$ n वें समूह के पदों का योग $= \frac{2n-1}{2}\,[2\,(2 - 2n + n^2) + 2n - 2] = (2n - 1)\,[n^2 - n + 1]$

$= 2n^3 - 2n^2 + 2n - n^2 + n - 1 = n^3 + (n - 1)^3$

15. If a, b and c are in arithmetic progression and a^2, b^2 & c^2 are in harmonic progression, then prove that either $a = b = c$ or a, b and $-\frac{c}{2}$ are in geometric progression.

यदि a, b, c स.श्रे. में हैं और a^2, b^2, c^2 ह.श्रे. में हो, तो सिद्ध कीजिए कि या तो $a = b = c$ या a, b, $-\frac{c}{2}$ गु.श्रे. में हैं।

Ans.

Sol. a, b, c are in A.P. $\Rightarrow 2b = a + c$

a^2, b^2, c^2 are in H.P. $\Rightarrow b^2 = \frac{2a^2c^2}{a^2 + c^2}$

$$\left(\frac{a+c}{2}\right)^2 = \frac{2a^2c^2}{a^2+c^2}$$

$(a^2 + c^2)\,[(a^2 + c^2) + 2ac] = 8a^2c^2$

$(a^2 + c^2)^2 + 2a^3c + 2a^2c^3 = 8a^2c^2$

$a^4 + c^4 + 2a^3c + 2a^2c^3 - 6a^2c^2 = 0$

$[(a^2)^2 + (c^2)^2 - 2a^2c^2] + 2ac\,[a^2 + c^2 - 2ac] = 0$

$(a^2 - c^2)^2 + 2ac\,[a - c]^2 = 0$

$(a - c)^2\,[(a + c)^2 + 2ac] = 0$

$(a - c)^2\,[a^2 + 4ac + c^2] = 0$

$a - c = 0$ or $a^2 + 4ac + c^2 = 0$

$a = c$ or $(a^2 + 2ac + c^2) + 2ac = 0$

$(a + c)^2 + 2ac = 0$

$(2b)^2 = -2ac$

$$b^2 = (a)\left(\frac{-c}{2}\right)$$

a, b, $-\frac{c}{2}$ are in G.P.

16. Let 'X' denotes the value of the product $(1 + a + a^2 + a^3 + ... \infty)(1 + b + b^2 + b^3 + ... \infty)$ where 'a' and 'b' are the roots of the quadratic equation $11x^2 - 4x - 2 = 0$ and 'Y' denotes the numerical value of the infinite series

$$(\log_b 2)^0 \left(\log_b 5^{4^0}\right) + (\log_b 2)^1 \left(\log_b 5^{4^1}\right) + (\log_b 2)^2 \left(\log_b 5^{4^2}\right) + (\log_b 2)^3 \left(\log_b 5^{4^3}\right) + ...\infty$$

where b = 2000. Find (XY).

दिया है कि 'X' श्रेणियों

$(1 + a + a^2 + a^3 + ... \infty)(1 + b + b^2 + b^3 + ... \infty)$

के गुणन को व्यक्त करता है जहाँ 'a' और 'b' द्विघात समी. $11x^2 - 4x - 2 = 0$ के मूल है तथा 'Y' श्रेणी

$$(\log_b 2)^0 \left(\log_b 5^{4^0}\right) + (\log_b 2)^1 \left(\log_b 5^{4^1}\right) + (\log_b 2)^2 \left(\log_b 5^{4^2}\right) + (\log_b 2)^3 \left(\log_b 5^{4^3}\right) + ...\infty$$

के आंकिक मान को व्यक्त करता है जहाँ b = 2000 तब XY का मान ज्ञात कीजिए।

Ans. $\frac{11}{15}$

Sol. $X = (1 + a + a^2 + ...)\,(1 + b + b^2 + ...) = \left(\frac{1}{1-a}\right)\left(\frac{1}{1-b}\right)$

$11x^2 - 4x - 2 = 0$

Now we form a quadratic equation whose roots are $\frac{1}{1-a}, \frac{1}{1-b}$

let x = a

$$y = \frac{1}{1-a} = \frac{1}{1-x}$$

$$x = \frac{y-1}{y}$$

$$11\left(\frac{y-1}{y}\right)^2 - 4\left(\frac{y-1}{y}\right) - 2 = 0$$

$5y^2 - 18y + 11 = 0$

$$X = \left(\frac{1}{1-a}\right)\left(\frac{1}{1-b}\right) = \frac{11}{5}$$

$$Y = (\log_b 2)^0 \left(\log_b 5^{4^0}\right) + (\log_b 2)^1 \left(\log_b 5^{4^1}\right) + (\log_b 2)^2 \left(\log_b 5^{4^2}\right) + ...$$

$$Y = \log_b 5 + 4(\log_b 2)^1 (\log_b 5) + 4^2 (\log_b 2)(\log_b 5) + ...$$

$$Y = \frac{\log_b 5}{1 - 4\log_b 2} = \frac{\log_2 b \times \log_b 5}{\log_2 b - 4} = \frac{\log_2 5}{\log_2 \frac{b}{16}}$$

At b = 2000

$$Y = \frac{\log_2 5}{\log_2 \frac{2000}{16}} = \frac{\log_2 5}{3\log_2 5} = \frac{1}{3}$$

$$XY = \left(\frac{11}{5}\right)\left(\frac{1}{3}\right) = \frac{11}{15}$$

17. The sum of the squares of three distinct real numbers, which are in GP is S^2. If their sum is αS, show that $\alpha^2 \in (1/3, 1) \cup (1, 3)$.

तीन भिन्न वास्तविक संख्याओं, जो गु.श्रे. में है, का योग αS तथा उनके वर्गों का योग S^2 है तो सिद्ध कीजिये $\alpha^2 \in (1/3, 1) \cup (1, 3)$.

Ans.

Sol. $a + b + c = \alpha S$, $a^2 + b^2 + c^2 = S^2$

$a^2 + b^2 + c^2 + 2(ab + bc + ca) = \alpha^2 (a^2 + b^2 + c^2)$

$2(ab + bc + ca) = (\alpha^2 - 1)(a^2 + b^2 + c^2)$

a, b, c are in G.P. $\Rightarrow$ a, ar, ar^2

$2a^2 (r + r^2 + r^3) = a^2 (\alpha^2 - 1)(1 + r^2 + r^4)$

$2r(1 + r + r^2) = (\alpha^2 - 1)(1 + r + r^2)(1 - r + r^2)$

$$\frac{2}{1 - r + r^2} + 1 = \alpha^2$$

let $\alpha^2 = y$

$$y = \frac{1+r+r^2}{1-r+r^2}$$

$(y-1)\,r^2 + (-y-1)r + (y-1) = 0$

Case–1 $\quad y \neq 1 \qquad D \geq 0$

$(1+y)^2 - 4(y-1)^2 \geq 0$

$y^2 + 3y - 1 \geq 0$

$$y \in \left(\frac{1}{3}, 3\right)$$

Case–2 $\quad y = 1 \Rightarrow \alpha^2 = 1$

$1 + r + r^2 = 1 - r + r^2$

$2r = 0$

$\Rightarrow r = 0$

But $r \neq 0 \qquad (\because$ a, b, c are in G.P.)

Hence $y = \alpha^2 \in \left[\frac{1}{3}, 1\right) \cup (1, 3]$

18. Find the sum to 'n' terms of the series

$$\frac{3}{1.2}\cdot\frac{1}{2} + \frac{4}{2.3}\cdot\frac{1}{2^2} + \frac{5}{3.4}\cdot\frac{1}{2^3} + \frac{6}{4.5}\cdot\frac{1}{2^4} + \ldots\ldots\ldots$$

श्रेणी $\frac{3}{1.2}\cdot\frac{1}{2} + \frac{4}{2.3}\cdot\frac{1}{2^2} + \frac{5}{3.4}\cdot\frac{1}{2^3} + \frac{6}{4.5}\cdot\frac{1}{2^4} + \ldots\ldots\ldots$ के n पदों का योग ज्ञात कीजिये।

Ans. $\left(1 - \frac{1}{(n+1)\,2^n}\right)$

Sol. $S = \frac{3}{1\cdot2}\cdot\frac{1}{2} + \frac{4}{2\cdot3}\cdot\frac{1}{2^2} + \frac{5}{3\cdot4}\cdot\frac{1}{2^3} + \frac{6}{4\cdot5}\cdot\frac{1}{2^4} + \ldots\ldots$

$$T_n = \frac{(n+2)}{n(n+1)}\frac{1}{2^n}$$

$$T_n = \left[\frac{(2n+2)-n}{n(n+1)}\right]\frac{1}{2^n}$$

$$= \left[\left(\frac{2}{n}\right) - \left(\frac{1}{n+1}\right)\right]\frac{1}{2^n}$$

$$T_n = \frac{1}{n\cdot 2^{n-1}} - \frac{1}{(n+1)2^n}$$

$$S_n = \sum_{r=1}^{n}\left[\frac{1}{r\cdot 2^{r-1}} - \frac{1}{(r+1)2^r}\right]$$

$$S_n = \left(\frac{1}{1\cdot 2^{1-1}} - \frac{1}{2\cdot 2^1}\right) + \left(\frac{1}{2\cdot 2} - \frac{1}{3\cdot 2^2}\right) + ... + \left(\frac{1}{n\cdot 2^{n-1}} - \frac{1}{(n+1)2^n}\right)$$

$$S_n = 1 - \frac{1}{(n+1)2^n}$$

19. Find the sum of infinite terms of the series : $\frac{3}{2.4} + \frac{5}{2.4.6} + \frac{7}{2.4.6.8} + \frac{9}{2.4.6.8.10} + ...$

श्रेणी $\frac{3}{2.4} + \frac{5}{2.4.6} + \frac{7}{2.4.6.8} + \frac{9}{2.4.6.8.10} + ...$ के अनन्त पदों का योग ज्ञात कीजिये।

Ans. 1/2

Sol. $S = \frac{3}{2\cdot 4} + \frac{5}{2\cdot 46} + \frac{7}{2\cdot 4\cdot 6\cdot 8} + \frac{9}{2\cdot 4\cdot 6\cdot 8\cdot 10} + ...$

$$T_r = \frac{2r+1}{2\cdot 4.....(2r+2)} = \frac{(2r+2)-1}{2\cdot 4.....(2r+2)}$$

$$T_r = \frac{1}{2\cdot 4\cdot .....2r} - \frac{1}{2\cdot 4\cdot .....(2r+2)}$$

$$S_\infty = \left(\frac{1}{2} - \frac{1}{2\cdot 4}\right) + \left(\frac{1}{2\cdot 4} - \frac{1}{2\cdot 4\cdot 6}\right) + \left(\frac{1}{2\cdot 4\cdot 6} - \frac{1}{2\cdot 4\cdot 6\cdot 8}\right) + ...$$

$$= \frac{1}{2}$$

20. Let S_k, k = 1, 2, ..., 100, denote the sum of the infinite geometric series whose first term is $\frac{k-1}{k!}$ and the common ratio is $\frac{1}{k}$. Then the value of $\frac{100^2}{100!} + \sum_{k=1}^{100}\left|\left(k^2 - 3k + 1\right)S_k\right|$, is.

यदि S_k किसी अनन्त गुणोत्तर श्रेणी का योग है जहाँ k = 1, 2, ..., 100, जिसका प्रथम पद $\frac{k-1}{k!}$ तथा सर्वानुपात $\frac{1}{k}$ है तो

$\frac{100^2}{100!}+\sum_{k=1}^{100}\left|\left(k^2-3k+1\right)S_k\right|$ का मान बताओं।

Ans. 3

Sol. $S_k=\frac{\left(\frac{k-1}{k!}\right)}{1-\frac{1}{k}}=\frac{\frac{k-1}{k!}}{\frac{(k-1)}{k}}=\frac{1}{(k-1)!}$

$$A=\frac{(100)^2}{100!}+\sum_{k=1}^{100}\left|\left(k^2-3k+1\right)S_k\right| \qquad ...(1)$$

$$=\frac{100^2}{100!}+\sum_{k=1}^{100}\left|\left(k^2-3k+1\right)\frac{1}{(k-1)!}\right|$$

$$=\frac{1}{99!}+\frac{1}{98!}+\sum_{k=1}^{100}\left|\frac{(k-2)(k-1)-1}{(k-1)!}\right|$$

$$=\frac{1}{99!}+\frac{1}{98!}+\sum_{k=3}^{100}\left(\frac{1}{(k-3)!}-\frac{1}{(k-1)!}\right)+(1-3+1)\times\frac{1}{(1-1!)}+(4-6+2)\times\frac{1}{(2-1)!}$$

$$=\frac{1}{99!}+\frac{1}{98!}+1+1-\frac{1}{99!}-\frac{1}{98!}+1+0$$

$$= 1 + 1 + 1 = 3$$

21. If a, b, c are positive real number such that $\frac{\log a}{b-c}=\frac{\log b}{c-a}=\frac{\log c}{a-b}$ then prove that $a^{b+c}+b^{c+a}+c^{a+b}\geq 3$.

यदि a, b, c धनात्मक वास्तविक संख्याऐं है इस प्रकार कि $\frac{\log a}{b-c}=\frac{\log b}{c-a}=\frac{\log c}{a-b}$ तो सिद्ध कीजिये कि

$a^{b+c}+b^{c+a}+c^{a+b}\geq 3$.

Ans.

Sol. $\frac{\log a}{b-c}-\frac{\log b}{c-a}=\frac{\log c}{a-b}=k$

$(\log a)(c-a)=(b-c)(\log b)$

$[a\log a+b\log b]=c[\log a+\log b]$

$\log a^a b^b = \log (ab)^c$

$a^a b^b = a^c b^c$

Similarly

$b^a c^a = b^b c^c$

and

$a^a c^a = a^b c^b$

$$\frac{a^{b+c} + b^{c+a} + c^{a+b}}{3} \geq \left(a^{b+c} \cdot b^{c+a} \cdot c^{a+b}\right)^{\frac{1}{2}}$$

$$= \left[\left(a^b c^b\right)\left(a^c b^c\right)\left(b^a c^a\right)\right]^{\frac{1}{2}}$$

$$= \left[\left(a^a c^a\right)\left(a^a b^b\right)\left(b^b c^c\right)\right]^{\frac{1}{2}}$$

$= a^a \cdot b^b c^c$

$\Rightarrow a^{b+c} + b^{c+a} + c^{a+b} \geq 3(a^a b^b c^c)$ _____(1)

$\log a = k\,(b - c)$

$a = e^{k(b-c)}$, $b = e^{k(c-a)}$, $c = e^{k(a-b)}$

$a^a b^b c^c = e^{k[ab - ac + bc - ab + ac - cb]} = 1$

Put value in (1)

$a^{b+c} + b^{c+a} + c^{a+b} \geq 3$

22. If positive square root of, $a^{\frac{1}{a}} \cdot (2a)^{\frac{1}{2a}} \cdot (4a)^{\frac{1}{4a}} \cdot (8a)^{\frac{1}{8a}} \ldots\ldots\ldots\ldots\infty$ is $\frac{8}{27}$, find the value of 'a'.

यदि $a^{\frac{1}{a}} \cdot (2a)^{\frac{1}{2a}} \cdot (4a)^{\frac{1}{4a}} \cdot (8a)^{\frac{1}{8a}} \ldots\ldots\ldots\ldots\infty$ का धनात्मक वर्ग मूल $\frac{8}{27}$ है तो 'a' का मान ज्ञात कीजिये।

Ans. $a = \frac{1}{3}$

Sol. $A = \left(a^{\frac{1}{a}}\right)(2a)^{\frac{1}{2a}}(4a)^{\frac{1}{4a}} \cdot (8a)^{\frac{1}{8a}} \ldots\ldots\ldots$

$$= a^{\left(\frac{1}{a} + \frac{1}{2a} + \frac{1}{4a} + \frac{1}{8a} + \ldots\ldots\right) \cdot 2\left(\frac{1}{2a} + \frac{2}{4a} + \frac{3}{8a} + \ldots\ldots\ldots\right)}$$

$$= a^{\frac{1}{a}\left[\frac{1}{1-\frac{1}{2}}\right] \cdot 2^{\frac{1}{a}\left[\frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots\ldots\right]}} \quad \text{_____(1)}$$

$$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots\ldots\ldots$$

$$\frac{S}{2}=0+\frac{1}{2^2}+\frac{2}{2^3}+.....$$

$$\frac{S}{2}=\frac{1}{2}+\left(\frac{1}{2^2}+\frac{1}{2^3}+.....\right)$$

$$S=1+\frac{1}{2}\left[1+\frac{1}{2}+.....\right]$$

$$=1+\frac{1}{2}\left[\frac{1}{1-\frac{1}{2}}\right]$$

$S = 1 + 1 = 2$

By (1)

$$A=a^{\frac{2}{a}}\cdot 2^{\frac{1}{a}\times 2}$$

$$A=a^{\frac{2}{a}}\cdot 2^{\frac{2}{a}}=(2a)^{\frac{2}{a}}$$

$$\sqrt{A}=(2a)^{\frac{1}{a}}=\left(\frac{8}{27}\right)$$

$$a=\frac{1}{3}$$

23. Find the sum of the series , $\frac{n}{1\,.\,2\,.\,3}+\frac{n-1}{2\,.\,3\,.\,4}+.....+\frac{1}{n\,(n+1)\,(n+2)}$.

श्रेणी $\frac{n}{1\,.\,2\,.\,3}+\frac{n-1}{2\,.\,3\,.\,4}+.....+\frac{1}{n\,(n+1)\,(n+2)}$ का मान ज्ञात कीजिये।

Ans. $S=\frac{n\,(n+1)}{4\,(n+2)}$

Sol. $\frac{n}{1.2.3}+\frac{n-1}{2.3.4}+.....+\frac{1}{n(n+1)(n+2)}$; $S_n=\sum_{r=1}^{n}\frac{n+1}{r(r+1)(r+2)}-\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}$

Now अब $\sum_{r=1}^{n}\frac{1}{r(r+1)(r+2)}=\sum_{r=1}^{n}\frac{1}{2(r+1)}\left[\frac{1}{r}-\frac{1}{r+2}\right]=\frac{1}{2}\left[\sum_{r=1}^{n}\frac{1}{r(r+1)}-\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}\right]$

$$= \frac{1}{2}\left[\sum_{r=1}^{n}\left(\frac{1}{r}-\frac{1}{r+1}\right)-\sum_{r=1}^{n}\left(\frac{1}{r+1}-\frac{1}{r+2}\right)\right]=\frac{1}{2}\left[\left(1-\frac{1}{n+1}\right)-\left(\frac{1}{2}-\frac{1}{n+2}\right)\right]$$

$$= \frac{1}{2}\left[\frac{n}{n+1}-\frac{n}{2(n+2)}\right]=\frac{n}{2}\left[\frac{1}{n+1}-\frac{1}{2(n+2)}\right]$$

Now अब $\quad S_n=(n+1)\frac{n}{2}\left[\frac{1}{n+1}-\frac{1}{2(n+2)}\right]-\left(\frac{1}{2}-\frac{1}{n+2}\right)$

$$=\frac{n}{2}-\frac{n(n+1)}{4(n+2)}-\frac{n}{2(n+2)}=\frac{2n^2+4n-n^2-n-2n}{4(n+2)}=\frac{n^2+n}{4(n+2)}=\frac{n(n+1)}{4(n+2)}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 24 to 25

The first four terms of a sequence are given by $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$.
The general term is given by $T_n = A\alpha^{n-1} + B\beta^{n-1}$ where A, B, α, β are independent of n and A is positive.

श्रेणी के प्रथम चार पद $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$ द्वारा दिये गये है।
श्रेणी का सामान्य पद $T_n = A\alpha^{n-1} + B\beta^{n-1}$ द्वारा दिया जाता है, जहां A, B, α तथा β, n से स्वतंत्र है तथा A धनात्मक है।

24. The value of $(\alpha^2 + \beta^2 + \alpha\beta)$ is equal to :

$(\alpha^2 + \beta^2 + \alpha\beta)$ का मान होगा :

(A) 1 (B) 2 (C) 5 (D) 4

Ans. B

Sol. $T_1 = 0, T_2 = 1, T_3 = 1, T_4 = 2$
$T_n = A\alpha^{n-1} + B\beta^{n-1}$
$T_1 = 0 = A + B \Rightarrow A = -B$
$T_2 = A\alpha + B\beta = 1 \Rightarrow A(\alpha) + (-A)\beta = 1 \Rightarrow A(\alpha - \beta) = 1$ ...(1)
$T_3 = A\alpha^2 + B\beta^2 = 1 \Rightarrow A(\alpha^2 - \beta^2) = 1$ ...(2)
$T_4 = A\alpha^3 + B\beta^3 = 2 \Rightarrow A(\alpha^3 - \beta^3) = 2$ ...(3)

$$\frac{(1)}{(2)}:\frac{A(\alpha-\beta)}{A(\alpha^2-\beta^2)}=\frac{1}{1}\Rightarrow \alpha+\beta=1$$

$$\frac{(1)}{(3)}:\frac{A(\alpha-\beta)}{A(\alpha^3-\beta^3)}=\frac{1}{2}$$

$$\frac{1}{\alpha^2+\beta^2+\alpha\beta}=\frac{1}{2}$$

$\alpha^2 + \beta^2 + \alpha\beta = 2$

$$\frac{(2)}{(3)}:\frac{A(\alpha-\beta)(\alpha+\beta)}{A(\alpha-\beta)(\alpha^2+\beta^2+\alpha\beta)}=\frac{1}{2}\Rightarrow\alpha+\beta=1$$

$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta = 1$
$\alpha\beta = 1 - (\alpha^2 + \beta^2 + \alpha\beta)$
$\alpha\beta = -1$
$(\alpha - \beta)2 = \alpha^2 + \beta^2 - 2\alpha\beta$
$= (\alpha + \beta)^2 - 4\alpha\beta$
$= 1 + 4$
$(\alpha - \beta)^2 = 5$

$$\left(\frac{1}{A}\right)^2=5\Rightarrow A^2=\frac{1}{5}$$

25. The value of $5(A^2 + B^2)$ is equal to :

$5(A^2 + B^2)$ का मान होगा :

(A) 2 (B) 4 (C) 6 (D) 8

Ans. A

Sol. $5\left(A^2+B^2\right)=5\times2\times\frac{1}{5}=2$

Paragraph for Question Nos. 26 to 28

Let x, y, z are positive reals and $x + y + z = 60$ and $x > 3$.

माना x, y, z धनात्मक वास्तविक है तथा $x + y + z = 60$ तथा $x > 3$ है।

26. Maximum value of $(x - 3)(y + 1)(z + 5)$ is :

$(x - 3)(y + 1)(z + 5)$ का अधिकतम मान होगा :

(A) (17)(21)(25) (B) (20)(21)(23) (C) (21)(21)(21) (D) (23)(19)(15)

Ans. C

Sol. $x + y + z = 60, \quad x > 3$

$$\frac{(x-3)+(y+1)+(z+5)}{3}\geq\left[(x-3)(y+1)(z+5)\right]^{\frac{1}{3}}$$

$$\left(\frac{60-3+1+5}{3}\right)^3\geq(x-3)(y+1)(z+5)$$

$(21)^3 \geq (x - 3)(y + 1)(z + 5)$

27. Maximum value of $(x-3)(2y+1)(3z+5)$ is :

(A) $\frac{(355)^3}{3^3.6^2}$ (B) $\frac{(355)^3}{3^3.6^3}$ (C) $\frac{(355)^3}{3^2.6^3}$ (D) None of these

$(x-3)(2y+1)(3z+5)$ का अधिकतम मान होगा :

(A) $\frac{(355)^3}{3^3.6^2}$ (B) $\frac{(355)^3}{3^3.6^3}$ (C) $\frac{(355)^3}{3^2.6^3}$ (D) None of these

Ans. A

Sol. $$\frac{(x-3)+\left(y+\frac{1}{2}\right)+\left(z+\frac{5}{3}\right)}{3} \geq \left[(x-3)\left(y+\frac{1}{2}\right)\left(z+\frac{5}{3}\right)\right]^{\frac{1}{3}}$$

$$\frac{\left(60-3+\frac{1}{2}+\frac{5}{3}\right)^3}{3} \geq \frac{(x-3)(2y+1)(3z+5)}{6}$$

$$\frac{(355)^3}{3^3 6^2} \geq (x-3)(2y+1)(3z+5)$$

28. Maximum value of xyz is :

xyz का अधिकतम मान होगा :

(A) 8×10^3 (B) 27×10^3 (C) 64×10^3 (D) 125×10^3

Ans. A

Sol. $$\frac{x+y+z}{3} \geq (xyz)^{\frac{1}{3}}$$

$$\frac{60}{3} \geq (xyz)^{\frac{1}{3}}$$

$$(20)^3 \geq xyz$$

$$8 \times 10^3 \geq xyz$$

Paragraph for Question Nos. 29 to 30

Given the sequence of number $a_1, a_2, a_3 \ldots\ldots\ldots\ldots a_{1005}$

which satisfy $\frac{a_1}{a_1+1} = \frac{a_2}{a_2+3} = \frac{a_3}{a_3+5} = \ldots\ldots\ldots = \frac{a_{1005}}{a_{1005}+2009}$

Also $a_1 + a_2 + a_3 + \ldots\ldots\ldots\ldots + a_{1005} = 2010$

संख्याओं की श्रेणी a_1, a_2, a_3 ..... a_{1005} दी गई है।

जो कि $\frac{a_1}{a_1+1}=\frac{a_2}{a_2+3}=\frac{a_3}{a_3+5}=\ldots\ldots\ldots=\frac{a_{1005}}{a_{1005}+2009}$ तथा

$a_1 + a_2 + a_3 + \ldots\ldots\ldots\ldots + a_{1005} = 2010$ को संतुष्ट करती है।

29. Nature of the sequence is :

श्रेणी की प्रकृति होगी :

(A) A.P. (B) G.P. (C) A.G.P. (D) H.P.

Ans. A

Sol. $a_1, a_2, a_3, \ldots\ldots\ldots, a_{1005}$

$$\frac{a_1}{a_1+1}=\frac{a_2}{a_2+3}=\frac{a_3}{a_3+5}=\ldots\ldots\ldots=\frac{a_{1005}}{a_{1005}+2009}$$

$a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_{1005} = 2010$

$a_1(a_2 + 3) = a_2(a_1 + 1) \Rightarrow \frac{a_2}{a_1} = 3 \Rightarrow a_2 = 3a_1$

$a_2(a_3 + 5) = a_3 (a_2 + 3) \Rightarrow \frac{a_3}{a_2} = \frac{5}{3} \Rightarrow a_3 = 5a_1$

$a_3(a_4 + 7) = a_4(a_3 + 5) \Rightarrow \frac{a_4}{a_3} = \frac{7}{5} \Rightarrow a_4 = 7a_1$

$a_1 + a_2 + a_3 + \ldots\ldots\ldots + a_{1005} = 2010$

$a_1 + 3a_1 + 5a_1 + \ldots\ldots\ldots + 2009a_1 = 2010$

$a_1[1 + 3 + 5 \ldots\ldots\ldots + 2009] = 2010$

$$a_1\left[\frac{1005}{2}\left[2\times1+(1004)\times2\right]\right]=2010$$

$$a_1=\frac{2}{1005},\qquad d=2a_1=\frac{4}{1005}$$

Series form an A.P.

30. 21st term of the sequence is equal to :

श्रेणी का 21वां पद होगा :

(A) $\frac{86}{1005}$ (B) $\frac{83}{1005}$ (C) $\frac{82}{1005}$ (D) $\frac{79}{1005}$

Ans. C

Sol. $T_{21} = a_1 + 20d = a_1 + 40a_1 = 41a_1$

$$T_{21} = \frac{82}{1005}$$

Paragraph for Question Nos. 31 to 33

Let V_r denote the sum of the first 'r' terms of an arithmetic progression (A.P.) whose first term is 'r' and the common difference is (2r – 1).

Let $T_r = V_{r+1} - V_r - 2$ and $Q_r = T_{r+1} - T_r$ for r = 1, 2, ...

माना V_r एक समान्तर श्रेणी (A.P.) के प्रथम 'r' पदों के योग को प्रदर्शित करता है जिसका प्रथम पद 'r' है एवं सर्वान्तर (2r – 1) है।

माना $T_r = V_{r+1} - V_r - 2$ तथा $Q_r = T_{r+1} - T_r$ जहाँ r = 1, 2, ...

31. The sum $V_1 + V_2 + ... + V_n$ is :

$V_1 + V_2 + ... + V_n$ का योग है :

(A) $\frac{1}{12}n(n+1)(3n^2 - n + 1)$ (B) $\frac{1}{12}n(n+1)(3n^2 + n + 2)$

(C) $\frac{1}{2}n(2n^2 - n + 1)$ (D) $\frac{1}{3}(2n^3 - 2n + 3)$

Ans. B

Sol. $a_1 + a_2 + a_3 + ... + a_r = v_r$

$a_1 = r, d = 2r - 1$

$$v_r = \left(\frac{r}{2}\right)\left[2\times r + (r-1)(2r-1)\right]$$

$$= \frac{r}{2}\left[2r + 2r^2 - 3r + 1\right]$$

$$v_r = \frac{r}{2}\left[2r^2 - r + 1\right]$$

$$v_r = r^3 - \frac{r^2}{2} + \frac{r}{2}$$

$$v_1 + v_2 + ...+ v_n = \sum_{r=1}^{n}\left(r^3 - \frac{r^2}{2} + \frac{r}{2}\right)$$

$$= \left[\frac{n(n+1)}{2}\right]^2 - \frac{n(n+1)(2n+1)}{2\times 6} + \frac{n(n+1)}{2\times 2}$$

$$= \frac{n(n+1)}{2}\left[\frac{n(n+1)}{2} - \left(\frac{2n+1}{6}\right) + \frac{1}{2}\right]$$

$$= \frac{n(n+1)}{2}\left[\frac{3n(n+1)-(2n+1)-3}{6}\right]$$

$$= \frac{n}{12}(n+1)\left(3n^2+n+2\right)$$

32. T_r is always :

(A) an odd number (B) an even number (C) a prime number (D) a composite number

T_r हमेशा :

(A) एक विषम संख्या है (B) एक सम संख्या (C) एक अभाज्य संख्या (D) एक मिश्र संख्या

Ans. D

Sol. $T_r = v_{r+1} - v_r - 2$

$$= \left[(r+1)^3 - \frac{(r+1)^2}{2} + \left(\frac{r+1}{2}\right)\right] - \left[r^3 - \frac{r^2}{2} + \frac{r}{2}\right] - 2$$

$$= \left[(r+1)^2 - r^3\right] - \frac{1}{2}\left[(r+1)^2 - r^2\right] + \frac{1}{2}[r+1-r] - 2$$

$$= \left[(r+1)^2 + r^2 + r(r+1)\right] - \left[\frac{2r+1}{2}\right] + \frac{1}{2} - 2$$

$=[3r^2 + 3r + 1] - r - 2$
$= 3r^2 + 2r - 1$
$= (3r - 1)(r + 1)$
Composite number

33. Which one of the following is a correct statement?

(A) $Q_1, Q_2, Q_3, \ldots$ are in A.P. with common difference 5
(B) $Q_1, Q_2, Q_3, \ldots$ are in A.P. with common difference 6
(C) $Q_1, Q_2, Q_3, \ldots$ are in A.P. with common difference 11
(D) $Q_1 = Q_2 = Q_3 = \ldots$

निम्नलिखित में से कौनसा कथन सही है?

(A) $Q_1, Q_2, Q_3, \ldots$ A.P. में है जिसका सर्वान्तर 5 है
(B) $Q_1, Q_2, Q_3, \ldots$ A.P. में है जिसका सर्वान्तर 6 है
(C) $Q_1, Q_2, Q_3, \ldots$ A.P. में है जिसका सर्वान्तर 11 है
(D) $Q_1 = Q_2 = Q_3 = \ldots$

Ans. B

Sol. $a = r$ and $d = 2r - 1$

$V_r = \frac{r}{2}\,(2r + (r-1)\,(2r-1)] = \frac{r}{2}[2r + 2r^2 - 3r + 1] = \frac{r}{2}(2r^2 - r + 1)$

$V_{r+1} = \left(\frac{r+1}{2}\right)(2\,(r+1)^2 - (r+1) + 1) = \left(\frac{r+1}{2}\right)(2(r^2 + 2r + 1) - r)$

$= \left(\frac{r+1}{2}\right)(2r^2 + 3r + 2) = \frac{2r^3 + 5r^2 + 5r + 2}{2}$

$T_r = V_{r+1} - V_r - 2 = 3r^2 + 2r - 1 = 3r^2 + 3r - r - 1 = (3r-1)\,(r+1)$ (composite no.)

$\sum V_r = r^3 - \frac{r^2}{2} + \frac{r}{2} = \frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4} = \frac{n(n+1)}{12}\,(3n^2 + 3n - 2n - 1 + 3)$

$= \frac{n(n+1)}{12}\,(3n^2 + n + 2)$

$Q_r = T_{r+1} - T_r = 3(r+1)^2 + 2(r+1) - 1 - 3r^2 - 2r + 1 = 3r^2 + 6r + 3 + 2r + 1 - 3r^2 - 2r + 1 = 6r + 5$

Common difference of Q_1, Q_2, Q_3 is 6 and they are in A.P.

Hindi. $a = r$ तथा $d = 2r - 1$

$V_r = \frac{r}{2}\,(2r + (r-1)\,(2r-1)] = \frac{r}{2}[2r + 2r^2 - 3r + 1] = \frac{r}{2}(2r^2 - r + 1)$

$V_{r+1} = \left(\frac{r+1}{2}\right)(2\,(r+1)^2 - (r+1) + 1) = \left(\frac{r+1}{2}\right)(2(r^2 + 2r + 1) - r)$

$= \left(\frac{r+1}{2}\right)(2r^2 + 3r + 2) = \frac{2r^3 + 5r^2 + 5r + 2}{2}$

$T_r = V_{r+1} - V_r - 2 = 3r^2 + 2r - 1 = 3r^2 + 3r - r - 1 = (3r-1)\,(r+1)$ (संयुक्त संख्या)

$\sum V_r = r^3 - \frac{r^2}{2} + \frac{r}{2} = \frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{12} + \frac{n(n+1)}{4} = \frac{n(n+1)}{12}\,(3n^2 + 3n - 2n - 1 + 3)$

$= \frac{n(n+1)}{12}\,(3n^2 + n + 2)$

$Q_r = T_{r+1} - T_r = 3(r+1)^2 + 2(r+1) - 1 - 3r^2 - 2r + 1 = 3r^2 + 6r + 3 + 2r + 1 - 3r^2 - 2r + 1 = 6r + 5$

अतः Q_1, Q_2, Q_3 समान्तर श्रेढ़ी में हैं जिनका सार्वअन्तर 6 है

MATCH THE COLUMN TYPE QUESTION(S)

34. Natural numbers are divided into groups as follows {1}, {2, 3}, {4, 5, 6}, {7, 8, 9, 10} …….. Now match the following :

Column - I	**Column - II**
(A) Sum of numbers in 100th group	(P) 125500
(B) Middle number in 501th group	(Q) 500050
(C) Last number in 1000th group	(R) 319601
(D) First number of 800th group	(S) 500500

प्राकृतिक संख्याओं का एक समूह निम्न प्रकार से विभक्त किया गया है, {1}, {2, 3}, {4, 5, 6}, {7, 8, 9, 10} …….. तो निम्न का मिलान कीजिये :

Column - I	**Column - II**
(A) 100वें समूह में संख्याओं का योगफल है	(P) 125500
(B) 501वें समूह में मध्य वाली संख्या है	(Q) 500050
(C) 1000वें समूह में अन्तिम संख्या है	(R) 319601
(D) 800वें समूह में प्रथम संख्या है	(S) 500500

Ans. A-Q; B-P;C-S;D-R

Sol. {1}, {2, 3}, {4, 5, 6}, {7,8,9,10}, ...

Number of digits in 100th group

$t_{100} = 1 + (100 - 1) \times 1$

$t_{100} = 100$

First term of 100th group

$S = 1 + 2 + 4 + 7 + ... + T_{n-1} + T_n$

$S = \quad 1 + 2 + 4 + ... + T_{n-2} + T_{n-1} + T_n$

$$0 = 1 + \underbrace{(1 + 2 + 3 + ...)}_{(n-1) \text{ term}} - T_n$$

$$T_n = 1 + \frac{(n-1)n}{2} = \frac{2 + n(n-1)}{2}$$

$$T_{100} = \frac{2 + 100 \times 99}{2} = 4951$$

$$\text{Sum of numbes in } 100^{th} \text{ group} = \frac{100}{2}[2 \times 4951 + 99 \times 1] = 500050$$

$$\text{(B) First tem of } 501^{th} \text{ goup} = \frac{2 + (501) \times 500}{2} = 125251$$

middle term of 501th group = 125251 + 249 = 125500

$$\text{(C) First term of 1000th goup} = \frac{2 + 1000 \times 999}{2} = 499501$$

Last term of 1000th goup = 499501 + 999 = 500501

(D) First number of 800th group $= \frac{2+800\times 799}{2} = 319601$

35. Match the following columns :

Column I	**Column II**
(A) The value of $(2^{1/4}\, 4^{1/8}\, 8^{1/16}\, 16^{1/32} \ldots)$ is	(P) 1
(B) S_n denote the sum of the first n terms of an A. P. If $S_{2n} = 3S_n$, then the value of S_{3n}/S_n is	(Q) 2
(C) The value of $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\right)}$ is	(R) 4
(D) If H is H. M. between p and q, then $\frac{H}{p}+\frac{H}{q} =$	(S) 6

निम्न Columns को सुमेलित कीजिए :

Column I	**Column II**
(A) $(2^{1/4}\, 4^{1/8}\, 8^{1/16}\, 16^{1/32} \ldots)$ का मान है	(P) 1
(B) S_n किसी स.श्रे. के प्रथम n पदों के योगफल को व्यक्त करता है यदि $S_{2n} = 3S_n$, तो S_{3n}/S_n का मान है	(Q) 2
(C) $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\right)}$ का मान है	(R) 4
(D) यदि H संख्याओं p तथा q के मध्य ह.माध्य है तो $\frac{H}{p}+\frac{H}{q} =$	(S) 6

Ans. A - Q, B - S, C - R, D - Q

Sol. (A) $2^{\frac{1}{4}}\cdot 2^{\frac{2}{8}}\cdot 2^{\frac{3}{16}}\cdot 2^{\frac{4}{32}}\ldots\ldots$

$$S = 2^{\left(\frac{1}{4}+\frac{2}{8}+\frac{3}{16}+\frac{4}{32}+\ldots\ldots\right)}$$

$$S' = \frac{1}{2^2}+\frac{2}{2^3}+\frac{3}{2^4}+\frac{4}{2^5}+\ldots\ldots\ldots \quad ...(1)$$

$$\frac{S'}{2} = 0+\frac{1}{2^3}+\frac{2}{2^4}+\frac{3}{2^5}+\ldots\ldots\ldots \quad ...(2)$$

$$\frac{S'}{2} = \frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}+\frac{1}{2^5}+\ldots\ldots\ldots = \frac{\frac{1}{2^2}}{1-\frac{1}{2}} = \frac{1}{2}$$

(B) $S_n = \frac{n}{2}\left[2a+(n-1)d\right]$

$$S_{2n} = \frac{2n}{2}\left[2a + (2n-1)d\right] = 3 \times \frac{n}{2}\left[2a + (n-1)d\right]$$

$4a + (4n - 2)d = 6a + (3n - 3)d$

$-2a + (n + 1)d = 0$

$2a = +(n + 1)\, d$ ...(i)

$$\frac{S_{3n}}{S_n} = \frac{\frac{3n}{2}\left[2a + (3n-1)d\right]}{\frac{n}{2}\left[2a + (n-1)d\right]} = \frac{3\left[+nd + d + +3nd - d\right]}{\left[+nd + d + nd - d\right]}$$

$$= 3\left[\frac{4nd}{2nd}\right] = 6$$

(C) $(0.2)^{\log_{\sqrt{5}}\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+.....\right)}$

$$= (0.2)^{2\log_5\left(\frac{\frac{1}{4}}{1-\frac{1}{2}}\right)}$$

$$= (0.2)^{2\log_5 \frac{1}{2}}$$

$$= \left(\frac{1}{5}\right)^{-2\log_5 2}$$

$$= 5^{2\log_5 2} = 5^{\log_5 2^2} = 2^2 = 4$$

(D) $H = \dfrac{2pq}{p+q}$

$$H\left[\frac{1}{p} + \frac{1}{q}\right] = H\left[\frac{p+q}{pq}\right] = 2$$